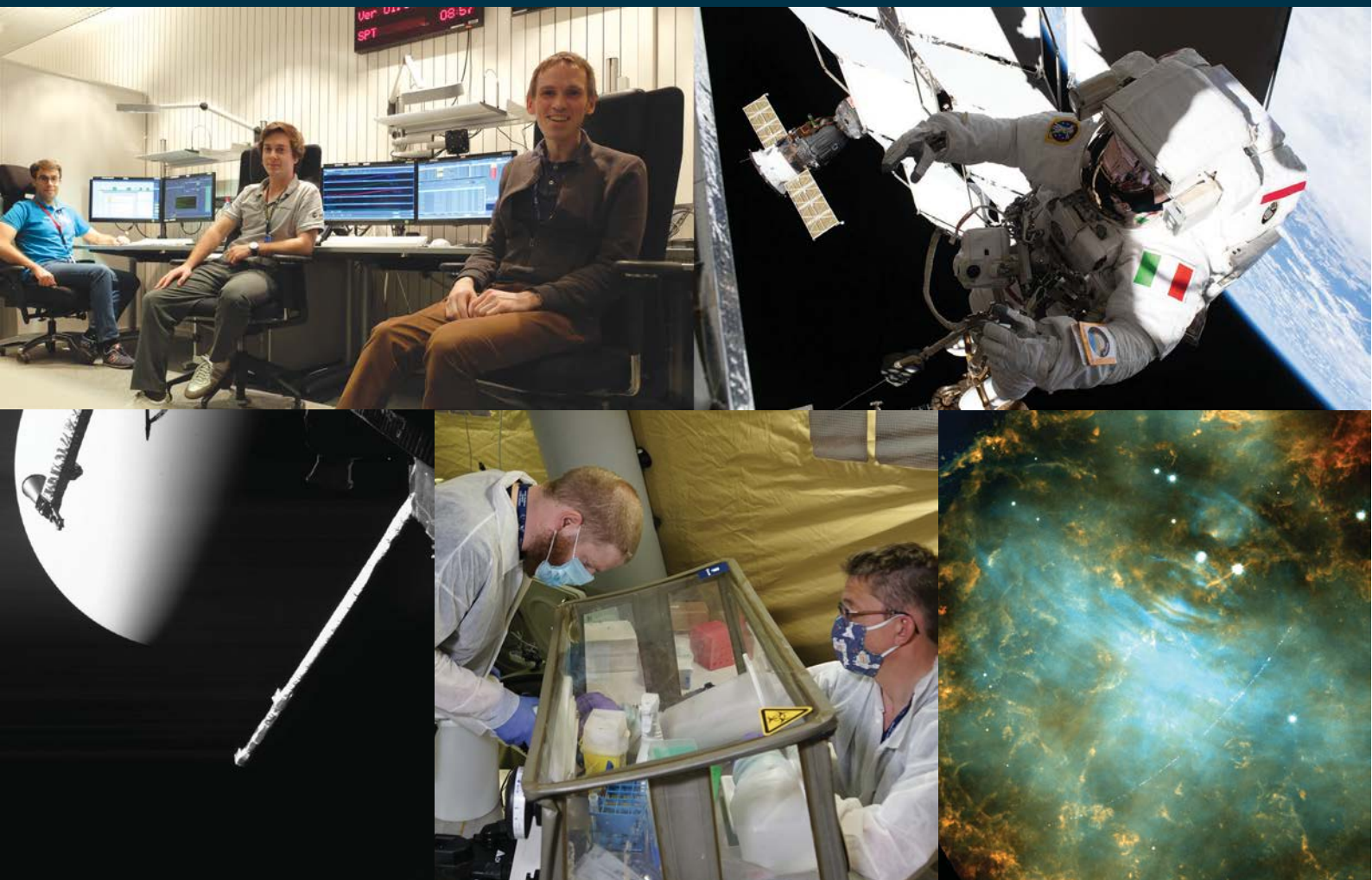


ESA ANNUAL REPORT 2020



The Report is published in accordance with the terms of the Convention for the establishment of the European Space Agency, Article XII 1(b), which require the Director General to make an annual report to the Council.

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ESA Annual Report 2020

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Foreword

2020 was marked by a COVID-19 pandemic which has changed our ways both of working and of conducting the meetings so integral to the functioning of the European Space Agency (ESA). Throughout that period, the Agency, its staff and Member State representatives demonstrated great resilience as well as an ability to adapt under these quite exceptional circumstances.

Succeeding Jean-Yves Le Gall as Chair of Council on 1 July was a challenge I was pleased to accept and one that I will always endeavour to meet with total commitment.

In 2020, despite the grave public health situation, the ESA Council continued to be very active, holding five regular sessions in either hybrid or fully virtual format. During that time, the principal focus of ESA activities was on beginning to implement the programmatic and political decisions taken at Space19+ – the ESA Council meeting at ministerial level – in Seville on 27-28 November 2019.

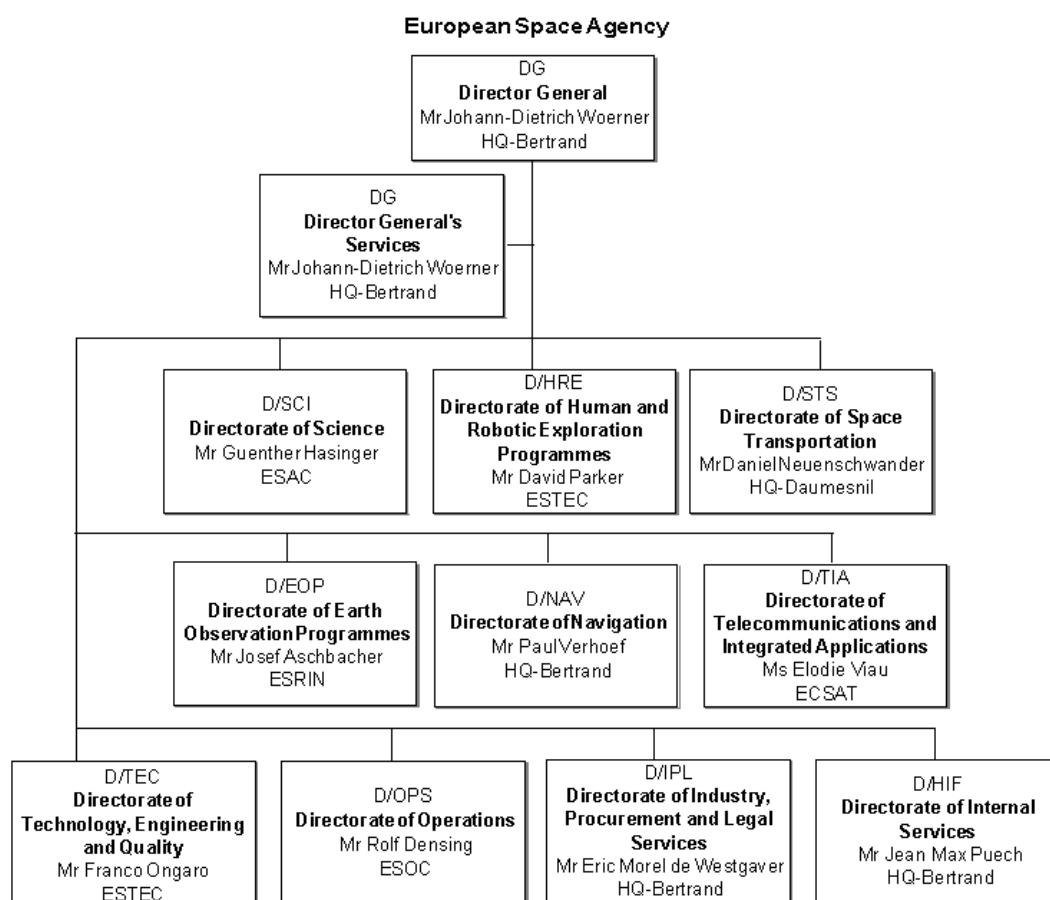
In addition, at its June meeting the ESA Council unanimously approved the Association Agreement between ESA and the Government of the Republic of Latvia (the Agreement entered into force on 27 July for a period of seven years) and at its December meeting, appointed Josef Aschbacher to succeed Johann-Dietrich Wörner as Director General of ESA, with Mr Aschbacher set to assume his new responsibilities in 2021.

Another important milestone was the 10th Space Council (joint and concomitant meeting of the Council of the European Union and of the Council of ESA at ministerial level) held in virtual mode on 20 November 2020. It adopted Resolution ESA/C-M/CCXCII/Res.1 (Final), “Orientations on the European contribution in establishing key principles for the global space economy”. Through this Resolution, Council stresses the need for European non-dependent technology and also recalls the importance of autonomous access to space and coordinated actions by ESA, Member States and the EU. With this latter aim in mind, 2020 also saw the start of the negotiations on the Financial Framework Partnership Agreement (FFPA), the aim of which is to establish the cooperation arrangements between the EU and ESA and take further steps towards reinforcing the sound and fruitful relations that already exist between the two organisations.

Anna Rathsmann

Council Chair

Management Structure



Status 1 September 2020

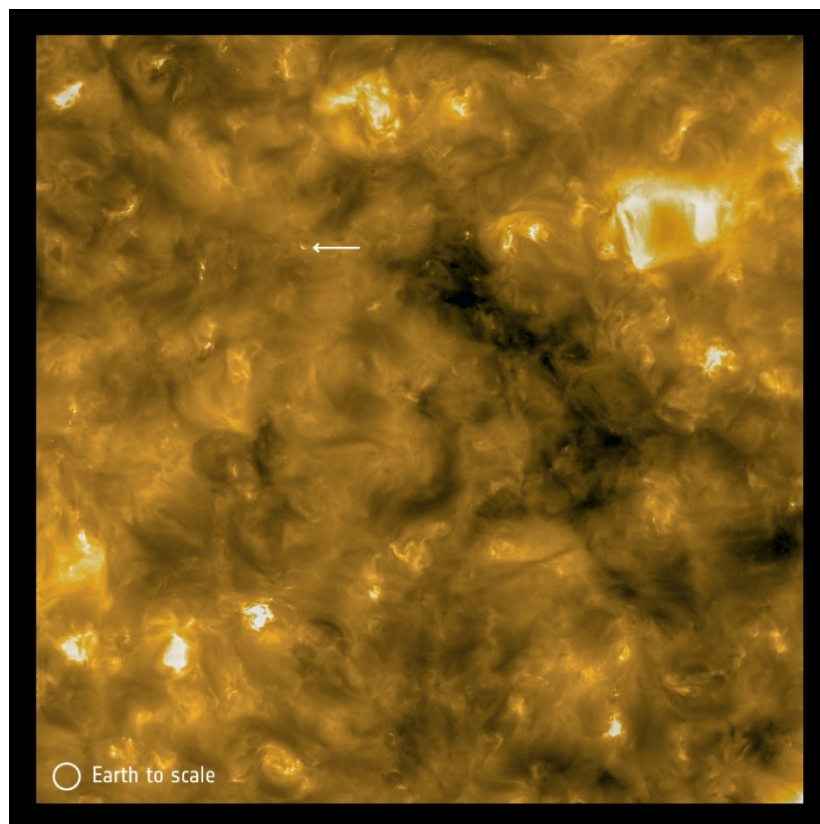
Programmes and Activities

Science & Exploration

Science

In 2020 two new missions in the Science Programme began delivering data to the scientific community: the small mission CHEOPS, launched on 18 December 2019 on a Vega rocket from Kourou and the medium mission Solar Orbiter, implemented in collaboration with NASA and launched on an Atlas V rocket from the Kennedy Space Center (KSC) on 10 February 2020. Both missions were successfully commissioned and are now in routine operations, with the first scientific results already having been announced.

Solar Orbiter will observe the Sun from close quarters, indeed getting closer to the Sun even than Mercury. Since its launch, the spacecraft has been travelling on a complex path: approaching the Sun requires substantial amounts of energy to reduce the spacecraft's orbital velocity and thus angular momentum around the Sun, resulting in a decreasing orbit perihelion. Both Solar Orbiter and BepiColombo (see below) make use of a number of flybys of Earth and Venus, thereby receiving a "gravitational kick" that slows down their orbit, making them plunge towards the inner Solar System, giving substantial additional "help" to their thrusters. The gravitational kick is also used to change Solar Orbiter's orbital plane, allowing it slowly to gain its vantage point from which to observe the Sun's polar regions, yet another "first" that will be achieved by the mission.



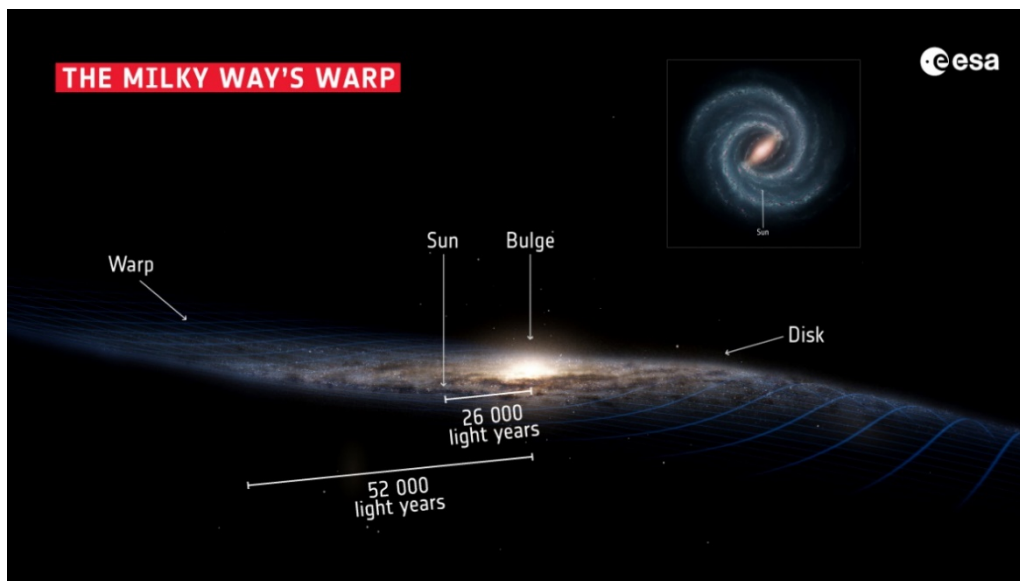
Solar Orbiter spots "campfires" on the Sun – Credits: ESA

Solar Orbiter flew by Venus in 2020, and while it was never designed to investigate the planet, this provided the opportunity to exercise the instruments able to be operated and to acquire useful scientific data on Venus's environment. Prior to its Venus flyby, Solar Orbiter had performed its first "close but not so close" approach to the Sun, at a distance of about half of the Sun-Earth distance, already making it the closest solar telescope ever to have observed the Sun. One notable feature of the first close-up observations has been the very small, short-lived flaring events that have been dubbed "campfires". The way in which the solar corona is heated to million degree temperatures has been a subject of lively scientific debate since the first observations of this hot plasma, some forty years ago. One long-standing hypothesis has been that the high temperatures are maintained through the conversion of energy stored in the magnetic field into heat taking place in very numerous, very small flaring events. Observation of the campfires provides strong evidence in that direction.

A new mission, Ariel, was added in 2020 to the Science Programme's portfolio, with a launch planned for 2029. Ariel, which will study the chemical composition of a large sample of extrasolar planets, is the third mission in the Science Programme dedicated to the study of exoplanets, showing the strong interest of Europe's scientific community in this topic. The three missions are highly complementary: while PLATO's focus is on the search for yet-to-be discovered rocky, temperate exoplanets, Ariel will analyse the atmospheric composition of known exoplanets, representing a natural "next step". The availability of CHEOPS, PLATO and Ariel, together with the access to JWST ensured by ESA's participation in this NASA flagship mission, will put European scientists in a unique position to lead in what is a competitive and coveted scientific domain.

BepiColombo, the newest planetary explorer in the Science Programme, will for the first time perform a detailed study of Mercury, the planet closest to the Sun. Following its launch in October 2018, the mission has been on a long circuitous trip towards Mercury, following the same principles mentioned above for Solar Orbiter (although using very efficient "solar electric propulsion"). BepiColombo performed, in 2020, flybys of Earth and Venus. While (again like Solar Orbiter) BepiColombo was never designed to investigate Venus, this again provided an opportunity to acquire useful scientific data on Venus' environment and atmosphere. BepiColombo is also a major example of international cooperation in the Science Programme, consisting of two probes, one European and one provided by Japanese agency JAXA, both being brought to Mercury by a European carrier module.

Gaia is a unique mission in the Science Programme, surveying the sky and mapping our Galaxy in three dimensions. A detailed map of our galaxy lends itself to scores of scientific applications, so that the mission, launched in 2013, continues to be the one giving rise to the largest number of scientific publications in the Science Programme, with over 1600 scientific papers published in the year 2020, the highest ever number for a space science mission. Gaia's scientific output will very likely increase further, with the publication of the new, much improved version of the Gaia catalogue (Early Data Release 3, EDR3) in December 2020. One notable result among many that have appeared in 2020 concerns the origin of the "warp" in the disk of our galaxy, the Milky Way. While it has been known since the 1950s that the disk is not flat, Gaia has shown that this warp is not static, but rather that it changes with time.



The Milky Way's Warp – Credits: ESA

The Science Programme has a number of missions currently being implemented, with two, JUICE and Euclid, approaching launch. JUICE will be the first mission to study from close up the moon of one of the Solar System's giant planets. These moons (JUICE's final destinations) are fascinating worlds in themselves, and some, such as Ganymede are prime sites for possible examples of extraterrestrial life, given the strong evidence of an ocean of liquid water lying under the thick crust of ice that forms its surface. JUICE is currently on track to be launched in 2022 and will also make use of planetary flybys to facilitate its long trip toward its remote destination.

Euclid is a space telescope that will create a detailed map of the Universe by observing an unprecedented number of galaxies to pin down the nature of the elusive force currently accelerating the Universe's expansion known as "dark energy". While the implementation of Euclid progressed significantly in 2020, an unexpected challenge has arisen in its "VIS" wide-field optical camera. Depending on the outcome of a detailed investigation of the anomaly, the mission is expected to be launched in 2022 or 2023.

Implementation of the PLATO mission progressed significantly in 2020, with a number of challenges having been overcome. PLATO will monitor the light of a large number of stars, looking for small, short dips in their light curve caused by brief partial eclipses ("transits") as planets orbit around these stars. Transit surveys have already been used by European and US probes to discover most of the exoplanets known to date (such as the CNES mission COROT and the NASA missions Kepler and TESS). PLATO's uniqueness will be in its observing brighter stars for a longer time than any previous missions, making it capable for the first time of observing planets similar — in terms of size and distance from the parent stars — to our own Earth.

The Science Programme continues to operate a fleet of productive spacecraft. The flagship XMM-Newton X-ray observatory remains highly productive as it enters its third decade of operations. For example, in 2020 it detected, for the first time, an energetic X-ray flare from a cool substellar object with roughly a third the temperature of the Sun and only some 5% of the Sun's mass. It is the first time such an object has shown an energetic flare in X-rays.

In part due to the exceptional concentration of launches in 2018–2022 (when six missions will be launched, representing a cadence in excess of one per year), and aggravated by the ongoing COVID- 19

crisis, the Science Programme is facing, a peak of expenditures in 2020–2022. While the Programme’s mid- to long-term affordability is not in doubt, the short-term cash flow challenges have been discussed with the Administrative and Finance Committee (AFC).

The Science Programme is not resting on its laurels, and is also currently planning its longer-term future: three candidates for the medium mission following Ariel were studied throughout 2020, EnVision (a radar mission to Venus), SPICA (an infrared telescope) and THESEUS (a mission dedicated to the observation of gamma-ray bursts). EnVision and THESEUS were found to fit within the boundaries of a medium mission, and a selection between the two of them will be made in 2021. Two flagship missions that will follow JUICE are also in an advanced study phase. Athena, an X-ray telescope of unprecedented size and performance, is approaching the final steps of its definition, and could start to be implemented in 2023. LISA, the first space-based gravitational wave antenna, is also in an advanced definition phase and could be ready for its implementation shortly after Athena.

For the still longer-term, a new planning exercise (“Voyage 2050”) is currently being carried out defining the scientific priorities for the Science Programme for the period following the launch of the LISA mission, up to 2050. The new long-term plan is being built following a broad community consultation, that began with a “Call for White Papers” issued in 2019.

Human Spaceflight and Robotic Exploration

Highlights

Luca Parmitano completed his six-month *Beyond* mission on the International Space Station (ISS) and returned to Earth, on the Kazakh steppes, in the Russian Soyuz MS-13 spacecraft on 6 February. From there he returned directly to the European Astronaut Centre where he was met by his family and colleagues and addressed the media. Afterwards, he worked with researchers to gather baseline data and undertook an extensive programme of rehabilitation supported by ESA experts.



Credits: ESA



Credits: ESA

The Human and Robotic Exploration Science Data Centre (SDC) in Madrid was officially inaugurated on 13 February. The event was well attended by Spanish academic and political representatives including Spanish Minister Pedro Duque. The SDC provides a high-quality service of knowledge distribution and preservation, for the benefit of the scientific community and the general public.

The joint ESA-Roscosmos project team evaluated all the activities needed for an authorisation to launch ExoMars. After due consideration of the recommendations provided by European and Russian

Inspectors General, ExoMars experts, and the ExoMars Tracking Committee, it was concluded that the tests necessary to make all components of the spacecraft fit for the Mars adventure needed more time to complete. It was decided to postpone the launch from July/August 2020 to between August and October 2022. The decision to delay the launch was announced publicly in a media event on 12 March with the ESA Director General, the Director of Human and Robotic Exploration and the ExoMars Team Leader.

The fridge-sized European Drawer Rack Mark 2 (EDR2) was launched aboard the Japanese HTV-9 cargo vehicle on 20 May. The new experiment facility is part of a comprehensive upgrade of the Columbus laboratory module. EDR2 is designed to run in parallel with the original EDR, providing even greater opportunities for science in space. After the end of the Fluid Science Laboratory (FSL) experiments by the end of 2021, the two racks will be swapped and EDR2 will be commissioned.

The negotiated draft ESA/NASA Mars Sample Return (MSR) flight elements Memorandum of Understanding (MoU) was signed by the NASA Administrator, Jim Bridenstine on 23 September and the ESA Director General on 5 October. ESA will provide the Earth Return Orbiter (ERO), the Sample Fetch Rover (SFR) and the Sample Transfer Arm (STA).

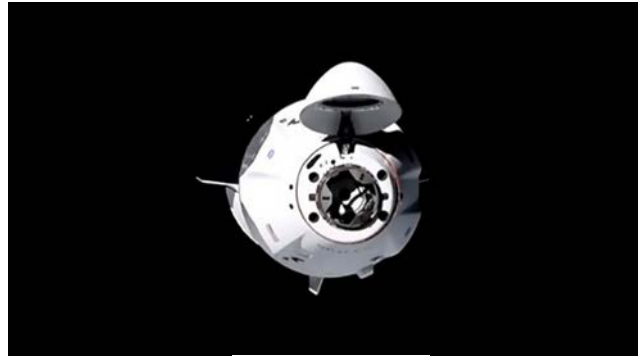
The historic Memorandum of Understanding between ESA and NASA concerning the cooperation on the civil Gateway was signed (remotely) by the ESA Director General and NASA Administrator on 27 October. The Memorandum of Understanding specifies the Partners' respective contributions, rights and obligations to the first human outpost in lunar orbit. The US also signed Gateway Memorandums of Understanding with the Canadian Government in mid-November, and with the Japanese Government at the end of December.

2 November 2020 marked exactly 20 years since the first crew took up residence on the ISS. Since then, 240 people including 18 ESA astronauts have lived and worked on the orbital outpost, carrying out essential research to benefit life on Earth.



To mark the 20th anniversary of continuous habitation of the ISS, ESA commissioned two graphic artists to illustrate the Station from two perspectives - Credits: ESA

Crew-1 Dragon, the first operational SpaceX mission to the ISS on behalf of NASA, arrived at the ISS on 16 November, 27 hours after launching from Kennedy Space Center (KSC) in Florida atop a Falcon 9 rocket. With the arrival of the four astronauts, the ISS crew was expanded to seven people, increasing the time available for scientific research.



Credits: NASA

The Atmosphere Space Interactions Monitor (ASIM), located outside the ISS since commissioning in June 2018, has been monitoring high-energy events that occur above clouds. An article about the ASIM instrument will be published in prestigious magazine *Science* in its January 2021 issue. In that article, the science team reports on the discovery that extremely energetic flashes of X-rays can be emitted from thunder clouds while at the same time illuminating a huge area of the lower ionosphere by UV radiation.

Astronauts

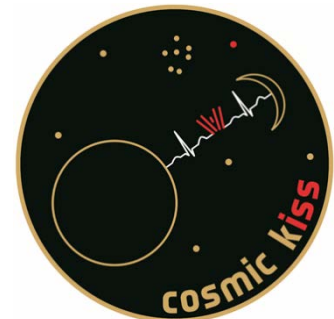


Credits: ESA

Thomas Pesquet was assigned to the second operational flight of SpaceX Crew Dragon spacecraft USCV2 due to launch to the ISS in April 2021. Thomas' second mission will be named *Alpha*, after Alpha Centauri, the closest stellar system to Earth.

Mattias Maurer was officially assigned to his first mission to the ISS, *Cosmic Kiss*. He will be launched aboard the third operational flight of SpaceX's Crew Dragon spacecraft (USCV3) no earlier than September 2021.

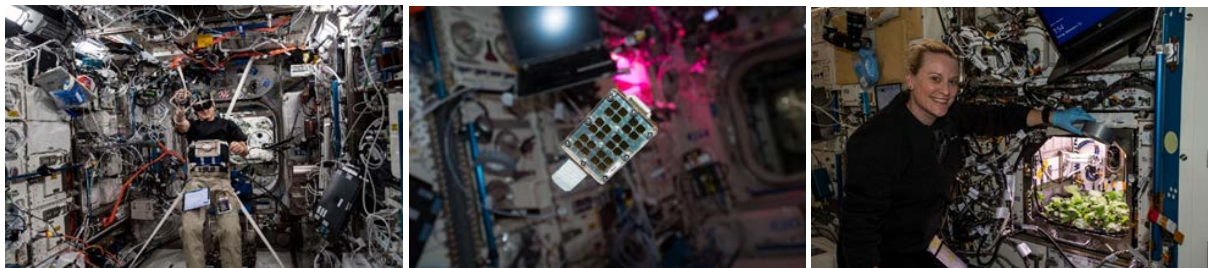
Matthias will live and work in orbit for roughly six months, carrying out science on behalf of researchers and international partners worldwide. His stay on orbit may well overlap with the latter part of Thomas Pesquet's mission.



Credits: ESA

ISS Experiments

A large number of experiments were performed on the ISS during 2020.



ISS experiments - Credits: ESA-NASA

Destination Mars

ExoMars

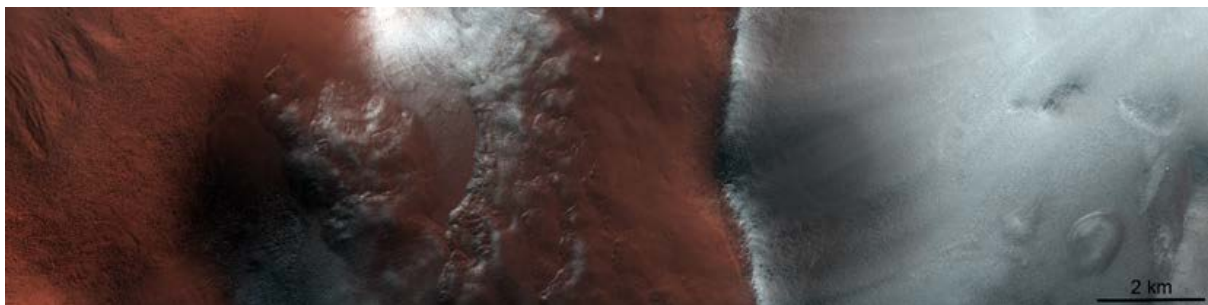
See entry about the ExoMars launch date under Highlights.

The Trace Gas Orbiter (TGO) spacecraft is healthy, and the new CaSSIS high-resolution camera software was updated and activated. CaSSIS began by taking two stereo pairs and ~250 monoscopic images per week, extending to one stereo pair per day.



Credits: ESA

TGO spotted new gas signatures at Mars, which can unlock new secrets about the Martian atmosphere, and will enable a more accurate determination of whether there is methane, a gas associated with biological or geological activity, at the planet. The spacecraft also spotted never-before-seen signatures of ozone (O_3) and carbon dioxide (CO_2), based on a full Martian year of observations by its sensitive Atmospheric Chemistry Suite (ACS). The findings were reported in two new papers published in *Astronomy & Astrophysics*.



Credits: ESA

The TGO was prepared to relay the MARS-2020 Perseverance mission data upon its arrival at Mars in 2021. The first UHF test for the mutual radio occultation experiment between Mars Express and the TGO was conducted in November and worked very well.

The Rover Module and the Spacecraft Composite (SCC) underwent integration and interface tests during 2020.

The parachute system had its High Altitude Drop Test (HADT-1b) in Oregon (US) in November, and an additional test, HADT2b for the qualification of the second main parachute is scheduled to take place in Kiruna (SE) in May 2021.

Mars Sample Return (MSR)

See entry under Highlights.

Destination Moon

As Europe prepares for future Moon missions, scientists have been studying the effects of lunar gravity to identify risks and create ways to keep astronauts healthy. ESA's space medicine team and its partners will use the planned Luna facility at the European Astronaut Centre (EAC) in Cologne (DE) to help with their research.

Powering Artemis to the Moon

The Multi-Purpose Crew Vehicle European Service Module 1 (MPCV-ESM1) is the first European contribution to the NASA-led Artemis programme while – programmatically – also serving as barter for ISS common operations costs provided by NASA to ESA. With a mass of over 15 tonnes at launch, the ESM is designed to provide the crew of Orion with propulsion, electrical power, thermal control, oxygen and water.

The (MPCV-ESM-1) Artemis 1 is at Kennedy Space Center (KSC). All open work was completed at the end of September, and the solar array final installation was completed at the beginning of October. Technicians finished encapsulating the ESM, and the acceptance and transfer of ownership to NASA took place in December, a historic moment.

Work on ESM-2 continued with an on-dock date of July 2021. COVID-19 constraints impacted work performed in Bremen by TASInI personnel. The Artemis 2 launch is scheduled to take place no earlier than the end of 2022.

The procurement of recurring equipment and subsystems for ESM-3 is in progress. TASInI delivered the primary structure to Airbus in Bremen on 12 October. The ESM-3 KSC on-dock date remains officially 1 December 2022. ESM-3 is not on the critical path for the proposed Artemis 3 human lunar landing, which depends on the development of a Human Landing Systems (HLS), for which contracts have not yet been placed by NASA.

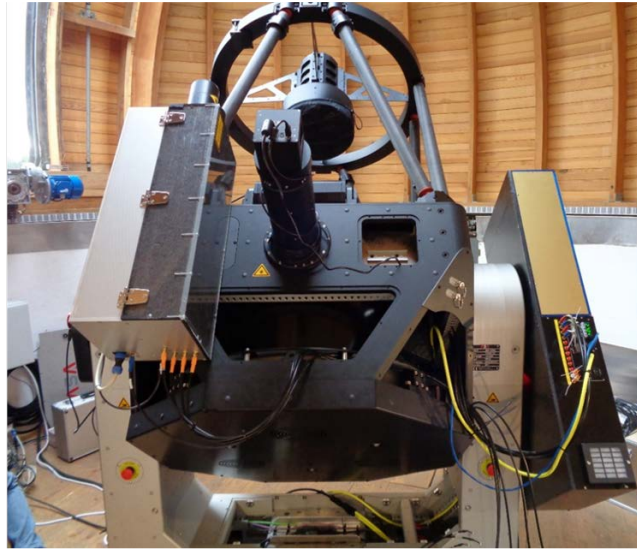


ESM-3 arrives in Bremen - Credits: ESA

Space Safety & Security

Space Safety

Core: all activities for pre-operational Space Weather service provision in ESA and Near-Earth Orbit (NEO) risk list maintenance, as well as hardware and software maintenance in all areas continued successfully under COVID-19 restriction measures. New portals have been deployed for the Space Weather and NEO services. The NEO coordination centre has achieved operational independence and the Laser and Optical Expert Centre has completed its acceptance. The ESA laser ranging station is ready for deployment.



ESA Laser Ranging Station (ELRS) ready for Deployment - Credits: ESA

Lagrange Mission: A new baseline for the Lagrange mission has been produced combining the results of the ESA CDF (Concurrent Design Facility) study and inputs from Airbus. This baseline substantially reduces the mission cost-at-completion estimate while maintaining the mission objectives. NASA signalled their support and strong desire to participate in the mission. An Internal Project Review (IPRev) was held on the Lagrange mission in August which gave some clear recommendations on the procurement process, contractual interfaces and the risk provisions to be considered for this purpose.



Lagrange Mission Consolidation - Credits: ESA

HERA: The Hera mission contract proposal was approved at the June meeting of the Industrial Policy Committee. The negotiations have been finalised, and the Phase B2/C/D contract was signed during the 51st PB-SSA meeting on 15 September at ESOC. The event with OHB management was hosted by the Director of Operations and received significant media coverage. The signature paved the way to a Preliminary Design Review which was successfully completed in December. The two CubeSat contracts were signed with GomSpace (LU) and Tyvak (IT) respectively.



Hera Contract Signature September 2020 - Credits: ESA

ADRIOS: The ClearSpace industrial proposal evaluation was completed by the board in May with a recommendation to proceed to negotiations. As part of an initial PATP (Preliminary Authorisation to Proceed), kicked off in July, ESA assessed the maturity of the service development through dedicated meetings, during and at the end of the PATP. The complete service was kicked off in November and a well-attended media briefing event was held virtually.



ADRIOS Contract Signature November 2020 - Credits: ESA

Security and Cyber Resilience

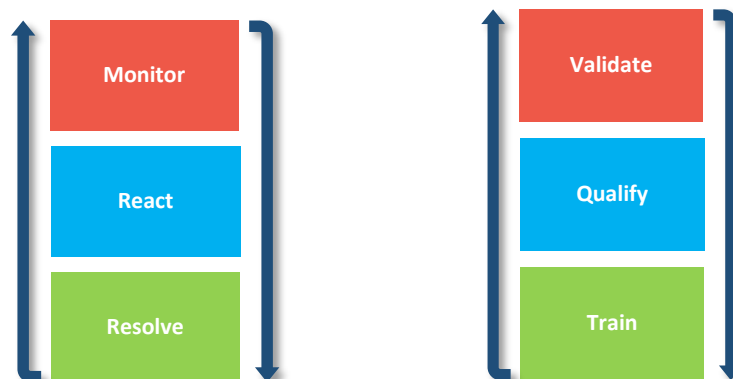
In an evolving global security context, it is our duty to acknowledge the importance of security within all aspects of the daily lives of European citizens and of European society. Space infrastructure has become a critical asset for the European economy and the security of European society, crucial to delivering the commodities and applications to which citizens have become accustomed. ESA's role in the European security effort is to strengthen its capability to integrate this new security context and adapt so as to guarantee full reliability in developing secure and robust space assets, missions and technologies.

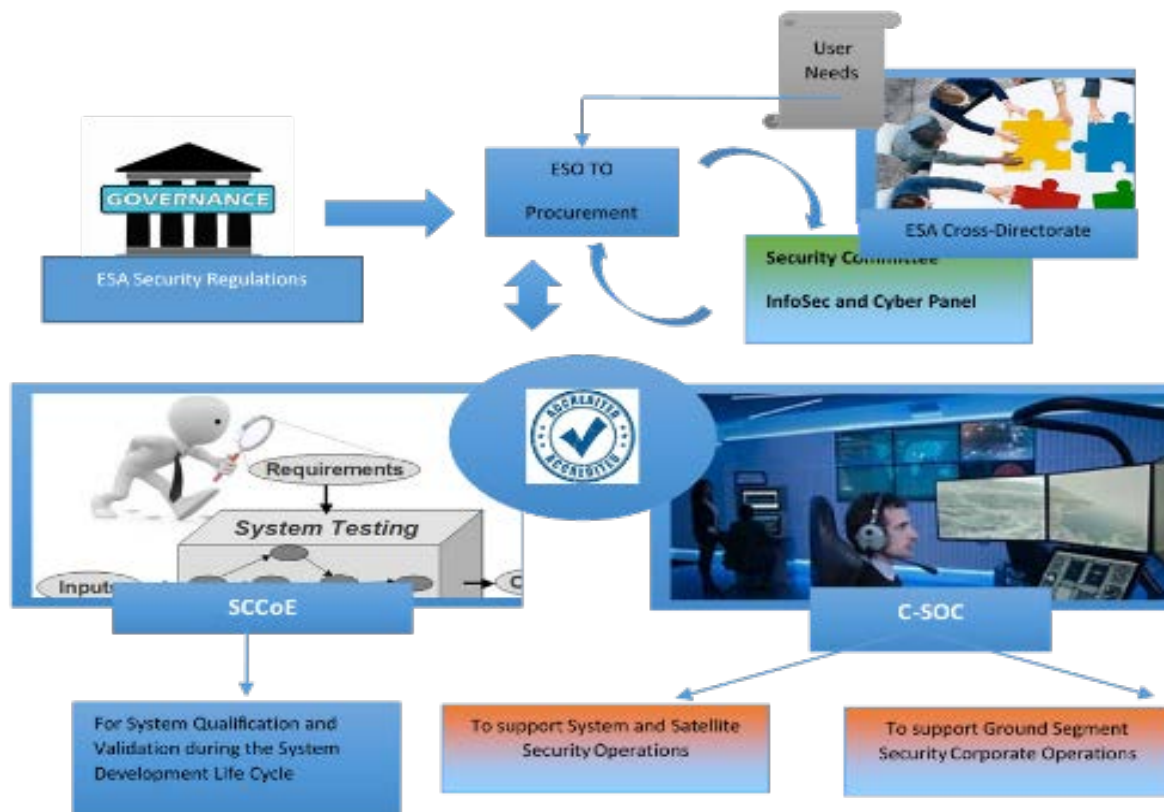
Cyber Resilience

Cyber safety and security has grown in importance due to a recognition of the criticality of space infrastructure in ensuring the availability of everyday applications and services and, in parallel, the exponential growth in innovation technologies that are available to potential attackers. Member States at Space19+ confirmed their investment in this field which, during 2020, fuelled the launch of two major complementary cyber resilience procurements, with the scope of designing, developing and certifying two cyber systems whose infrastructures will be centralised at ESEC, for which the ESA Security Office is responsible for the site from the technical perspective, with key functionalities also distributed at ESRIN, ESOC, ECSAT and ESTEC according to each site's main expertise.

The Industrial Policy Committee in its 323rd Session on 30 June 2020 approved the procurement proposals for these initiatives which will provide guidance, decision-making, culture-shaping, a cyber-HUB for Europe, and intelligence information sharing through two main drivers:

- **Security Cyber Centre of Excellence (SCCoE)**, an innovative tool, providing a unique capability in Europe. It will perform validation and testing of space systems through a synthetic environment, including the validation of security operating procedures and critical components, against up-to-date complex cyber threat scenarios. It will also represent the focal point for a Security Information Sharing capability, training and centralisation of forensic services/expertise as well as developing a distributed risk analysis process capability;
- **Cyber Security Operations Centre (C-SOC)**, complementing the capabilities of our state-of-the-art Computer and Communications Emergency Response Team (CERT), the C-SOC will provide an ESA-wide cyber monitoring and management capability. The Cyber Security Operations Centre (C-SOC) will monitor and track relevant information and events with the objective of maintaining the overall Agency security posture. The C-SOC will detect security incidents and support the readiness of the organisation's defensive capabilities. The C-SOC will be the ESA Super SOC coordinating all cyber functionalities at ESA and representing an essential tool not only for ESA, but for all Member States and third parties.





Credits: ESA

The C-SOC Phase 1 contract was concluded in 2020 having achieved the objective of reaching the SRR phase by contract closure in January 2021. The SRR will feed directly into the C-SOC Phase 2, which is expected to be placed by September 2021 and will cover the development phase. The SCCoE contract was signed in December 2020 with kick off of the project scheduled for January 2021.

The two Cyber Systems (SCCoE and C-SOC), through a cross-directorate team, will operate in synergy as preventive measures to support the ESA directorates during the ESA development life cycle of complex systems and/or to support the Satellite Operations and Ground Segment corporate activities.

The ESA Security vision is to provide ESA, in the short–medium term, with two new preventive-responsive capabilities:

- a cyber validation and qualification tool to support the development of any Space and Corporate system in accordance with the ESA Security Framework able to guarantee the robustness of the system against the threat landscape;
- a cyber preventive analysis tool to support the Ground Segment corporate activities and to support the operation of the satellites, monitoring any potential cyber attempt.

The goal of proceeding with the two procurements in parallel was to be ready in 2020 with a first preoperational configuration and to achieve full service for 2023.



Credits: ESA

Corporate Security

The ESA Security Framework is the backbone upon which a sound security system is built and it is imperative that it evolves in accordance with the space security context. The ESA Security Agreement, ratified in the parliaments of all ESA Member States, represents the foundation stone for European collaboration in security for space. The ESA Security Regulations, elaborated in conjunction with all ESA Member States and endorsed by the ESA Council provides its legal basis, grounding the ESA Security Framework. The ESA Council, in its 290th session in June 2020, approved the update of the ESA Security Regulations, after which the revised ESA Security Directives were issued on 1 July 2020 as a DG Instruction to all staff, grounding the ESA Security Framework and providing the building blocks and means for its implementation in ESA.

Reviewing and improving the coverage, completeness and correctness of security risk management across ESA business such that security risk is accurately understood and able to be properly mitigated to preserve ESA's security risk tolerance when realising ESA business objectives.

Space Programme

The need to have a homogeneous approach for the accreditation and certification of ESA projects, based on the foundation of a Security Risk Management Process, meets a global vision to build up space systems and infrastructure able to adequately protect the availability, integrity and confidentiality of the information managed. Within this scope, the ESA Security Regulations address this responsibility to the ESA Security Office, which represents not only the ESA Security Accreditation Authority for classified and unclassified projects, but is also tasked with supervising and controlling security implementation activities. The goal for the ESA Security Office (ESO) is to bring ESA up to a level at which it is fully compliant with the ESA Security Regulations and is a trustworthy and reliable security partner.

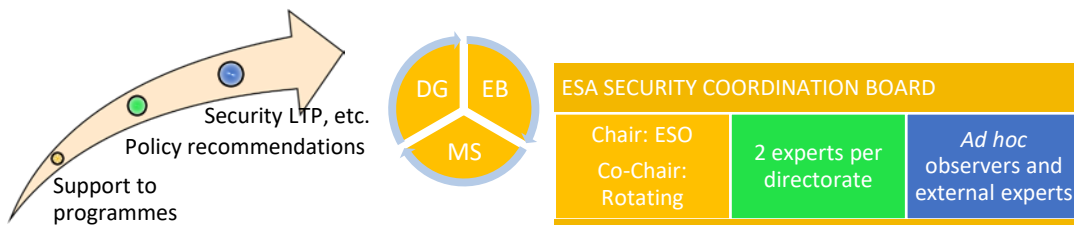
In 2020, ESO began collaborating with and supporting the Directorates of Human and Robotic Exploration (HRE), Earth Observation (EOP), Telecommunications and Integrated Applications (TIA), Navigation (NAV), Operations (OPS) and is in bilateral exchange with all ESA directorates in order to ensure an appropriate security assurance level for all ESA space projects and activities

Security Strategy

In order to equip ESA to continue to fulfil its role as a leading global player and symbol of excellence in space technology, ESA's security strategy will drive the Agency's vision, taking the global security context and political evolutions into account, in such a way as to provide ESA's Member States and partners with pioneering space applications and missions and steer ESA in the right direction in the interests of its Member States and stakeholders, feeding into the ESA Long Term Plan (LTP) and the ESA Council at ministerial level.

The **ESA Security Coordination Board (ESCB)**, managed by ESO and bringing together the representatives from all corners and areas of expertise in ESA, convened for the first time on 3 November 2020 to analyse the evolution of the global strategic space security framework, in terms of policy, threats, space security applications and technologies to propose long-term space security solutions. Its output will be the Security Long-Term Plan (S-LTP) which will ultimately feed into ESA's Long-Term Plan supporting the subsequent strategic orientations to be taken by Member States at ministerial level (including joint ESA-EU Space Councils).

The Security Strategy pillar, concentrating on policy, regulatory and technological aspects alike, will share with SEC, and provide the Director General with the most accurate and useful analyses affecting security and the evolution of ESA. Through this dynamic, security aspects can be anticipated and fully integrated in ESA's overall long-term strategy, its policy-making and programme management.



Applications

Earth Observation

EO and COVID-19

In 2020, Earth Observation programmes and projects were greatly affected by the COVID-19 pandemic. Nevertheless, all ESA Earth Observation missions remained fully operational and continued to provide uninterrupted data streams. The situation also presented a number of opportunities to showcase the benefits of Earth observation, opportunities which were communicated and pursued across a range of activities. Among other things, the Directorate of Earth Observation (EOP) compiled a dossier indicating examples where ESA satellites are used to monitor COVID-19 indicators.

In terms of practical action, one important example of COVID-19 related activities was the establishment of the ESA-European Commission “Rapid Action on Coronavirus and Earth observation” (RACE) Task Force. It uses a dashboard concept and monitors and analyses the situation facing economic operators as well as indicating trends in human activity indicators.



RACE Dashboard – Credits: ESA

RACE activities are coordinated with complementary international cooperation. For example, another dashboard was set up between ESA, NASA and JAXA to demonstrate the joint capabilities of the three agencies in terms of observing the environmental and economic impact of COVID-19 from space. This dashboard features a range of economic, agricultural and water-related indicators.

An additional measure for counteracting the impact of COVID-19 was to temporarily increase the ESA InCubed funding share for all new activities and projects contracted until the end of 2020. However, Member States were free to set their own co-funding levels.

Copernicus

The Copernicus Sentinel-6 *Michael Freilich* (formerly *Sentinel-6A*) satellite was successfully launched aboard a SpaceX Falcon 9 rocket on 21 November 2020. Using the latest radar altimetry technology, it will provide a new overview of ocean topography and advance the long-term record of sea-surface height measurements.

It is worth noting that the launch date has not been affected by COVID-19. Named after NASA's former Earth Science Director, Sentinel-6 *Michael Freilich* is the eighth Sentinel in orbit. It represents a joint effort between agencies and institutions such as ESA, the EU, EUMETSAT, NASA, NOAA and CNES.



Copernicus Sentinel-6 Michael Freilich at liftoff – Credits: ESA



Copernicus Sentinel-6 Michael Freilich on display – Credits: ESA

The Sentinel-6 mission comprises two identical satellites launched sequentially. Five years after the launch of the first satellite, Copernicus Sentinel-6B will take over duties from its predecessor when launched. The mission as a whole will ensure continuity of data until at least 2030.

Early 2020 saw the completion of the Phase A/B1 main industrial studies for the Copernicus Sentinel Expansion Missions, formerly known as Sentinel High Priority Candidate Missions (HPCM). The Phase B2/C/D&E1 offers for all missions were received in early spring. Despite difficult working conditions, all ten offers were evaluated and prepared for Industrial Policy Committee approval on 1 July. All resulting industrial contracts were signed by the end of 2020, representing an overall value of more than €2.5 billion. Work on the Next Generation of the existing Sentinel families continues nominally.

On 6 December 2020, negotiators from the Council of the European Union and the European Parliament reached a provisional political agreement on the proposed Regulation establishing the next EU Space Programme. A financial envelope of €14.8 billion (at current prices) is provided for the Programme, including €5.42 billion for Copernicus overall and €4.1 billion for the Copernicus Space Component. This represents a funding shortfall of €2.7 billion relative to the budget specified by the Copernicus Space Component Long Term Scenario.

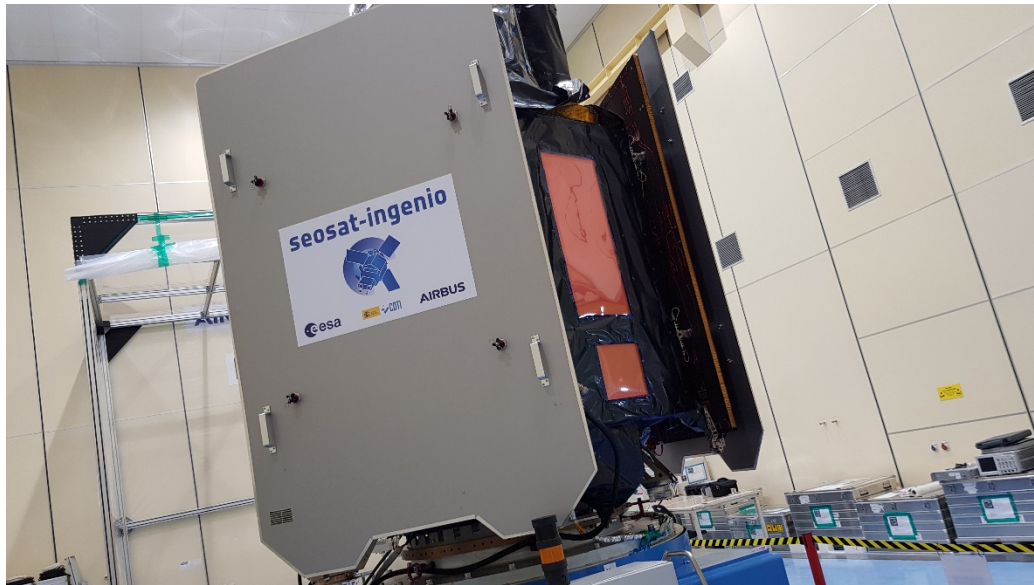
An update to the Line-to-Take document, dubbed LTT-2, was prepared at the request of ESA's Director General and his counterpart from the European Commission's Directorate-General for Defence Industry and Space (DEFIS). The LTT-2 constitutes a joint understanding between ESA and the Commission and aims to guide both organisations in terms of the next steps in defining and implementing the Copernicus Space Component.

Concerning other political aspects, the "Brexit agreements", containing the necessary provisions for the UK to continue to participate in Copernicus, have now been finalised.

More generally, the information provided by Copernicus continues to see massive levels of user uptake. 392 000 users have registered on the open access hub and 308 PB of data had been downloaded by the end of 2020. These numbers continue to grow and are only the tip of the iceberg, as a great deal of data is used directly by the Copernicus services or mirrored on other sites.

Earth Explorers / Science / Other Missions

The Spanish high-resolution imaging mission SEOSAT-Ingenio was lost aboard the failed Vega flight VV17 on 17 November 2020. SEOSAT-Ingenio was designed to provide high-resolution images of Earth's land cover. It would have benefitted society in many ways, including in terms of monitoring land use, helping plan urban development and managing water resources. It would also have helped map natural disasters such as floods, wildfires and earthquakes. However, it should be noted that another mission objective has been achieved: that of building up a high level of expertise in space systems in Spain, which has been demonstrated by the signature of the industrial contract with the Spanish industrial prime for the Copernicus Expansion Mission LSTM.



SEOSAT- Ingenio – Credits: ESA

The SEOSAT team was the recipient of a 2020 special award, honouring its outstanding achievements in the context of mission development and capability enhancement within the Spanish space industry.

The Earth Explorers in orbit continue to deliver crucial data for science, within the remit of their original scope and beyond. The wind mission Aeolus is a significant example of this. Aeolus data are now assimilated by the weather services ECMWF, DWD, Météo-France and the Met Office. CryoSat celebrated its 10-year launch anniversary on 8 April.

The 75 M€ space segment contract for the Altius mission with QinetiQ (Belgium) was signed in January 2020. With the contract in place, QinetiQ will lead a consortium of companies, from Belgium, Canada, Luxembourg and Romania. Altius, planned for launch by the end of 2023, will measure stratospheric ozone and other atmospheric trace gases.

The three Earth Explorer 10 candidate missions Daedalus, Harmony and Hydroterra were studied in Phase 0 in preparation for the selection by PB-EO on which mission would enter Phase A. The Call for Earth Explorer 11 was released on 25 May 2020. Subsequently, a workshop with proposers was held in October to explain the proposal guidelines.

Progress on the Earth Explorers under development is nominal. As for EarthCARE, the acceptance test campaign of the PFM Cloud Profiling Radar in non-redundant configuration, with the High-Power Transmitter (HPT)-A replaced by a dummy transmitter, has been successfully completed. Regarding HPT-A repair and a full HPT spare, a way forward has been agreed, based on joint ESA-JAXA technical working group recommendations.

The FSSCat mission carrying Φ -sat 1 was launched aboard a Vega Small Spacecraft Mission Service (SSMS) proof of concept flight on 3 September 2020. Φ -sat 1 is an experiment to demonstrate the capability of artificial intelligence on EO satellites. It is the first Deep Convolutional Neural Network to fly in orbit and will perform cloud masking on hyperspectral images. It has successfully passed its commissioning.



Myriad 2 VPU from Intel for Φ -sat 1 – Credits: CERN

Four Scout candidate missions have been readied for selection by PB-EO. ESP-MACCS was selected as the first Scout mission in November 2020. The selected missions will be developed over three years for launch in 2024.

Meteorological Missions

Progress on Meteorological Missions is nominal. As for Meteosat Third Generation (MTG), the Qualification Acceptance Reviews are planned for June 2022 and May 2023 for the satellite MTG-I and MTG-S PreFlight Models. Rider 4 to the MTG contract, corresponding to the Financial Hardship Claim received in July, was signed in December 2020. Regarding MetOp-SG, in view of the impact of COVID-19, the Flight Acceptance Reviews are now scheduled for September 2023/June 2024 and the launches for January/October 2024 for satellites A1/B1, respectively.

As for the Arctic Weather Satellite (AWS), the industrial proposal to the Phase B/C/D/E Request for Quotation was received and evaluated in October and the contract proposal approved by the IPC in November 2020.

Transversal / Partnerships / International Cooperation

Another edition of the ESA Phi-Week was held from 28 September to 2 October as a virtual event. It was attended by over 2100 participants and saw, among other things, the announcement of the “EOP Artificial Intelligence-enhanced Quantum Initiative for Earth Observation - QC4EO”, a joint initiative between ESA and the European Organization for Nuclear Research (CERN).

Meetings with ECWMF and EUMETSAT were held on the subject of Digital Twin Earth, the contribution to the European Commission “Destination Earth” initiative. Digital Twin Earth is a next-generation European Earth observation data facility deploying emerging technologies to provide rapid access data storage and advanced cloud data processing.

The Mid-Term Review for the Global Monitoring of Essential Climate Variables Initiative (also known as Climate Change Initiative, CCI) was held in November. A new CCI website was launched as well.

ESA and NASA agreed to set up a Strategic Partnership in the field of Earth Sciences in December. This partnership will have a long-term perspective, and full coherence with the work of the Joint Programme Planning Group will be ensured.

The International Charter Space and Major Disasters celebrated its 20th anniversary in October. It was activated 55 times in 2020.



ESA EO Phi-Week – Credits: ESA

Telecommunications and Integrated Applications

Reinforcing the competitiveness of the European satellite communication industry

Implementation of the new ARTES 4.0 programme, approved at Space19+, began in 2020. ARTES 4.0 provides various means to strengthen ESA support to the competitiveness of European industry and the space industry at large and promote commercialisation. The ARTES programme, by means of a broad portfolio, continued to grow for the fourth consecutive year, with industry also co-funding the industrial activities.

The programme focusses on three new strategic directions, namely Space for 5G, Space Systems for Safety and Security (4S), and Optical Communication – ScyLight. Within these Strategic Programme Lines, new projects are also fostered in partnership with industry, whether ESA- or industry-initiated.

Six new Partnership Projects – Sunrise Phase 2, Novacom I, Novacom II, HydRON, SAGA and Iris – were initiated with more in preparation. The BASS and Core Competitiveness Programmes have been well subscribed and growth linked to market needs is predicted to support competitiveness in the upstream and downstream sectors.

“Space for 5G” Strategic Programme Line

Under the “Space for 5G” Strategic Programme Line, ESA is promoting the key role satellite telecommunications can play in the deployment of 5G services and contributes to the digitalisation of industry and society. Unique opportunities for the deployment of hybrid satellite-terrestrial or converged networks are emerging through the current digitalisation of industry and society – the common set of 5G standardisation bringing together terrestrial and satellite systems – and the

emergence of collaborations between mobile network operators and satellite operators. This also provides new, more appealing commercial business cases for operators and industrial users.

ESA's key priorities are supporting industry and engaging with key vertical markets, including transport (maritime, aviation and land), logistics, public safety, media and entertainment, e-health, and agriculture, to demonstrate the benefits of integrating 5G and satellite in their connectivity portfolio. Close to 100 5G-related activities are under way. A 5G/6G Hub has been deployed at ESA ECSAT (in Harwell, UK) connecting 5G industrial communities and offering a collaborative space to tackle the challenges and opportunities.

“Optical Communication-ScyLight” Strategic Programme Line

Under the Optical Communication-ScyLight” Strategic Programme Line, ESA is fostering European leadership in optical and quantum communications, a cutting-edge technology at the frontiers of knowledge and one not without technological challenges. System concept approaches as well as technology and product development are being addressed with the aim of unleashing the full capabilities of optical communication technologies at large.

ESA is building the world's first High-throughput Optical Network or HydRON. In 2020, the Phase A/B study for the HydRON Demonstrator was prepared. As the next step, the HydRON demonstrator payloads and minimum viable product will be defined, addressing experiments, measurement campaigns and system aspects for novel technologies.

“Space Systems for Safety and Security” (4S) Strategic Programme Line

Advanced satcom solutions will provide support in making Europe's digital infrastructure ubiquitous, more secure and robust, while also making it possible to bridge the digital divide and addressing resilience and sovereignty concerns in respect of critical applications such as air traffic management, finance, industrial processes and essential government services. Two studies were kicked off in 2020, to provide vision and technical/programmatic proposals for next generation satcom secured connectivity. Those studies will assist the European public stakeholders such as ESA, the European Commission, EUROCONTROL and EMSA to define the best way forward.

ESA is also supporting the European Commission in its exploration of Govsatcom and EuroQCI, building blocks towards a system of secure space connectivity. An ESA role in the space segment and related ground mission segment would help the Commission define the best programmatic and technical approach, while ensuring complementarity with the long-term efforts of Member States through ARTES.

The European Commission, ESA and industry agreed on a draft User Requirements Document for the EU's EuroQCI initiative in October. In parallel, the ARTES SAGA project will demonstrate all required technologies for the EuroQCI Space Segment and validate different space-based QKD services.

Through the Iris Project, ARTES is addressing Satellite Communication for Air Traffic Management. The industrial consortium is working on the definition of the Iris Global Solution phase building on the Iris IOC results. Validation of the Iris system has moved into the implementation phase and a commercial agreement with the first airline was agreed, with pre-commercial Iris flights planned for 2021.

Generic Programme Line - Partnership Projects

The contract for Novacom I/Onesat was signed with Airbus Defence and Space in May. The objective is to enable competitive European software-defined satellite platforms. The contract for a second

Partnership Project, Novacom II/Space Inspire, was signed in November with Thales Alenia Space. The project will make it possible to develop and validate innovative European subsystems and equipment for the Space Inspire software-defined satellite.

Satellite Automatic Identification System (SAT-AIS)

The ESAIL Satellite for maritime Automatic Identification System (AIS) was successfully launched on Vega SSMS VV16 on 2 September, delayed due to COVID-19. exactEarth will integrate ESAIL into its fleet of satellites, providing maritime services to customers including fisheries, customs, search and rescue, environmental protection agencies and the European Maritime Safety Agency.

LuxSpace identified a commercial mission opportunity and a new microsatellites platform product line and Phase B was kicked off in February 2020 to develop Triton-X. A Phase C RFQ (request for quotation) was issued in 2020 with qualification of the platform planned for mid-2023.

European Data Relay System (EDRS)

EDRS-C joined the EDRS service in July 2020. EDRS satellites (A and C) provide full commercial service for all four Copernicus Sentinel satellites – Sentinel-1A, Sentinel-1B, Sentinel-2A and Sentinel-2B – with more than 40000 successful optical intersatellite links performed.

Next Generation Platform (Neosat)

The first Spacebus Neo satellite built by Thales Alenia Space was launched on 16 January. In parallel, development of Airbus DS's Eurostar Neo product line proceeded, with mechanical qualification of the complete structure with the first Eurostar Neo satellite Eutelsat Hotbird 13F scheduled for launch in 2021.

The two primes have sold a total of 15 Neosat-based satellites, with more sales expected, providing a return on investment for Participating States exceeding €20 for every euro invested in the Neosat Partnership Project.

Electra

The project objective is to develop, launch and validate, in orbit, a satellite system based on an innovative "full electric propulsion" platform in the 3-tonne launch mass range. At system level, the focus was on the closure of the platform Critical Design Review, with only a few issues remaining open. Regarding the mission implementation, a restricted ITT was issued on 25 November.

Quantum

Quantum is the first software-defined satellite, preparing the way for the next generation of satellites that can be reprogrammed in orbit. The Quantum satellite Flight Model Completion Review (FMCR) was held with Eutelsat and Airbus at the end of July. The satellite is in storage at Airbus Toulouse pending a launch on Ariane 5, now planned for mid-2021.

Govsatcom Precursor – Pacis 3

The Pacis 3 objective is to develop the innovative elements of a secure governmental communication payload and to demonstrate in-orbit pooling and sharing concepts by 2024. Following the successful X-band Transmit/Receive Direct Radiating Arrays and Ka-band antenna Preliminary Design Reviews (PDRs), development of the antenna elements Engineering and Qualification Models (EM/EQM) has been initiated.

Pioneer

xSpancion: a contract was signed in November 2020 with Clyde Space (UK) for production of 10 nanosatellites used to debug the production line, and validate missions with 4 nanosatellites for Inter-Satellite Link demonstration.

FARADAY 2G Project: the aim of this project is to enhance capabilities of In-Space (UK) through a scalable and modular nanosat in the 50–150kg range; the corresponding PDR (Preliminary Design Review) was successfully concluded in 2020.

Saas project: the first ten cubesats developed by Spire Global (UK) have been launched and in-orbit data is being provided.

QKDSat is a partnership with ArQit Ltd (UK) whose aim is to provide commercial services for the distribution of highly secure encryption keys using quantum-optical transmission techniques through satellite links. The System PDR was concluded in November.

“Business Applications and Space Solutions” Generic Programme Line

The Programme manages over 150 activities and 300 start-ups through an extensive network of business incubation centres (BICs) across ESA Member States. A Space Solutions socio-economic impact analysis conducted in 2020 looking back over the 10 years of achievements determined that, with 368M€ of ESA investment, the Programme including BICs funded around 2000 companies, helped companies to create over 5900 direct jobs, generate €1.4 billion in revenue and secure more than 970M€ of private investment.

The investment network, too, continued to grow during the pandemic. Two ESA Investor Forums were held, with a total of 18 companies pitching to raise over 65M€ from a group of private investors and corporations, 20 in the first and 35 in the second forum.

A Green Dossier, with more than 90 activities undertaken in the BASS programme that contribute to the achievement of environmental goals, showcases how space data is instrumental to the monitoring of environmental parameters and feeding economically sustainable solutions.

“Core Competitiveness” Generic Programme Line

The ARTES CC Programme provides broad support for the general development of disruptive technologies, products and services that serve as building blocks to maintain and improve the competitiveness of ESA Member States' satcom industries.

The introduction of mega-constellations, the race to Very High Throughput Satellite (VHTS) systems, the take-off of the Internet of Things and Machine-to-Machine market are opportunities implemented through ARTES CC. 2020 saw a significant ramping up in the volume of activities, reflecting the strong interest and engagement of industry as well as the support from delegations.

Multi-Directorate Initiatives

Moonlight Initiative - Lunar Communication and Navigation Services

Moonlight is a multi-domain initiative across D/TIA, D-HRE and D/NAV, investigating a Public Private Partnership for the delivery of communication and navigation services around the Moon by European private actors. The Phase A/B1 Outline Proposals were received as well as contract(s) in view of a contract signature in Q1 2021.

Navigation

Travel restrictions imposed by the COVID-19 pandemic, limited access to classified documentation in general, including within ESA and industry, as well as to Galileo control centres and other deployment sites have had an impact on the integration, deployment and test activities including for the Galileo Batch 3 Satellites. The schedule of activities had to be reviewed and consolidated and delays of several months were incurred at system level. During the COVID-19 first wave, priority was given to supporting the GSA in operations and service provision of the Galileo and EGNOS systems. The public health situation worsened after summer 2020 as the second wave of the pandemic hit, causing further delays in the deployment and test activities of upgrades of the ground control and ground mission segments. Batch#3 Satellite activities were further impacted by the COVID-19 measures put in place by the various companies. Tight coordination within ESA and between ESA and industry meant it was possible to mitigate the impact of the second wave on work schedules. However, several months of impact at system level and on other major Programme milestones could not be avoided. Requests received from industry have been addressed on a case by case basis with the European Commission and GSA.

Over the reporting period, 22 Galileo satellites were in operation delivering excellent quality initial services, including in particular the Search and Rescue Return Link service. Concerning the 19 July 2019 service interruption, the inquiry board's recommendations were implemented in the course of 2020 in order to fit optimally with the Full Operational Constellation (FOC) roadmap (within sanitary constraints). In particular, through successful coordination with the European Commission and the GSA it was possible to narrow down the activities needed to achieve the system design qualification for the Galileo First Generation (G1) Full Operational Capability (FOC) for each of the Galileo services.

ESA received *force majeure* notifications from most of the EGNOS and Galileo industrial prime contractors. Although industry was in the process of returning to site, nominal productivity was not fully attained and significant issues remained due to the sanitary restrictions at national levels (notably affecting international travel), industrial levels (affecting productivity notably in AIT and classified activities), and at hosting sites (affecting deployments).

The next Galileo launch (L11) originally planned for the end of 2020 has been moved to the end of 2021 (slot from the end of November until the end of December). ESA expected its related activities to be completed at the beginning of 2021 (satellite and ground segment qualification), but both operational qualification and deployment on the operational chains by the GSA as well as the availability of the launcher have remained under constant review.

The Galileo Second Generation (G2) system review and procurement of the G2 batch#1 satellites (G2B1) were also interrupted by the COVID-19 crisis. These processes were resumed by mid-May. The detailed transition scenario from G1 to G2 were analysed together with the European Commission. The Invitation to Tender Request for the G2B1 Best and Final Offer was completed, enabling the G2G programme to commence its next phase.

In this context, an ESA-internal, corporate project review (IPRev) allowed the ESA Director General to commit to the delivery of the G2G system infrastructure programme in compliance with the programmatic requirements and boundary conditions, provided the planned resources are made available as part of the Financial Framework Partnership Agreement (FFPA) negotiations. The schedule risk associated with the procurement of the G2B1 satellites was judged to be acceptable and to not require any Batch-3 Spare (B3S) satellites. Under these circumstances, the first G2 satellite launch is

scheduled for the end of 2024. Contract signature with the planned-for two contractors (out of three bidders) was expected by the end of January 2021.

In 2020, G2 Ground Segment and System Support reached Phase B2 and was rescheduled to be compatible with the satellite launch schedule. The first block will be a G1 Ground Segment extension, planned to be able to operate the first new G2 satellites, and is expected to be operational by 2024 in time for the first launch, before a completely new technology G2 ground segment will take over operations of all G1 and G2 satellites in orbit at the moment of transfer.

The COVID-19 crisis delayed the upgrade versions of the EGNOS system under development (V2.4.2A and V2.4.2 B) and consequently a new schedule had to be devised. A new schedule for the development of the EGNOS future generation system infrastructure (EGNOS V3) was also consolidated in Q3-20 in order to cope with development delays.

Upstream GNSS R&D activities, funded by the H2020 budget and delegated to ESA, faced delays in the implementation of the agreed activities due to changes in industrial participation conditions related to Brexit.

The COVID-19 crisis had limited impact on the implementation of the yearly NAVISP Workplan.

Following the successful subscriptions to NAVISP Phase 2 at Space19+, the programme has been able to expand further and with promising results, building on the achievements of Phase 1 of the programme.

A new multi-domain cooperation has been initiated between D/HRE, D/TIA, and D/NAV to define and develop Positioning Navigation and Timing capabilities for lunar exploration (the Moonlight programme) and has started to be assessed with Member States.

In parallel with all these activities, implementation of the re-organisation of the Navigation Directorate has continued, the key positions have been filled and priority recruitments have proceeded at a near-nominal pace. This re-organisation was necessitated by the ramping up of the Galileo 2nd Generation Programme.

Considerable work was undertaken during the year in the context of the FFPA negotiations and agreement with the EU on the implementation of the EU satellite navigation programmes (EGNOS, Galileo, GNSS R&D) for the next 7 years. This agreement will have a dimensioning impact on NAV activities.

Enabling & Support

Space Transportation

Exploitation

Three Ariane 5, two Vega and two Soyuz launches were carried out from Europe's Spaceport in 2020, carrying a total of 64 payloads from commercial and institutional customers. European institutional payloads or public private partnerships launched include Eutelsat Konnect, SSMS-POC and CSO-2.

Europe's launch service provider, Arianespace, reported the signature of launch service contracts with three customers for a total of seven satellites in 2020.

Ariane

2020 saw three Ariane 5 flights take place: VA251, VA 252 and VA253.

After a new Ariane 5 configuration with extended tanks in the upper stage and a new lighter Vehicle Equipment Bay VEB-H was qualified in 2019, the first flight with this new configuration took place on Ariane flight VA253 in August.

The system qualification of the Ariane 5 ECA launcher for the JWST mission was completed at the end of 2020. This paved the way towards the Launch System Technical Qualification Review (LTQR) in spring 2021 as last development gate. Following a successful LTQR, JWST is due to be launched in autumn 2021.



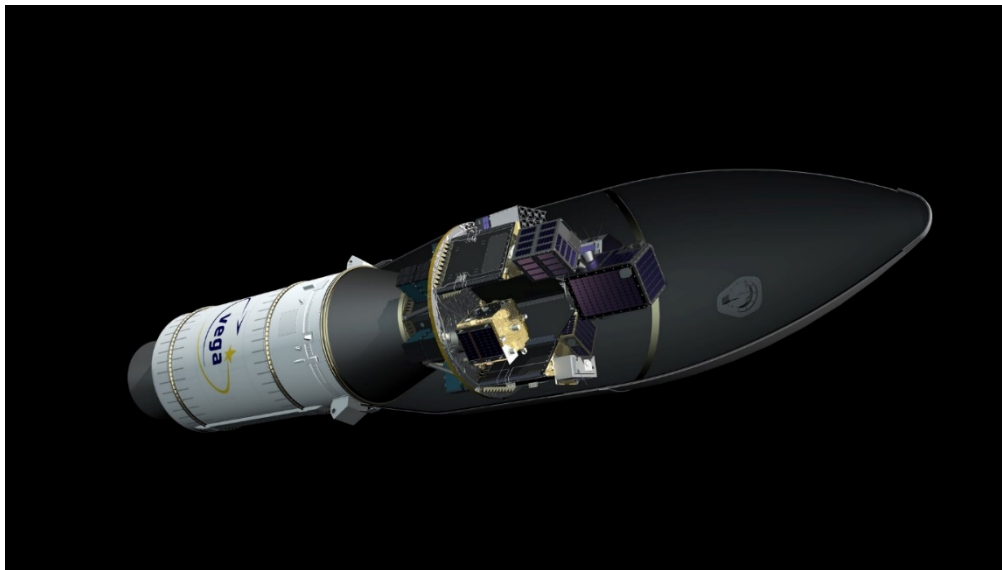
Launch of VA253 – Credits: ESA

Vega

On 2 September, Vega's return to flight mission VV16 took place from Europe's Spaceport in French Guiana. It was the first Proof of Concept flight under the Light satellites Low-cost Launch opportunities (LLL) initiative using the Small Satellites Mission Service (SSMS) dispenser.



VV16 Launch – Credits: Arianespace



Artist's view of Vega VV16 with SSMS and SAT-AIS – Credits: ESA

ESA's LLL initiative was approved by the ESA Council meeting at ministerial level in 2016 in order to prepare the way for routine services for light satellites using the European launch vehicles Vega/Vega-C and Ariane 6.

On 17 November, eight minutes after the liftoff of Vega mission VV17, and following the first ignition of the engine of the AVUM upper stage, a deviation of trajectory was identified, resulting in the loss of the mission. An Independent Inquiry Commission (IEC) co-chaired by ESA and Arianespace identified a problem related to the integration of the fourth-stage AVUM Thrust Vector Control system as the most likely cause of the loss of control of the launcher. Based on the IEC recommendations, ESA and its industrial partners took corrective actions to allow Vega launches to resume in 2021.

Development and future preparation

Ariane 6

Launcher System: Preparation of the Launcher System qualification is progressing at system level and subsystem levels. In particular, the mechanical qualification tests of the Vinci Thrust Frame were successfully performed and the qualification campaign for the Auxiliary Power Unit is well on track with the last tests to be performed in 2021.

The first Ariane 6 upper stage has been assembled in Bremen, Germany, coupling the launcher's tanks with the equipped engine bay of the re-ignitable Vinci engine and the Auxiliary Power Unit for the first time. The upper stage is currently undergoing fluid and electrical tests, before leaving for hot firing tests on the newly developed P5.2 test bench at DLR Lampoldshausen, Germany.

In parallel, the first Ariane 6 core stage and the second Ariane 6 upper stage are under preparation. These specimens will be shipped to Europe's Spaceport from Les Mureaux (France) and Bremen (Germany) respectively, for the combined tests campaign to be started in 2021.

In addition, the core stage and upper stage for Ariane 6's maiden flight are being prepared: the oxygen tank of the core stage using innovative friction stir welding technology has been welded, while the Vinci and Vulcain 2.1 engines successfully passed their acceptance tests.

Launch Base: At Europe's Spaceport, activities to finalise the Ariane 6 launch base are progressing. Initial integration testing with full-scale "mock-ups" of the P120C strap-on booster and of Ariane 6's central core took place in the new mobile gantry. Testing of the cryogenic arms linking the launch pad to the launch vehicle are also ongoing at Fos-sur-Mer in France, before shipping to French Guiana.

The Launch Base Critical Design Review close-out session was held on 11 December. The new version of the Control Bench Launch Pad's Core (version Core 3.0) was delivered to French Guiana.



Ariane 6 Launch Base – Credits: ESA

Launch System: Activities have been converging on the preparation of the Launch System Critical Design Review (CDR) Close-Out which will be kicked off in early 2021. Activities are focussing on the definition of the Launch System configuration to be used for the Launch System Combined Tests and for the Maiden Flight Campaign, the validation of the above configuration, including the Operational Products and finally the adequate mitigation of all technical risks identified so far.

Vega C

Step 1 of the Vega C Launch System Ground Qualification Review, focused on the Maiden Flight configuration, was kicked off on 10 December. Step 2 is planned to be held in October 2021.

The qualification at system and subsystem level is currently under finalisation. In particular, the second stage activities are ongoing, the Zefiro 40 SRM qualification was achieved, including Z40 motor firing test, as well as the IMC burst test and the nozzle and flexible joint mechanical testing. In parallel, Z40 FM1 casting has been performed and integration activities have progressed nominally.

Activities for the modification of the mobile gantry for Vega C, such as modifications of the fluids networks as well as connection and installation of the new control bench are running within the time slots available between the flights, and are planned for completion in the last slot just before the start of the Vega C Maiden Flight campaign.

Concerning Vega Evolution, the first Development Model of the M10 cryogenic engine was developed, with the aim of undergoing fire tests in 2021.

P120C

The second qualification model of the P120C solid rocket motor completed its hot firing on 7 October with a final test to prove its readiness for flight. With this successful third static fire, all the propulsion elements of the launcher system have completed their qualification tests.



P120C – Credits: ESA

Space Rider

On 9 December, two contracts with industry were signed at Palazzo Chigi in Rome, Italy, in the presence of Italian Government representatives. The first contract is for delivery of the Space Rider flight model including the re-entry module and the AVUM orbital service module, and the second contract covers the delivery of the ground segment by Italian co-prime contractors: Telespazio and Altec.

Activities are on track for the first flight of Space Rider in the third quarter of 2023 from Europe's Spaceport in French Guiana.



Space Rider – Credits: ESA

Preparation of the future

Prometheus

The Prometheus engine is a precursor for a new liquid rocket engine family that has been designed for low cost, flexibility, and reusability. The project aims to design, produce, and test an advanced low-cost 100-tonne class LOx/LCH₄ or LH₂ potentially reusable engine. The engine is designed to have recurrent costs of 1M€ and targets flexibility of operation through variable thrust, multiple ignitions, compatibility with main and upper stage operation, and minimised ground operations before and after flight. Its production features additive manufacturing for the engine components, the introduction of a full electric command system, and the on-board rocket engine computer (REEC) for engine management and monitoring.



© ArianeGroup Holding

Prometheus – Credits: ArianeGroup

The reference design and industrial file for the Prometheus methane engine was finalised in 2020, paving the way for a first engine test in 2021. Modifications are being made to the P5 test bench at the DLR test facility in Lampoldshausen, Germany, in order to implement new methane infrastructure and to further adapt the bench for the testing of Prometheus.

Phoebus Demonstrator

MT Aerospace and ArianeGroup have been contracted to develop “Phoebus”, a Prototype of a Highly Optimised Black Upper Stage. This low-cost lightweight Phoebus demonstrator introduces carbon composite materials for the tanks containing the cryogenic propellants such as liquid hydrogen and oxygen, and for other primary and secondary structures. The goal is to achieve optimum mass savings while ensuring propellant compatibility with these new materials, at a lower production cost.

Phase A/B1 was kicked off on 4 April 2019. Phase A/B1 covers the period until the System Requirements Review (SRR) has been completed, leading to the start of Phase B2 in 2021.

Themis Reusability Demonstration

In an incremental, risk-controlled approach, the Themis programme is preparing for in-flight demonstration, in a follow-on phase, of a full-scale reusable Prometheus-powered protoflight vehicle in the 2023-24 time-frame, for lower composite application. In addition to technology and competence acquisition, the demonstration will validate the global performance objectives (Low RC, NRC, P/L, reliability, time to market) elaborated within the coordinated system studies. Taking stock of the progress made with the Prometheus engine demonstration, the Themis/Reusability Element was decided on at Space19+. This element further advances the activities begun at ESA or at national level and the methodology changes implemented within ETID and Prometheus towards a more agile, risk-taking approach promoting cost-driven innovation and quick design-manufacturing-test loops. Following contract signature for the initial phase on 15 December 2020, the hardware-rich and “test & learn” approach is being pursued with the following tests having been performed (full-scale tanks LN₂ tests, structural health monitoring, fluidic and electrical equipment) and preparation of next steps.

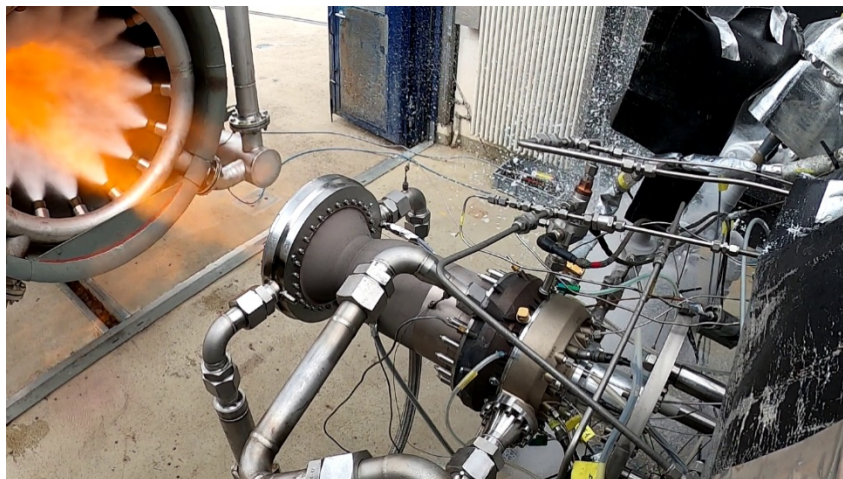


Themis – Credits: CNES

Expander-cycle Technology Integrated Demonstrator

The Expander-cycle Technology Integrated Demonstrator, or ETID, is a full-scale integrated demonstrator for an upper-stage rocket engine. ETID is a precursor of the next generation of 10-tonne rocket engines. Results from the project should pave the way towards a lightweight, high-performance rocket stage, while also indicating methods of substantially reducing its serial production cost, thereby honing its market competitiveness.

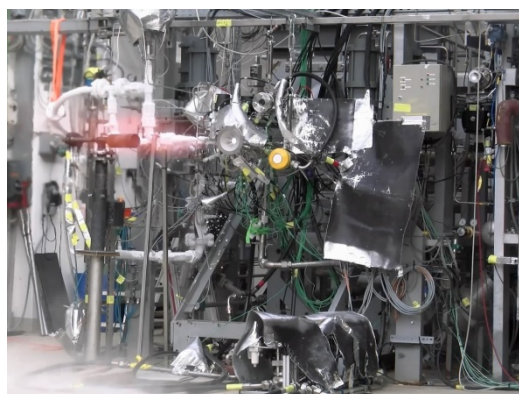
The ETID project Final Key Point took place in December, closing out the very successful demonstration and presenting the latest analysis of the results of the additive-manufactured thrust chamber hot-fire tests on the P8 stand earlier in the year at DLR Lampoldshausen, Germany. ETID has contributed to combustion chamber technology advancements, for the benefit of all thrust class engines.



ETID – Credits: ESA

Storable Propulsion Demonstrator

The Berta engine, intended to demonstrate storable environmentally-friendly propulsion in the 1–10kN thrust range, saw the completion of the thorough testing of its throttleable injection technology. Activities intended to prepare extensive testing at P2 DLR Lampoldshausen, Germany, for the benefit of the ASTRIS and HRE projects, are to be carried out in 2021.



BERTA Test – Credits: ESA

Advanced technologies

ESA selected system-driven critical technologies to be further matured, bringing also the prospect of validating and embarking technologies on large-scale demonstrators. The technologies relate to low-cost avionics, propulsion, launcher structure, and reusability technology roadmap support.

Commercial Space Transportation services and support to Member States

ESA issued a permanently Open Call for Proposals as part of Boost!, the new programme on Commercial Space Transportation Services. This programme emerged in response to ESA's long-term vision to support private economic operators in their undertakings to develop new European space transportation services, encompassing launch opportunities, in-space transport services up to return from space capabilities. Through Boost!, ESA concluded the first three co-funding contracts in 2020 in support of the pre-commercial development of new launch services using micro-launchers. In addition, through Element 2 of the Commercial Space transportation services and support programme, ESA helped institutions in Norway, Portugal and the United Kingdom to deploy national spaceport initiatives and the related regulatory frameworks.

Europe's Spaceport – Guiana Space Centre

In 2020, ESA Director General and Philippe Baptiste, President of CNES, signed the contract for the maintenance of the Guiana Space Centre (CSG) launch range in operational condition over the 2020–2024 period, including Core Launch Range Renewal (CLRR) activities. The contract implements the CSG Resolution approved at Space19+ by the ESA Council meeting at ministerial level. ESA will thus help maintain Europe's spaceport facilities in safe and working order, as a main element of the guaranteed access to space for all European customers, and in particular for European institutional missions.



Ariane 6 launch complex at Europe's Spaceport – Credits: ESA

The contract also includes CLRR activities, notably investment in the modernisation and enhancement of Europe's spaceport, with a view to providing increased flexibility. In 2020, CLRR activities focussed on preparing industrial procurements, including developing technical requirements and sharing future opportunities with industry and delegations, in particular through CSG Industry Days, held in October.

The CLRR investments do not only cover needed end-of-life overhauls for systems such as water and air conditioning underground networks as well as road networks, telemetry or tracking systems. They will also serve to revamp the operational concept introducing new working methods and technologies such as automation, digitalisation, remote monitoring and control. Energy transition also lies at the heart of this modernisation with a significant reduction in CO2 footprint and energy consumption through the use of a smart grid and the shift towards renewable energy.

Operations

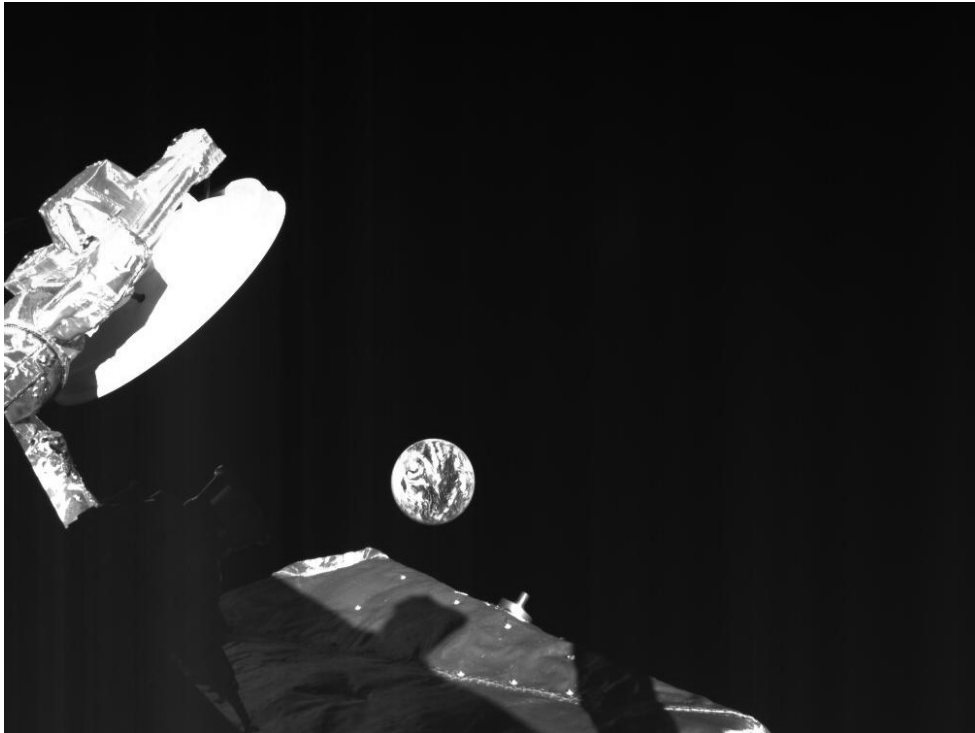
All flying satellites have been continuously monitored and controlled at all points in time. Thanks to the efficiency of the teams involved, ad-hoc adaptations have been invented and implemented so that the number of remote operations has been significantly improved. Simulations and rehearsals have been performed under COVID-19 regulations, with reduced attendance within the control rooms and without having industry specialists and relevant project personnel on site. Despite the separation of teams, very good progress has been achieved in building up the teams to support LEOPs (Launch and Early Orbit Phases), minimising residual risks for the execution of the actual critical operations.

- 10 February: Solar Orbiter launch & commissioning (Atlas-5)
- 10 April: BepiColombo Earth flyby
- 15 October: BepiColombo Venus flyby
- 21 November: Sentinel 6A LEOP (Falcon9)
- 27 December: Solar Orbiter Venus flyby

- Tracking critical operations:
 - o July: C-Mars
 - o 16 September: Vega return to flight VV
 - o November: ChangE-5 separation
- ESA's OPS-SAT "Space Lab" commissioning was completed and service provision got under way. This 3U cubesat is the first of its kind, with the sole purpose of testing and validating new techniques in mission control and on-board satellite systems. It has already successfully demonstrated a number of "first time ever" advanced concepts and technologies, paving the way to future ESA missions. In addition, OPS-SAT is providing a unique and unprecedented service to the wider space and academic community, providing a flying in-orbit laboratory for validating new concepts in the form of onboard experiments. The take-up of this service has been highly successful with over 200 experiments registered and increasing.



Operations under COVID-19 - Credits: ESA



BepiColombo Earth flyby on 9/10 April 2020 - Credits: ESA



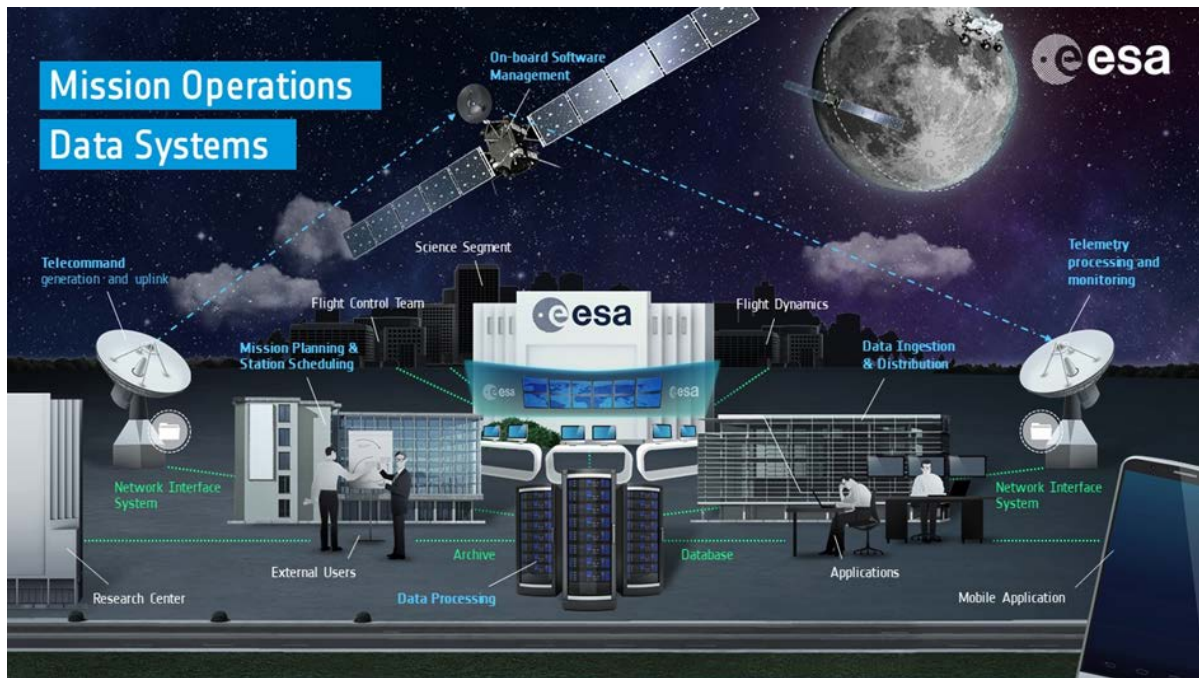
BepiColombo Venus flyby on October 15 – Credits: ESA



Solar Orbiter Venus flyby on 27 December 2020 - Credits: ESA

Ground Systems Engineering

- Under ESA's Mission Operations Infrastructure (MOI) investments, the following cornerstone projects progressed successfully:
 - o The New Norcia deep space antenna (NNO-3) ITT was issued on 3 April, the start of an important procurement that will allow ESA to prepare for an increased demand for support from space missions in the areas of science, exploration and space safety.
 - o Production continued of the X- and Ka-band cryogenically cooled feed systems aimed at increased data reception capability. The first deployment is planned for Cebreros in April 2021.
 - o The deployment of the new ESA Laser Ranging Station (ELRS) has progressed satisfactorily with the completion of the infrastructure design review in Tenerife (Spain) in April, targeting deployment in June 2021.
 - o Next-generation control system application EGOS-CC successfully passed its Implementation Review with the Readiness Review scheduled for Q3/2021. Its deployment for selected flying missions and ESA's tracking station network (ESTRACK) has been planned and initiated.
 - o In addition, the next generation of mission operations data systems is being natively designed to be multi-mission, to support distributed operations, and to incorporate innovative technologies such as artificial intelligence and augmented/virtual reality.
- Successful delivery and deployment of the Galileo Geodetic Reference Frame Facility (operational infrastructure) took place at the Galileo Control Centre in Oberpfaffenhofen, despite COVID-19 constraints.



Mission Operations Data Systems - Credits: ESA

Basic activities

Introduction

ESA's sustained success is underpinned by the Agency's Basic Activities, which complement the mandatory element of the ESA budget, together with the Science Programme. They are mandatory because they are essential to ESA's achievements.

Essentially, Basic Activities form part of the ESA "membership fee" and comprise a portfolio of ESA undertakings that remain transparent most of the time because they simply work – and work well. They include early-stage innovation and technology development (through the Discovery and Preparation and the Technology Development Element), core technical expertise, support of standardisation and harmonisation efforts, essential infrastructure such as labs and ground stations – as well as ESA's own satellite Test Centre.



Investing in the basics of space ([link](#)). Calculating the deployment of Mars Express' MARSIS radar antenna – Credits: ESA

Discovery and Preparation

With the Discovery element, ESA is exploring and investing in an open innovation approach into early-stage concepts and early technology projects together with universities and research centres, industry and specifically SMEs. In addition to the permanently open Discovery element channel on the Open Space Innovation Platform ([OSIP](#)), in 2020 the Discovery element ran dedicated campaigns on Marine Litter, Harbour to Harbour Autonomous Shipping, Lunar Caves, Off-Earth Manufacturing and Construction and Model-Based System Engineering, resulting in 160 co-funded research, small study and early technology development activities.

The Preparation element prepares and enables future mission and programmes through pre-Phase A and Phase-A studies and dedicated system analyses of payload or underpinning processes to establish robust trade-offs for mission designs. In 2020, ESA launched its first-ever open call for new mission ideas and concepts, resulting in over 200 mission ideas that are under consolidation and will lead to new Concurrent Design facility (CDF) pre-Phase A feasibility analysis in the course of 2021.

The Advanced Concepts Team (ACT) – ESA's internal research think-tank tasked with monitoring and performing research on advanced space concepts and technologies to prepare for disruptive innovation – has further enlarged its research network and its open science activities, specifically via new scientific competitions attracting a large number of global scientists.



ESA's Open Space Innovation Platform - Credits: Sacha Berna for ESA

DISCOVERY AND OSIP



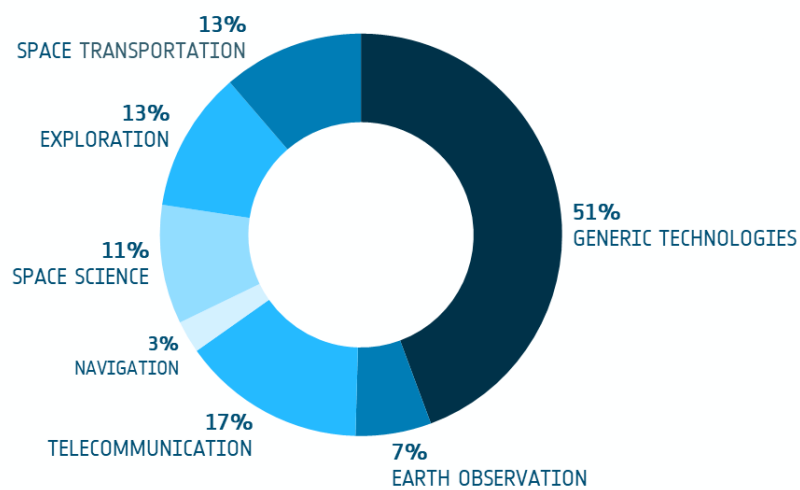
Overview of 2020 Discovery and OSIP – Credits: ESA

Technology Development Element

By testing the feasibility of promising new technologies, the Technology Development Element (TDE) gives Europe the ability to look ahead to plan and define future space missions and activities, helping “future-proof” the European economy.

Long-term planning is inherently risky when it comes to space, but the TDE reduces the degree of risk by demonstrating the reliability of a given technology long before any mission comes to be based around it. The high-precision, high-performance technologies that result often find wider markets elsewhere, strengthening European competitiveness as well as its technology base. The TDE is the only ESA technology programme supporting all of ESA’s fields of activity across the entire spectrum of technical disciplines, providing the technological nucleus for most future developments.

TDE COMPLETED ACTIVITIES PER DOMAIN IN 2019



TDE completed activities per domain in 2019 – Credits: ESA

Core Technical Expertise

The space industry is in the midst of rapid, fundamental change, and TEC is responding in turn. Guided by ESA’s Technology Strategy, 2020 saw it strengthen its competences in emerging fields, including digital engineering, AI and cyber security. One notable output was its first roadmap for model-based system engineering, which was elaborated together with European stakeholders including space agencies, research organisations, universities and the private sector.

Technical Infrastructure

Advanced missions need advanced testing. To save time and money, TEC offers industry-unique infrastructure, which would not make economic sense to develop multiple times in multiple places for separate missions. TEC has pooled resources to make world-class test facilities available to all European space (and indeed non-space) companies. The labs’ technical excellence also contributes to a larger ESA goal: the diffusion of space technology and knowledge into the wider economy. Also available is TEC’s 3000 m² satellite test centre, the largest in Europe.

2020 saw the start of TEC’s Open Lab initiative, intended to streamline third-party access to TEC laboratories. At present, demand for lab time far outstrips supply due to limitations in personnel.

Nonetheless, opening up access to TEC facilities is a way to raise the competitiveness of small-medium enterprises, start-ups and small-capitalisation companies without competing with existing markets. Looking to the future, ESTEC's Test Centre began expansion, commencing construction of a new 350 m² cleanroom.



Testing for Ireland's first satellite, EIRSAT-1, at ESA's Hertz antenna test chamber ([link](#)) (Courtesy of ESA).

2020 ESTMP publication

Published together with the European Commission, the 2020 Edition of the European Space Technology Master Plan (ESTMP) provides a comprehensive overview of space technology R&D across Europe, as well as the status of the latest European Space Technology Harmonisation Roadmaps. In addition, more than 30 European countries and Canada present their national organisations and plans for space technology development as well as national budgets. Copies of the document can be requested by contacting estmp@esa.int.



ESTMP 2020 Edition overview – Credits: ESA

Standardisation

The European Cooperation for Space Standardisation (ECSS) 4.0 initiative seeks to enhance the usability of current standards. A task force involving Eurospace, SMEs, ESA, CNES and DLR introducing a low risk/low cost rapid delivery strategy-based system by identifying core ECSS requirements has been kicked-off.

New ECSS work items included a digitalisation platform for the ECSS system and ECSS standards for advanced technologies such as 3D-printing, long-term storage and use of commercial off-the-shelf (COTS) components.

Technical and Quality Management

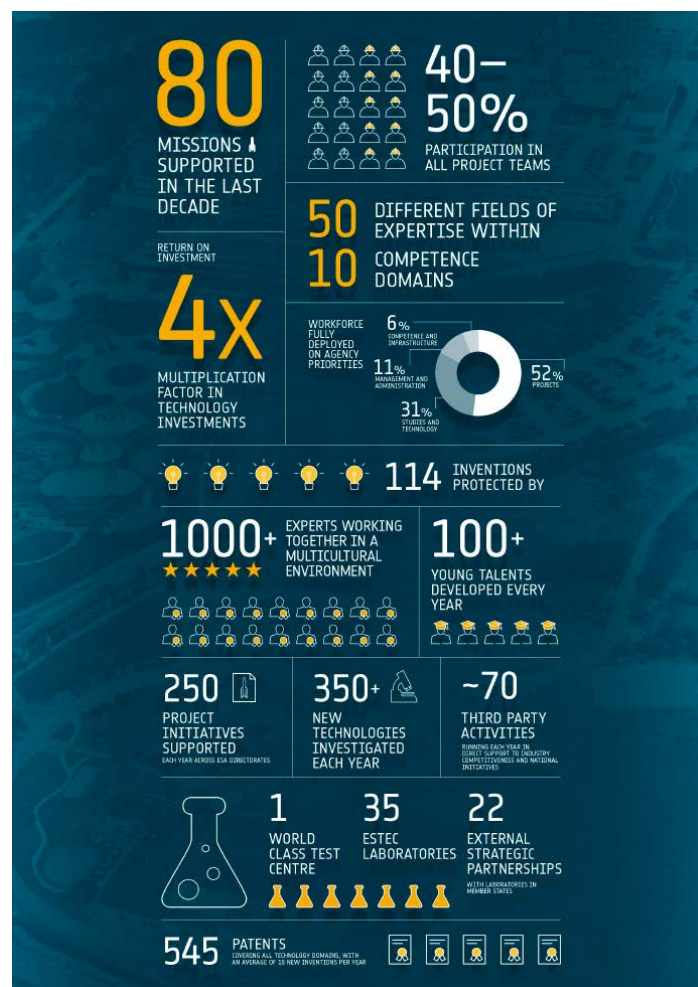
What/Who is TEC

TEC is the Directorate responsible for the Space Engineering activities in the main technical areas, as well as quality management of the Agency's products and activities and technology transfer activities.

TEC supports missions with technical expertise and by ensuring enabling technologies are available at the very point when new ESA and European missions need them – one definition of innovation being “the right technology at the right time”.

The technology fostered through TEC's R&D programmes is also a key element of competitiveness, to underpin the European space industry's continued viability – considering around a third of its turnover comes from commercial markets.

TEC also oversees the coordination of European standardisation efforts to boost industrial cost effectiveness and the general technical state of the art, while also levelling the playing field for smaller companies – opening up new opportunities to compete based on standardised working and interfaces.



TEC in numbers – Credits: ESA



TEC supporting missions tests: Maxwell Test Chamber ([link](#)) (Courtesy of ESA)

COVID-19 and business continuity

When COVID-19 struck, most ESTEC personnel switched to online working. Onsite testing did not stop, with more than 80 test campaigns including testing for Galileo atomic clocks and JUICE payloads as well as MPCV, PLATO, METOP SG, EarthCARE and Proba-3 items. The Test Centre resumed work on Galileo satellites, along with Sentinel-1's solar array and a JUICE payload.

R&D projects continued on a socially distanced basis and technical workshops shifted online. Notable firsts included the refinement of ESTEC's oxygen plant, producing oxygen and metal from moondust. ESTEC's Test Centre began work on an extension.



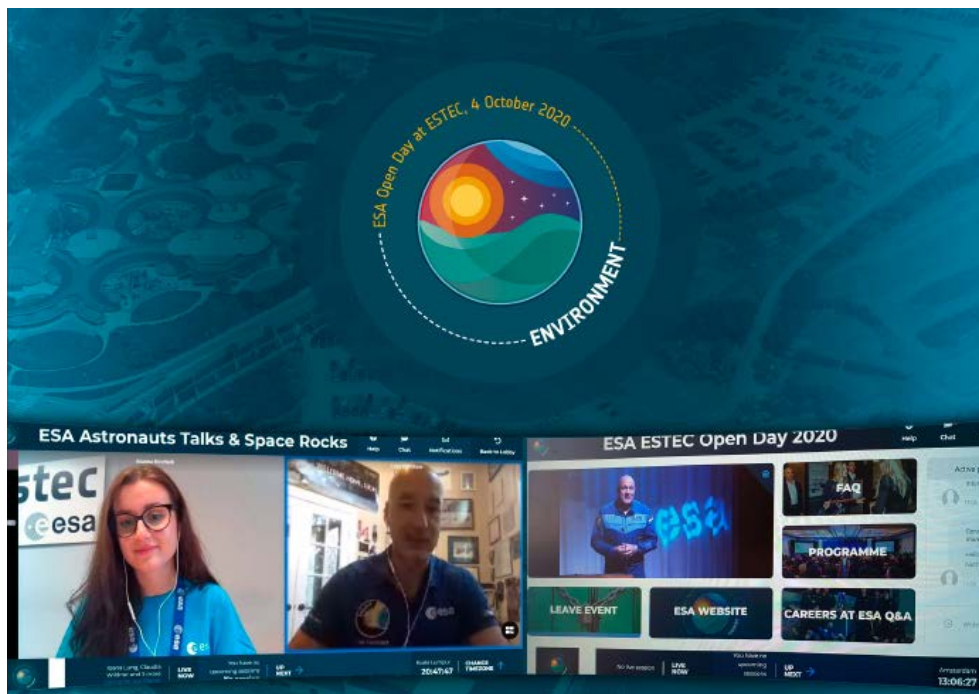
ESTEC Materials and Electrical Components Laboratory ([link](#)) – Credits: ESA



Expanding ESTEC's Test Centre (September 2020) ([link](#)) – Credits: ESA

ESA Open Day at ESTEC 2020

ESA's Open Day at ESTEC is an annual fixture, which shifted online in 2020. A total of 6125 users from 92 countries attended – with families and friends watching together – plus 1385 more viewers via ESA Web TV. Some 89 ESA volunteers from five directorates participated, with partners including NL Space.



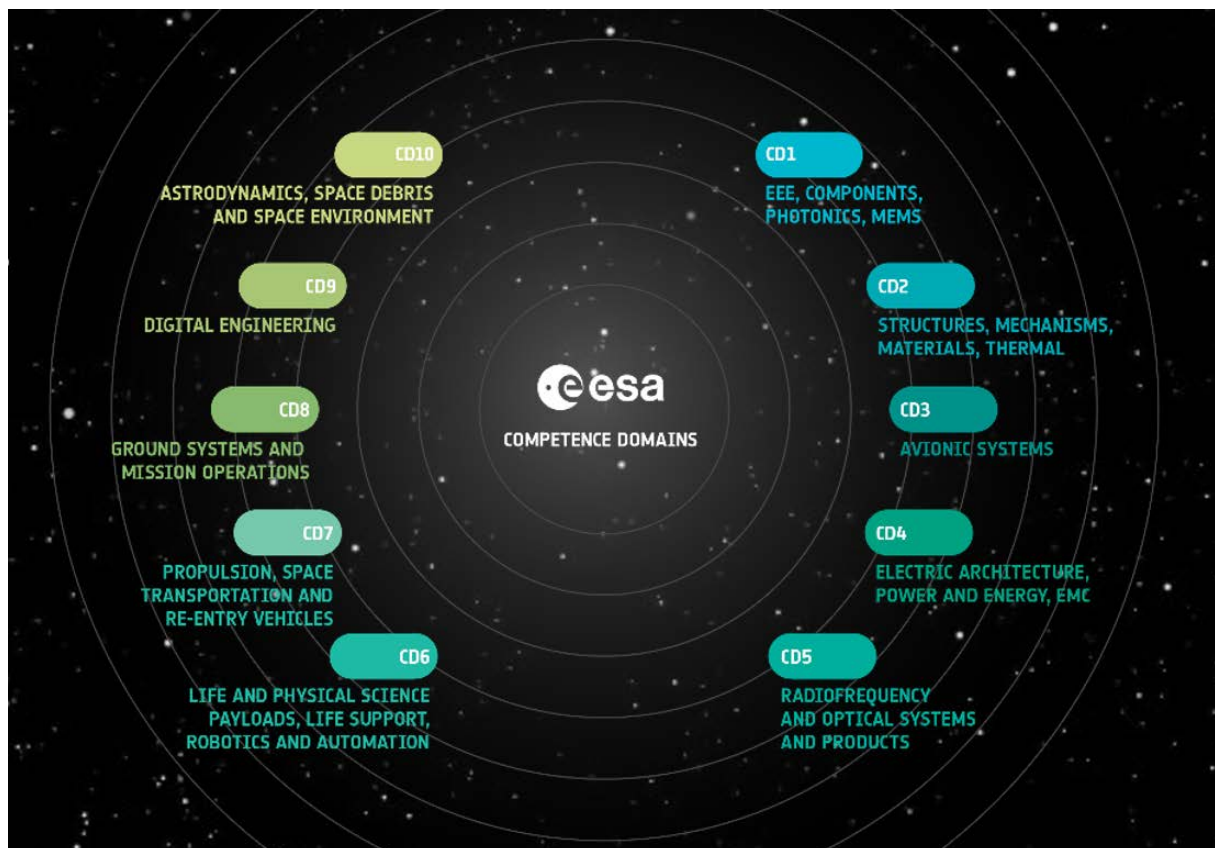
ESTEC virtual open day, 4 October 2020 – Credits: ESA

Supporting missions

The Directorate provides hands-on engineering skills and specialist expertise to all project teams across the Agency. Priorities in 2020 included critical test support for JUICE (with the Mechanical Systems Lab performing environmental tests for several JUICE payloads), MTG and Sentinel-4, along with major efforts to de-risk Galileo Second Generation's antenna design and passive hydrogen maser atomic clock.

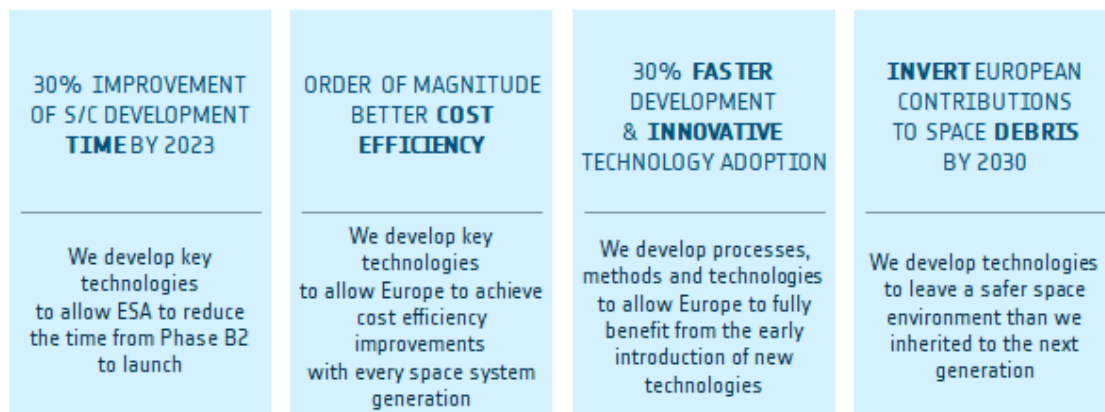
ESA Technology Strategy: Implementation

The ESA Technology Strategy guides the planning, development and implementation of all technology development activities prepared, conducted and coordinated by ESA. It takes into account both the technical expertise of ESA, grouped into 10 competence domains, and the insight for future needs for space missions.



ESA 10 Competence Domains – Credits: ESA

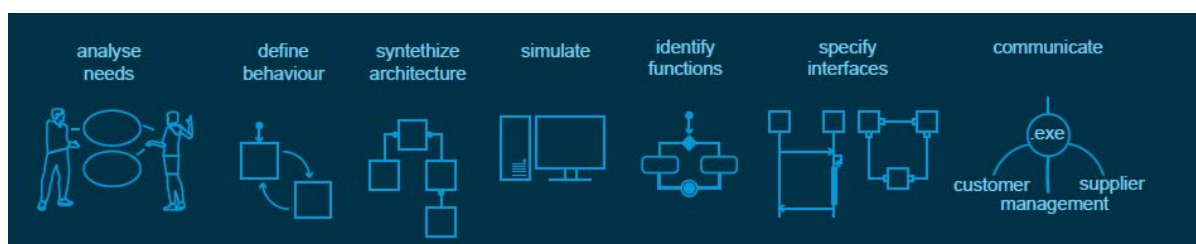
This analysis of user needs and technology innovation has resulted in priority technology themes for R&D planning.



2020 PRIORITIES: DIGITALISATION & COTS

ESA Technology Strategy targets and 2020 priorities – Credit: ESA

- Model-Based for System Engineering (MBSE):** initiatives carried out within industry and in the framework of ESA's R&D programmes have been brought together under the common MB4SE initiative. At the start of 2020, a call for ideas on MBSE was published on ESA's [OSIP](#). This resulted in four studies and 12 early Technology Developments initiated for a budget of €2.5m notwithstanding some ideas directed through other channels such as open ITTs, inclusion in the MB4SE Harmonisation roadmap, and research co-sponsorship.



MBSE initiative platform – Credits: ESA

- Commercial Off-The-Shelf (COTS):** The ESA 11-point workplan strategy was still progressing well in 2020, aimed at ensuring efficient utilisation of COTS in future ESA missions while continuing to foster a strong European space-qualified component manufacturing community. An [OSIP](#) campaign opened in Spring 2020 resulted in 52 selected proposals to proceed to the next stage of evaluation while others have been identified as candidates for technology programmes.

Cross-Cutting Initiatives:

- Advanced Manufacturing:** identifies new materials and processes, and spins in disruptive materials and manufacturing processes from non-space industrial sectors. In 2020, ECSS-Q-ST-70-80C DIR1 "Processing and Quality Assurance Requirements for Metallic Powder Bed Fusion Technologies for Space Applications" was issued.
- Digital Design-to-Produce (D2P):** develops, spins-in and demonstrates core technologies, which enable a digital engineering process flow from design to verification, operations and data exploitation. The main goal is to reduce schedule and cost by designing space systems

specifically for manufacturability, integration and verification. Many activities have begun, focused on the application of Model Based Systems Engineering and visualisation techniques.

Sustainability/green:

- o **Clean Space/Sustainability** (since 2020 part of the Space Safety Programme (S2P)) addresses the sustainability pillar of the Technology Strategy by developing technologies to guarantee the future of space activities by protecting the environment.
- o **Design for Removal technologies in Copernicus Expansion missions:** implementation in all 6 Copernicus Expansion missions kicked off in 2020 of specific Design for Removal requirements that simplify their deorbit by a servicer in case of a failure in orbit. It is the first time such an approach has been implemented within ESA and is pioneering these technologies worldwide, as a result of fast-track co-funded R&D by EOP, TEC and OPS.
- o **Active Debris Removal and In-Orbit Servicing:** ADRIOS is developing technologies to remove debris from orbit while demonstrating in-orbit servicing technologies. The resulting Clear Space-1 mission will remove a Vega launcher adapter from orbit.

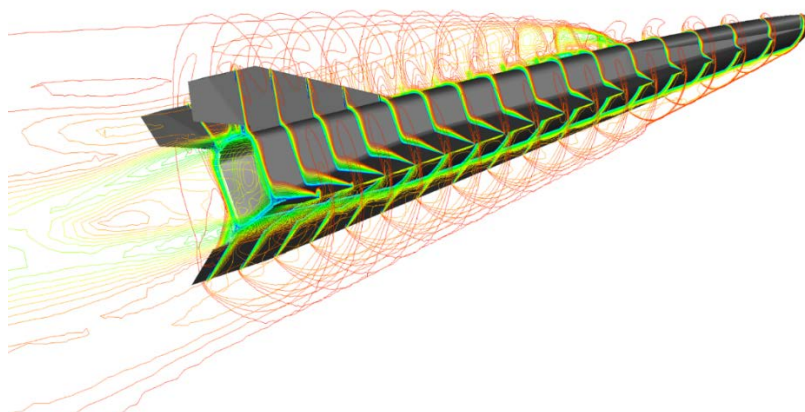
TEC Programmes

The TEC Directorate directly manages, on behalf of ESA, the TDE (Technology Development Element) and GSTP (General Support Technology Programme), and supports, through dedicated funds allocation, the development of smallsats and cubesats to test miniaturised technologies in orbit.

Technology Development Element

In 2020, 127 activities were initiated in the TDE, with 51M€ of industrial commitments in total.

Civil high-speed transportation has been hampered by excessively high fuel consumption or a too low cruise efficiency, but a radical new vehicle concept, HEXAFLY, aims to create a free-flying high-speed cruise vehicle with a multi-disciplinary platform where several breakthrough technologies can be mounted on board for testing. TDE's role in this international endeavour was to assess the shape of HEXAFLY-INT, which has been gradually designed in several iterations since 2014, to see just how aerodynamic it truly is.



HEXAFLY-International simulation – Credits: ESA

GSTP

Through the optional [General Support Technology Programme](#), ESA, Participating States and industry work together to convert promising engineering concepts into a broad spectrum of useable developments. The [GSTP Annual Report 2019](#) was published in 2020.

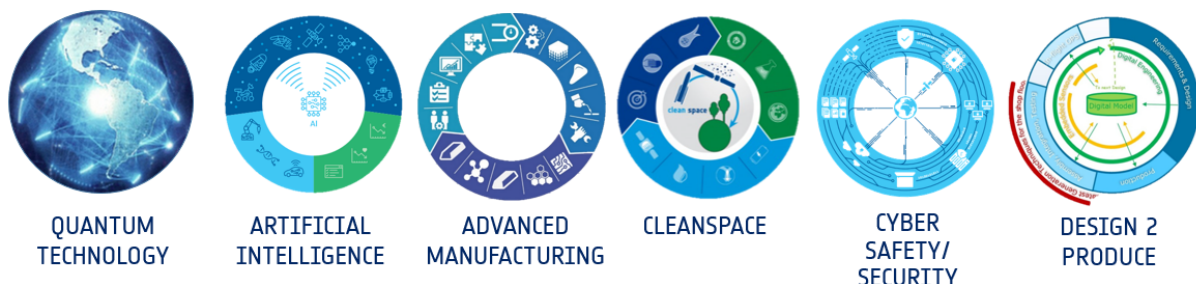
GSTP performs its activities under three distinct elements (see figure below) and one component - the Precise Formation Flying demonstration mission.



GSTP three elements: Develop, Make, Fly – Credits: ESA



GSTP framework evolution – Credits: ESA



Space is critical to the development and deployment of these new quantum technologies.

ESA has been at the leading edge AI research related to space and is key to tackling the 'data overflow' crisis we are facing.

The AM cross cutting initiative aims to create new high performance Space products by actively reducing the limitations imposed by the traditional manufacturing processes/concepts.

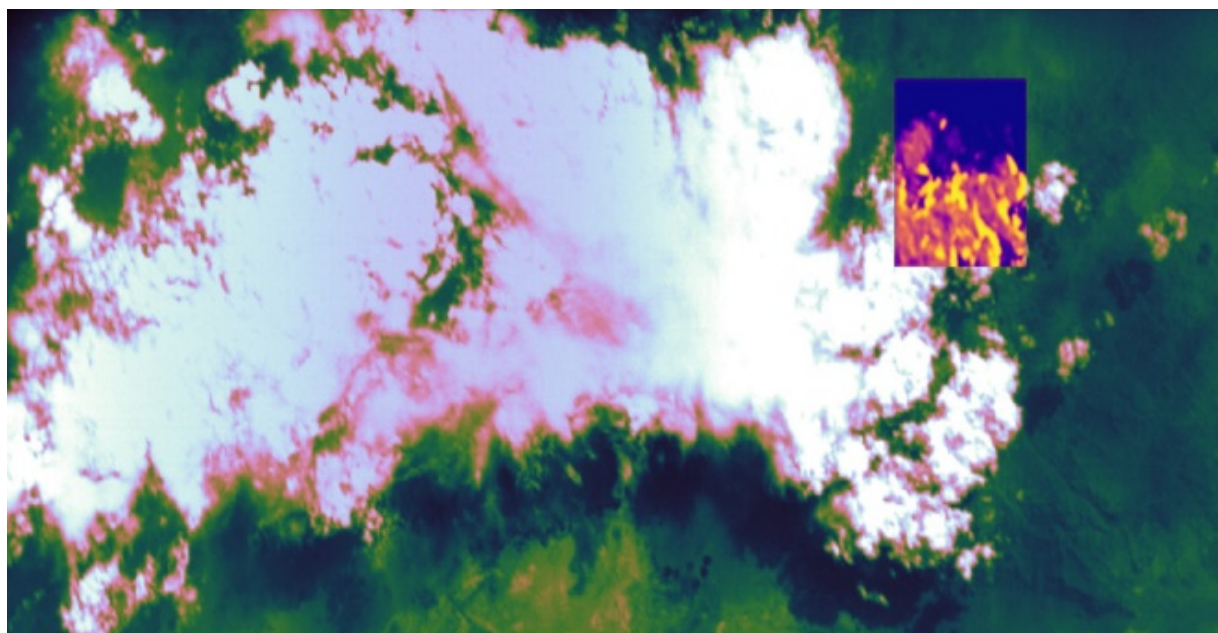
Since 2012, the Clean Space initiative has addressed the entire lifecycle of the Agency's projects, from conceptual design until preparation for active debris removal.

Space systems, interwoven with critical terrestrial infrastructure, are not immune to cyberattacks. Cybersecurity therefore has become a critical requirement for next generation space systems.

With this cross-cutting initiative ESA is aiming to develop techniques to improve the space system end-to-end development process with the goal to reduce engineering lead time and cost.

Cross-cutting Technologies – Credits: ESA

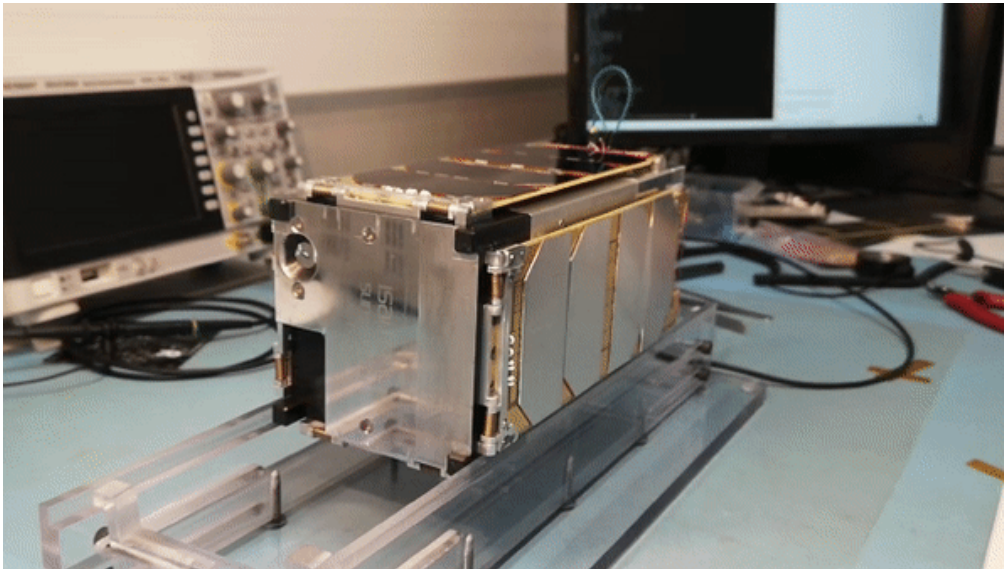
- For the first time in history, an image was processed in space using AI. Along with the launch of the Simba and PICASSO cubesats, September also saw HyperScout 2 fly on the FSSCAT cubesat – all three developed by GSTP with Belgium and the Netherlands, respectively. HyperScout 2's onboard processing capabilities were extended with the Myriad 2 Vision Processing Unit (VPU) from Intel – a GSTP Element 1 supported activity.



The deep neural network algorithm identified the clouds in an image of part of the Earth's surface. The capability to process images using artificial intelligence on a satellite opens up possibilities for a large number of applications – Credits: ESA/ cosine

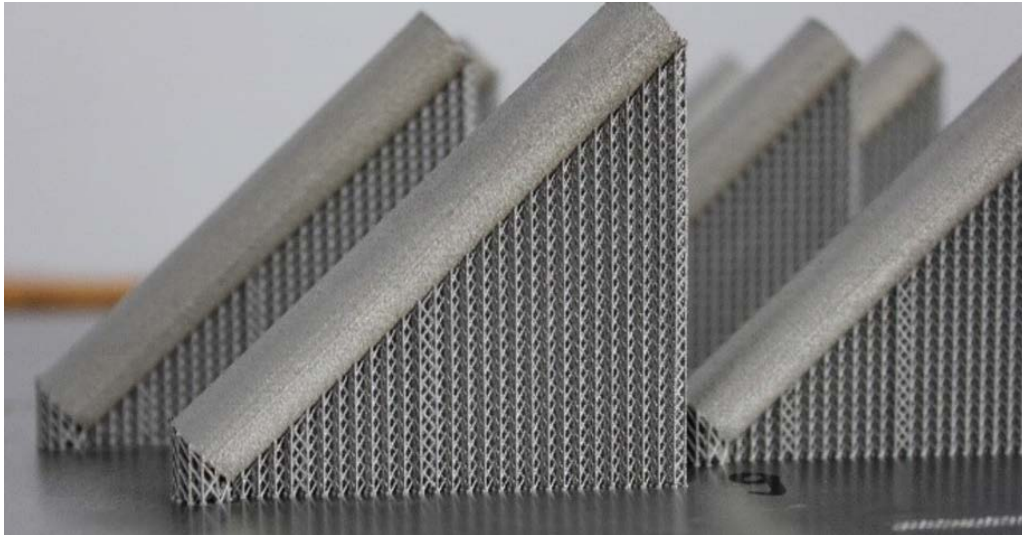


The Myriad 2 VPU from Intel – Credits: CERN



The SUNSTORM cubesat – Credits: ESA

- ESA is preparing a GSTP-developed X-ray flux monitor for operation in space – bringing Europe closer to autonomy over its space weather forecasts. By measuring X-ray emissions from the Sun, we can predict when space weather events might occur and prepare for them. The monitor will launch in 2022 on board the SUNSTORM cubesat.
- A successful activity in Germany kick-started several SMEs to begin using additive manufacturing (AM). Now activities in Romania will develop SMEs' ability to design, manufacture and test parts using AM, based on parameters and establishing processes that meet space industry requirements.



Parts developed using Additive Manufacturing under the GSTP Framework – Credits: ESA

Small missions

- TEC has supported small missions of other directorates, including multiple technology developments for Space Safety and Security's Hera mission, notably its autonomous navigation system.



Hera mission implementation contract signed ([link](#)) – Credits: ESA

- TEC has also commenced its ADRIOS (Active Debris Removal/ In-Orbit Servicing) initiative ([link](#)). The first mission to result will be Space Safety and Security's ClearSpace-1 in 2025.

SMALL MISSIONS STATUS

PROJECTS IN ORBIT

- Proba-1 small sat will remain operational until 20 years anniversary in october 2021
- Proba-2 small sat is operational for spaceweather,
- Proba-V small sat has completed the operational mission and has entered the experimental phase,
- PICASSO cubesat with the VISION payload launched with VV16 is in commissioning phase with BISA,
- SIMBA cubesat launched with VV16 is in commissioning phase with RMI



PROJECTS IN DEVELOPMENT

- Proba 3 Formation Flying mission development is progressing in phase D for a launch end 2022,
- μ HETSat mission is in phase D with a launch planned in 2021,
- RadCube cubesat is in Phase D with a launch planned in 2021,
- PRETTY cubesat is at CDR stage for a launch planned in 2021,
- Sunstorm cubesat is in Phase D for a launch planned in 2021,
- GOMX-5 cubesat is in Phase C with a launch planned in 2022,
- PV-CC cubesat is in Phase C with a lunch planned in 2022.

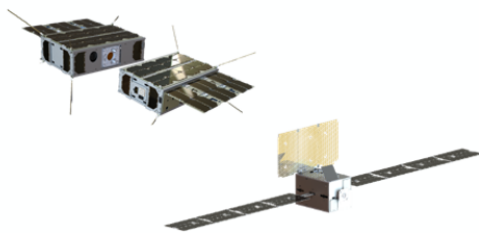


Small mission status: in orbit and in development – Credits: ESA

SMALL MISSIONS PREPARATION

PROJECTS IN PREPARATION

- RACE and M-ARGO for robotics and deep space demonstrations have completed Phase A,
- LUMIO and VMMO are in evaluation and launch opportunities (to Moon orbit) are being discussed



CDF ACTIVITIES are performed in support to small missions preparation:

- AMACS: Advanced Manufacturing technologies for nanosat/cubesat,
- ELAINE: 16 U cubesats constellation for spectrum mapping,
- SMARTIES: Small Mars mission ARchiTecture Study,
- MARTINI Cubesat-class Exploration missions in Mars orbit,
- ELCANO: constellation of cubesats / microsat for Navigation.



Small missions in preparation – Credits: ESA

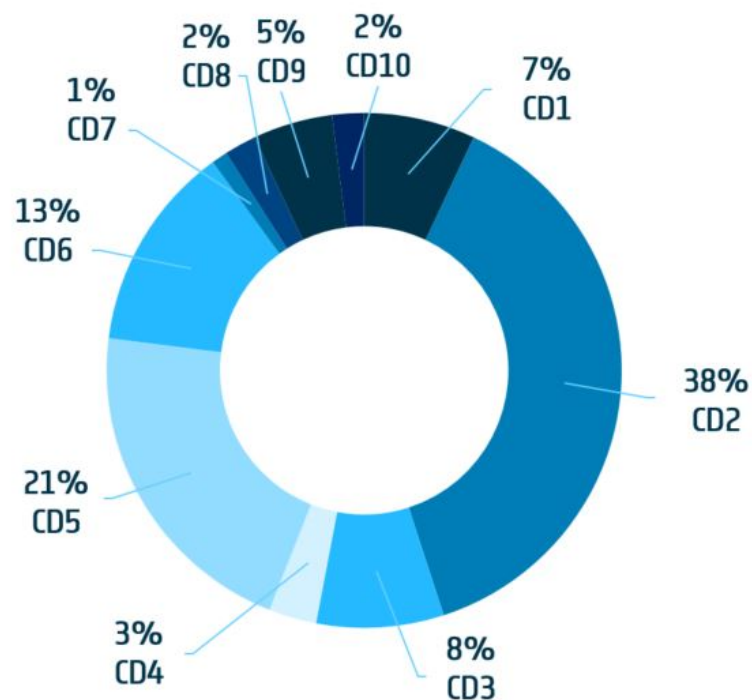
Technology Transfer and Patents

TEC's technology transfer strategy adds value through the identification of space technology or know-how, financing demonstration activities and providing access to technical support.

In 2020, a total of 73 applications to the Technology Transfer Announcement of Opportunities were received. A total of 6 feasibility studies, 20 proofs of concept and 3 demonstrator proposals were selected.

- 50% of 2019–2020 financed activities mature for the next stage of the Technology Transfer process
- 50% of 2009–2018 financed activities resulted in a transfer

2019-2020 Activities financed by CDs



2009-2020 activities financed by CDs – Credits: ESA

More information about the status of projects financed by the TTP can be found at the following [link](#).

Resources Management, Corporate Activities and External Relations

Human Resources, Facility Management and Informatics

Human Resources

Some considerations on the impact of the COVID-19 crisis

Since February/March, the efforts of the Human Resources (HR) Department have essentially been driven by the objective of ensuring business continuity and healthy working conditions, which translated into having almost all staff (save a few whose activities on site were required) telework as from mid-March, and relaxing some conditions associated with the teleworking scheme (for instance, some expatriated staff were authorised to work from their home country in order to take care of their families).

A Lessons Learned questionnaire was circulated to all staff in June, seeking their views on the teleworking experience, on the tools placed at their disposal and on how the crisis was being managed by ESA. Very positive answers were received on this last point, and the vast majority of respondents expressed a significant desire to work from home and maintain a certain level of flexibility after the crisis, alternating on-site work with telework periods according to business needs. On this basis, the Human Resources Department has had several discussions with staff representatives in order to define a new teleworking policy which would contribute to the modernisation of ESA as an organisation, able to attract new talents in the coming years. Although onsite presence is and will remain a priority to ensure collaborative work, team cohesion and the integration of newcomers, it is proposed to ease access to teleworking and reinforce performance management by objectives and a culture based on autonomy and trust.

ESA Medical Teams

At the beginning of the pandemic, ESA medical teams implemented protocols to conduct contact tracing, risk assessments and quarantine measures. Significant efforts were made to issue health communications and instructions in the rapidly changing epidemiological situation. ESA medical officers played a key role in advising local and central crisis management teams on medical matters. There was a special focus on making recommendations on teleworking in order to prevent contamination on site. National lockdowns required medical personnel on all sites to shift to remote consultations, to increase their availability to respond to various urgent and non-urgent requests and to provide greater support for all medical and psychosocial matters. To boost the psychological support available to colleagues normally working at ESA sites, ESA arranged access to other psychological support services.

Internal HR Communication

Throughout the COVID crisis, internal communication has been of the utmost importance. Since the onset of the pandemic, communication specialists within HR have co-ordinated with senior management and the COVID crisis teams to communicate regularly on the decisions taken. They developed a dedicated intranet site, articles, posters, booklets and other communication material to keep all employees informed of the various measures. ESA was thus able to adjust to the changed working conditions quickly and efficiently, while maintaining a high degree of productivity and motivation. This was reflected in the abovementioned questionnaire circulated among ESA staff in June where the organisation's COVID-related communications were rated positively. The vast majority

of the survey participants approved of the way ESA had managed the crisis so far in terms of information (92%), resilience and reactivity (95%), precautionary decisions taken (93%) and compliance with host nations' special requirements (88%).

Learning & Talent Development (L&TD)

The pandemic situation and consequent immediate teleworking conditions for all ESA staff led to a cessation of all internal courses organised on site as from March. In view of these lasting conditions, gradual re-engineering of the courses into a virtual format began, building on experience gained both internally with new collaborative tools and externally with training providers, the latter also experimenting with new approaches to encourage staff participation. As a result, nearly 90% of ESA internal courses were offered again throughout the second half of the year, except where practical hands-on training could not be translated into this virtual context. Despite not offering the same learning impact as a face-to-face event, participants expressed satisfaction as reflected in the high evaluations obtained after the training. In parallel, the specific challenges encountered by staff and managers when dealing with remote working also triggered the development of new learning opportunities, helping staff to work from home, adopt resilience strategies and cope with virtual meetings and remote teams. To compensate for the suspension of certain types of specific internal training, staff could also request funding for external online resources.

Other HR activities in 2020

Recruitment across ESA

In 2020, a total of 336 vacancies for permanent, fixed-term and advanced recruitment positions were published, of which 150 were internal only and 186 were both internal and external.

Due to the COVID-19 crisis, ESA adjusted the selection process. Starting in March, face-to-face interviews were suspended, and all interviews conducted online, using videoconferencing platforms. This transition proved very successful, since the actual number of recruitments increased both for external (118 in 2020 vs. 87 in 2019) and internal (144 in 2020 vs. 81 in 2019) candidates.

Talent Acquisition

The staff "retirement wave" has put the need for proactive search for talent into sharper focus as ESA faces increasing challenges to attract sufficient suitable applications while implementing diversity criteria. In line with long-standing practice within industry, a dedicated Talent Acquisition Service was established in 2020. The team is responsible for conducting employment marketing measures to strengthen ESA's talent outreach capacity. The success of the Young Graduate Trainee campaign (see below) accompanied by marketing activities tailored to the profiles of potential candidates has shown initial signs of the Service's usefulness.

Entry-Level Programmes

ESA is offering a variety of entry-level programmes, such as Young Graduate Trainees (YGT), Research Fellows (RF), National Trainees (NT), and Interns. Each programme has its specific purpose and advantages for both ESA and trainees. In 2020, ESA hosted 257 YGTs, 74 NTs, and 90 RFs. Notwithstanding the uncertainties arising from the COVID-19 pandemic, the recruitment of YGTs was maintained at a constant level. In terms of diversity, in particular, the 2020 campaign was highly successful and the number of applications increased substantially to reach 7827 applications. With 23% of the applications coming from female candidates, 46% of all YGTs hired were women. As for

geographical representation, 85% of all YGTs taken on were from under-represented ESA Member States whereas only 38% of applications came from those countries.

A new project was begun in 2020 with the creation of a Junior Professional Programme, complementing the existing entry-level programmes but focusing in particular on strengthening the diversity of ESA's permanent workforce. To this end, the Programme aims for succession planning and the career development of young professionals. Subject to the necessary approvals, the Programme should be implemented as of 2022.

Performance Management Reform

Despite the COVID-19 situation, implementation of the Performance Management Reform is progressing on schedule and has entered a major milestone i.e. the goal-setting exercise. Setting goals and expectations in line with the grade level and post description of each staff member is instrumental to assessing the performance of staff at the end of the year. The next milestone of the reform in June 2021 is the transition to the new salary scales and launch of the first mid-year performance review in the summer. Particularly during this pandemic, it is crucial to allow for flexible adjustment of goals in line with changing circumstances, which is the purpose of the mid-year review.

Throughout the entire implementation of the reform, staff and managers are being helped by means of intensive training, awareness sessions and many other communication and change management measures.

Social Security

A newly created "Adult Beneficiary Fee" was planned to be implemented for the first time in January 2021, addressing the provision of social security coverage for entitled spouses and other dependants. Sessions were held to inform staff members and pensioners of the introduction of this new fee.

Internal Justice Reform

Following extensive negotiations between the Administration and the Staff Association, a major reform relating to the Agency's internal justice system came into force, with the approval of the AFC and Council.

The reform aims, on the one hand, to strengthen ESA's dispute settlement system, notably by enhancing the efficiency of the Advisory Board as well as by professionalising and rendering the procedure more transparent, and on the other, to improve the granularity of disciplinary action to be taken in cases of misconduct.

Diversity & Inclusiveness

A word from ESA's Chief Diversity Officer: In 2020, in the framework of the Diversity and Inclusiveness initiatives, very positive results were achieved in continuing to attract and retain a gender-balanced and geographically diverse workforce. The Agency has especially reinforced its commitment to gender balance, having achieved certification from external auditors EDGE Certification following an Agency-wide employee survey and audit. ESA's ongoing commitment to diversity and inclusiveness relating to people with a disability has also generated a number of actions; the launch of the Parastronaut Feasibility Project is world-leading in its aim to widen accessibility to human spaceflight, while ESA also collaborated with the IAE, ESO and SKAO to create the first SARA 2020 workshop, designed to explore the links between space science, astronomy, inclusion and disability.

Estates and Facilities Management

2020 was an unprecedented year for ESA's Estates and Facilities Department. The COVID-19 pandemic required rapid reconfiguration of ESA's establishments to permit the business continuity of onsite critical activities; this included spacecraft operations at ESOC, corporate IT functions at ESRIN and Test Centre campaigns at ESTEC. In addition to this, the Department has also been responsible for Agency crisis management policy and crisis support to senior decision makers. It therefore had key roles in both the Central Crisis Management Team and those at all of ESA's main sites and centres.

ESA's response to the pandemic was based upon long-established crisis teams led centrally by the Director General and locally by Heads of Establishment. Critical tasks had been identified together with the necessary key workers to carry them out. In line with national guidelines, a minimum onsite posture was adopted and then expanded to include "priority one" and then "priority one delta" activities. Supported by a resilient central IT infrastructure, the overall response proved remarkably robust.

In its more classical function, the estates teams implemented COVID-19 protection measures at least to the standards mandated by host nations and in accordance with the requirements of ESA's medical advisors. Services, not least cleaning, security and engineering were adapted to ensure that essential hygiene and access control arrangements were implemented. Best practice in such aspects as ventilation and cooling in relation to COVID-19 was constantly monitored and suitable environments created. Personal protective equipment (PPE) was rapidly procured and distributed in line with emerging requirements.

The longer-term impact has centred significantly on investment activity. The ESA HQ renovation project incurred a four-month delay as construction activity and manufacturing of components was curtailed. Given the very broad impact of COVID-19 this was at the lower end of expected outcomes. Elsewhere in the Agency, steps were implemented to recover delays in other estates programmes, and this was ongoing at the end of 2020.

Information Technology

The IT Department was a key success factor in ESA's smooth shift to teleworking of almost all staff and contractors from one day to the next. The Department consistently supported both connectivity during the COVID-19 pandemic and the resultant teleworking scenarios, while continuing to develop and complete ongoing projects. These included preparation for the transition towards esa365. New applications providing extended functionality for users migrating to esa365 are being introduced and validated in order to enable the transition away from the Lotus Notes email and applications solution.

COVID-19 Teleworking

The IT Department has been in the front line in ensuring that the Agency can continue to work without interruption while working from home during the COVID-19 crisis. Service continuity was prioritised to provide an uninterrupted remote service to thousands of employees and contractors who suddenly had to begin teleworking. Remote assistance was provided, and IT colleagues were onsite where necessary as essential workers, adopting new, safe procedures as required. A dedicated website was created as a reference point for teleworking for all users, to provide clear information on the new way of working.

The IT Department increased ESA's teleworking capabilities. The Webex solution was redeployed to support the poly-conferencing infrastructure, Skype for Business support for all users was increased,

Alcatel OpenTouch software was released to connect the ESA telephony network over the internet, and the Agency's VPN strengthened. During this time, the IT Department sustained more connections at any one time than ESA has ever had to offer, with approximately 4400 simultaneous connections every day. At the same time, an Endpoint Detection and Response Infrastructure – a cyber security monitoring tool to mitigate the increased threats from the COVID-19 situation – was released successfully.

Digital Security during Teleworking

Security has been a constant priority for the IT Department. Initiatives to improve the Agency's network security and educate users were important to the Department's activities in 2020, particularly with respect to the challenges resulting from remote working. This also included the successful surveillance of the digital infrastructure for security compliance with ISO 27001 standards.

esa365 – Microsoft 365 for the Agency

Despite the increased effort required to support the entire workforce while teleworking, the IT Department continued with its mid- and long-term goals to support ESA's working needs. Thus, concerning the esa365 project – the purpose of which is to roll out Microsoft 365 Agency-wide – the technical configuration was completed as was the migration of a group of pilot users to validate business use cases, with the general migration wave set to take place in 2021 over some nine months. The implementation of esa365 requires forward-working improvements to introduce new software and backward-compatibility efforts to preserve the information in the IBM Domino tools currently used by ESA.

ESA's Business

ESA Council meeting at ministerial level documents went digital with the ESA blue-docs system being further enhanced in 2020. A digital subscription tool also delivered Space19+ outcomes and deliberations to stakeholders. Finally, a project to consolidate ESA's corporate digital archives was prepared and rolled out.

New Human Resources Software Functions

New functions for requesting and approving leave and flexi time, along with digital reporting on learning modules and critical system updates were developed and implemented for ESA's Human Resources software. This was particularly important given ESA's teleworking requirements during the pandemic.

E-Tendering Developments

A number of new modules were developed for the Agency's tendering tool esa-star. A Publication module has replaced EMITS, giving external entities an easy-to-use application for browsing tender information, procurement news and other ESA documents. The ESA-Match module is the Agency's new matchmaking platform for economic operators to showcase their skills and establish business relationships through direct contact with other entities. ESA-Star ECM (Entity Capability Mapping) is a tool designed to support ESA's harmonisation process. ASTRA – a new offering that will replace Lotus Notes Easy Rate applications – was also deployed.

Procurement and Industrial Policy

Procurement

The Procurement Department main objectives for 2020 were to ensure the contractual implementation of the programmatic decisions of Space19+ including, for instance, the procurement of the 6 Copernicus HPCM missions and the E3P missions.

Nevertheless, to avoid an economic calamity in addition to the COVID-19 pandemic, the first months of 2020 were devoted to securing the most timely and effective support to the European space industry.

On 23 March, a business continuity plan was released on EMITS addressing:

- (i) the continuation of the procurement process as planned, through adapted meetings (fully or partially virtual) of the ESA delegate bodies (IPC, in particular);
- (ii) the full digitalisation of the tendering process, through the esa-star system and allowing virtual meetings for Tender Evaluation Boards;
- (iii) the reduction of time-to-contract, mainly through virtual meetings for contract negotiations and electronic signature of contracts and CCNs;
- (iv) the reduction of time-to-payment, based on the full digitalisation of the handling of invoices, leading to payments substantially ahead of contractual due dates;
- (v) the exceptional entitlement of partial and advance payments throughout the industrial supply base.

These measures, initially intended to be in place for three months, were further extended and remained in force at the end of 2020. With 1470 contracts issued in 2020, 24% more than in previous years, thanks to the devotion and professionalism of all the staff and contractors involved, the Agency was warmly praised for its initiative as something quite unique in the European and international public procurement context.

Despite these difficult circumstances, the procurement approaches for the implementation phases of Mars Sample Return - Earth Return Orbiter, Mars Sample Return - Sample Fetch Rover, Gateway- I HAB, Gateway - ESPRIT, Mars Sample Return - Sample Transfer Arm and ARIEL were all approved.

In parallel, the full procurement was completed for flagships and key programmes like Copernicus HPCM, ERO, I-HAB and ESPRIT.

Particularly, for Copernicus HPCM, following the bringing forward of the relevant competitive ITT in summer 2019, six mission contracts totalling €2.5 billion were signed in the second half of 2020.

Following an innovative procurement procedure, the service contract for the ADRIOS (Active Debris Removal In-Orbit Servicing) project was signed with the Swiss start-up, ClearSpace.

In December, the CSG contract between ESA and CNES was signed, laying down the detailed conditions for the execution of CSG Launch Range activities and the Core Launch Range Renewal programme activities over the period 2020–2024. Also, a procurement relating to the so-called New European Space Transportation Solutions initiative was initiated in December, focussing on satisfying future European space transportation service needs for European institutional users.

Industrial Policy and Audit

Due to the COVID-19 crisis, the 2020 High Level Forum with industry was postponed until 2021. Instead, a virtual preparatory event was held so that representatives from SMEs, midcaps and Large System Integrators could report on the impact of the crisis and of the measures taken by ESA to support them. Moreover, the first online edition of the ESA Industry Space Days was organised in September 2020, bringing together a record 1900 participants from 1000 companies through two days of virtual presentations and business-to-business meetings. The COVID-19 crisis has led ESA to monitor even more closely the status and health of Europe's space industry and its different supply chains, thus leading to the further development of a European space industry observatory, in collaboration with Member States and industry.

In 2020 the Industrial Audit Services were able to achieve record cost savings amounting to 217M€ generated by the audit results agreed with industry. These results have an accumulating effect on savings in the last three years of 448M€.

In October 2020 the ESA Council approved the "Formal Review of Geographical Return 2015–2019 Period" document confirming that results better than the overall minimum criteria decided were secured for the Member States at the end of 2019 (ref. ESA/IPC(2020)45, rev.1). The resulting Resolution including proposed industrial policy measures for the Period 2020–2024 (ESA/C/CCXCI/Res.1 (Final)) was also confirmed by Council in November, where the new geo-return lower limits were set and a proposal for the alignment of geo-return domains to programmes was established. During 2020, the Executive began working on implementing the ambitious Action Plan established in the Resolution with a view to ensuring their timely completion in 2021.

ESEC

Against the backdrop of the developing COVID-19 pandemic and under the guidance of ESA's central business continuity team, ESEC was able to protect its workforce and on this basis succeeded in maintaining operational activities in all domains.

Looking at the development of activities, the year was marked by the ramping up of preparatory efforts for the establishment at ESEC of the hub for ESA's cyber resilience efforts. A new Belgian Government reasserted Belgium's support for the deployment of related activities at ESEC. On a practical level, ESEC activities in this domain span from a thorough review of the current and future site security framework, via contributions to the procurement preparation for the cyber resilience activities to the determination of infrastructure needs and definition of the related procurement efforts.

Support to the operation of EU GNSS continued flawlessly including the move of the core IOT support area into a state-of-the-art infrastructure and the preparations for the hosting of a new EGNOS NLES.

Operation of the fleet of PROBA satellites (PROBA-1, -2 and -V) was performed while PROBA-V moved into a new, experimental mission phase. The PROBA team also provided support to the preparation of the PROBA-V companion satellite(s) as well as the PROBA-3 and ALTIUS projects that will be operated from ESEC.

The education team successfully adapted and moved a number of its training offers online. Wherever possible, support to test and validation of educational satellite projects continued under strict safety precautions both in-situ at European universities and on-site at ESEC.



ESEC site Redu, Belgium – Credits: ESA

Finance, Planning and Control

Faced with the sudden and unexpected COVID-19 situation, the Department mobilised rapidly, transitioning immediately to a 100% remote working model, developed business continuity measures to ensure financial flows to industry, adapted the 2021 budget process, and implemented additional measures to ensure ESA treasury and pension fund asset performance.

Financial Accounting and Operations

ESA successfully closed the 2020 accounts according to the accelerated timetable in the revised Financial Regulations. For a ninth consecutive year, ESA obtained a "clear" audit opinion, a major achievement providing maximum reassurance on the quality of its financial management and reinforcing the foundations of trust and confidence.

In line with the Audit Commission recommendation to further strengthen ESA's internal control system to meet audit readiness objectives and the revised timetable, ESA successfully completed a third interim closure. All accounting closure and audit activities, and fixed asset inspections, were successfully performed through virtual collaboration involving multiple internal and external players.

Financial accounting processes were further digitalised and financial accounting QMS and Financial Instructions were updated.

ESA accurately managed and accounted for some 54 500 transactions. Through highly digitalised processes, ESA efficiently operated and interacted with suppliers and stakeholders. Supplier payments were processed fully electronically incorporating special COVID-19 measures designed to ensure

industry cashflows. In particular, the average time-to-payment decreased to ten calendar days compared to ESA's normal 30-day contractual terms. Further digitalisation developments were carried forward in invoice validation, using AI-based tools.

Corporate Planning, Budgeting and Controlling

Planning and Controlling implemented the Space19+ ministerial decisions and were instrumental in consolidating and assessing multi-year financial plans of all programmes and activities, leading to the recommendation of cost plans by all Programme Boards and Council's adoption of the 2021 budgets.

The specifically adapted 2020 process included an additional July planning exercise to submit revised cost plans to September programme boards and committees, incorporating COVID-19 impact assumptions. ESA's Medium-Term Financial Plan was prepared, reflecting the COVID-19 measures in support of industry in 2020 and confirming the sustainability of programmes and activities for the 2020–2022 period. Continuous engagement with Member States helped to optimise 2020 and 2021 income following a historically high one-time budget in 2020.

Relevant QMS and Financial Instructions were updated, while continued effort was devoted to digitalisation including an Agency-wide planning and control tool and extensions to the Digital Subscription tool used by Member States at Space19+.

The team contributed to the FFPA discussion with the European Commission, leading the "ESA Cost structure and remuneration" working group, supporting the AFC Advisory Group, and supporting directorates' preparation of remuneration estimates.

Cost

ESA's total cost in 2020 amounted to 5127M€ of which 3908M€ related to ESA programmes and activities and 1219M€ to Third Party activities.

Mandatory Activities represented 16% (793M€), 59% was accounted for by Optional Programmes (3000M€), 2% by activities associated to the General Budget (110M€), 24% by programmes financed by Third Parties (1219M€) and 0.1% by others (4.7M€).

Mandatory Activities are composed of the Science Programme and the Basic Activities.

Basic Activities represented 5% of ESA total cost in 2020 (249M€) and include, among other elements, the Discovery, Preparation and Technology Development Programme, Core Technical Expertise, Common Infrastructure, Education, Earthnet and Heritage Space.

In 2020, the cost of the Science Programme amounted to 545M€ (11% of ESA total cost). The main activities were the development of JUICE, Euclid, PLATO and SMILE as well as the preparation of future missions (Voyage 2050). Solar Orbiter was successfully launched on an Atlas V from Cape Canaveral on 10 February and in November, the SPC adopted the Ariel mission as the fourth Medium mission in the Cosmic Vision plan, with a launch planned for 2029. In addition, the Science Programme funded the operations of ten satellites in orbit, among which GAIA reached an all-time record of scientific publications in a single year.

The execution of programmes including the study, development, launch and operations of satellites and spacecraft as well as the development, exploitation and maintenance of launchers and other space systems is implemented via the Agency's Optional Programmes, in which Member States participate through voluntary subscription.

In 2020, €929m, 18% of ESA total cost, were accrued on Space Transportation programmes (excluding the Guiana Space Centre, the CSG). The execution of the programmes was impacted in 2020 by a combination of COVID-19 effects and specific programmatic delays mainly in the development area. The development of the Ariane 6 Launcher System has reached its final phase, and the Ariane 6 Launch Base Critical Design Review close-out was held in December. The Vega C Launch System Ground Qualification Review was kicked off in December. For Space Rider, the main procurements for Flight segment bridging Phase C, D/E1 and Ground segment Phase D/E1 have been released and the activities are progressing. In addition, the Future Launchers Preparatory Programme continued to make strides in various technologies and integrated demonstrators according to their objectives.

The Ariane exploitation activities supported three successful launches of Ariane 5 during the year. There were some delays due to the unavailability of the CSG launch range, impacted by COVID-19 and technical anomalies. Vega exploitation supported two launches of Vega during the year. VV17 failed in November due to a manufacturing issue in the AVUM.

In Earth Observation, 756M€ were accrued in 2020 for ESA Programmes & Activities (representing 15% of ESA total cost). The Copernicus Sentinel-6 Michael Freilich satellite was successfully launched on 21 November, providing systematic measurements of sea levels. The Copernicus HPCM contracts (CO2M, CRISTAL, LSTM, CIMR and CHIME) were all signed in 2020. Development work continued on Earth Explorers: EarthCARE's ATLID was delivered to the prime for integration, and both Biomass and Flex are nearing their respective Critical Design Reviews (CDRs). Altius completed its Preliminary Design Review (PDR) in September. In addition, following PB-EO approval, implementation of the first and second SCOUT missions was initiated.

Earth Observation Third-Party activities amounted to 516M€ (42% of Total ESA Third Party Programmes), mainly related to cooperation with the EU on Copernicus for a total of 337M€. The remaining cost is linked to the cooperation with EUMETSAT on MetOp Recurrent (76M€), Meteosat Third Generation (€71M) and SEOSAT (25M€). It should be noted that on the EO Meteorological Programmes, out of the 110 M€ new subscriptions received for Meteosat Third Generation, €30m were contributed by EUMETSAT via the MTG Recurrent Third Party programme, thus allowing the programme to run up to completion.

Cost for Human Spaceflight, Microgravity and Exploration reached 638M€, the highest amount since 2015, and 12% of ESA total cost. A majority of the cost accrued is related to Orion-ESM as a result of the implementation of risk correction measures on ESM1 and fast delivery of recurring equipment and subsystems on ESM3. The accruals also include completion of Long Lead Items (LLIs) and Electrical Electronic and Electro-mechanical (EEE) components for ESM Batch A and reflect the significant volume of partial milestone accruals agreed in the framework of the COVID-19 measures to support industry. For ExoMars, the pace of activities has intensified in preparation for the 2022 launch, including the completion of the Engineering Test Model (ETM), compatibility tests between the Descent and Rover modules and parachute testing. Following signature of the Phase C/D contract for the Earth Return Orbiter (ERO) mission, in the framework of the Mars Sample Return project, activities progressed significantly including completion of the mission design consolidation for the S-PDR.

Telecommunications and Integrated Applications grew to a record 421M€ in 2020, representing 8% of ESA total cost. The main objective of the Programme Domain remains European and Canadian Competitiveness on the worldwide telecommunications market. This was implemented through the continuation of Generic Programme Lines Core Competitiveness, Partnership Projects, Business Applications and Space Solutions as well as the crucial Future Preparation. In addition, three Strategic

Programme Lines were successfully initiated: 5G/6G, Space Systems for Safety and Security (4S) and Optical Communications boosting inter alia private and public connectivity ambitions for highly secure services while at the same time supporting competitiveness and commercialisation. The Konnect satellite launch took place in January under the Spacebus Neo platform. The ESAIL satellite (under SAT-AIS) was launched in early September.

In 2020, cost for Technology Support represented 3% of ESA total cost (130M€) of which 18M€ were for PROBA-3 development. The remaining costs were applied to R&D activities.

Cost for the Space Safety Programme amounted to 59M€ in 2020 (1% of ESA total cost). A new baseline for the Lagrange mission was produced combining the results of the ESA Critical Design Facility study and input from Airbus reducing the cost-at-completion estimate while maintaining the mission objectives. NASA expressed a strong wish to participate in the mission and the Independent Project Review made clear recommendations on the procurement process and the risk provisions to be considered. The Hera mission Phase B2/C/D contract was signed during the September meeting of PB-SSA and the Preliminary Design Review (PDR) was successfully concluded in December. The Preliminary Authorisation to Proceed of ADRIOS was initiated in July leading to a successful kick-off in November while significant work was carried out on development plans, requirements, launcher compatibility through mission analysis and rendezvous and capture definition.

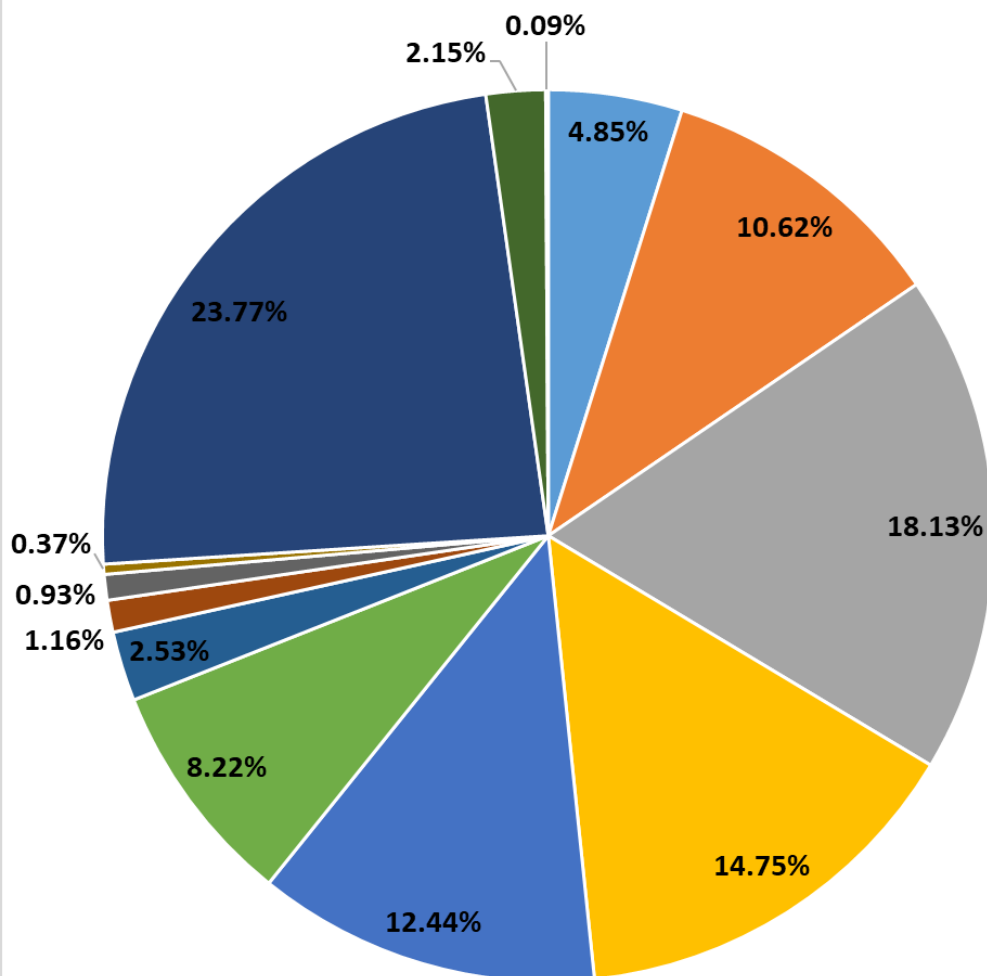
Activities for the Prodex programme, representing 0.9% of ESA's total cost for 2020, continued throughout the year with institutes and industries participating in over 250 ESA activities in the domains of Space Science, Earth Observation, Microgravity and Exploration (scientific instrumentation and exploitation).

Navigation programmes represented 0.4% of ESA's total cost in 2020 (19M€). The NAVISP Programme has now incubated more than 162 activities. At Space19+, Ministers gave a clear vote of confidence in the NAVISP programme by subscribing to Phase 2 with a significant enhanced yearly level of activities. All the Element 1 activities defined are ongoing as scheduled. Procurement of Element 2 activities increased at pace. 85% of the total Element 2 Phase-1 and 45% of the total Element 2 Phase-2 available envelopes for industrial activities have been committed. The Element 3 Call for Proposals generated numerous activities corresponding to more than 90% of the total Element 3 Phase-1 and 30% of the total Element 3 Phase-2 available envelopes for industrial activities. NAVISP is also supporting the definition of PNT requirements for lunar exploration in collaboration with D/TIA and D/HRE, aiming for a long-term cooperation with NASA on the subject, as a possible outcome of the next ESA Council meeting at ministerial level.

Navigation Third-Party programmes represented 51% of Total ESA Third Party Programmes (628M€) related to activities funded by the European Union through the Galileo and EGNOS Delegation Agreements and the EGNOS GSA Working Arrangement. Galileo System performance is excellent and stable. 22 spacecraft are operational, 24 satellites are available for Search And Rescue Service, and two satellites (launched in 2014 in elliptic orbit) have been integrated in the constellation as auxiliary satellites since 30 November. Galileo is moving towards FOC with the launch of two Batch#3 satellites planned in 2021 on Soyuz, and ten additional Batch#3 satellites currently in production at OHB. Although the COVID-19 crisis has caused a two-month delay in the planned schedule of activities, the further development of EGNOS V2, the current generation under operation, has progressed at a good pace. A successful qualification review of EGNOS V242-A took place, allowing its operational deployment by the operator.

Cost (€million)

- Basic Activities €249 million
- Scientific Programme €545 million
- Space Transportation €929 million
- Earth Observation €756 million
- Human Spaceflight, Microgravity & Exploration €638 million
- Telecommunication and Integrated Applications €421 million
- Technology Support €130 million
- Space Safety €59 million
- Prodex €48 million
- Navigation €19 million
- Third Party & ECSA €1,219 million
- Associated to General Budget €110 million
- Miscellaneous €5 million



Income

ESA Mandatory Activities (including the General Budget and Science Programme) and activities associated with the General Budget are financed by Member and Participating States' contributions. The ESA Convention stipulates that the Agency's mandatory activities are financed using a contribution scale based on the national income of the Member States over the previous three years. In 2020, they represented 15% of total ESA income.

ESA Optional Programmes are financed by Member and Participating States' contributions, and for some programmes, also by other entities, including the European Union and EUMETSAT. Participating States declare a voluntary subscription on a multiannual basis. In 2020, they represented 58% of total ESA income.

Third Party Programmes, managed by ESA, are fully financed by third parties. In 2020, they represented 27% of total ESA income.

	2020	
	Mandatory Activities (%)	Optional Programmes (%)
Austria	2.24%	0.89%
Belgium	2.80%	6.01%
Czech Republic	1.09%	1.04%
Denmark	1.83%	0.47%
Estonia	0.14%	0.07%
Finland	1.36%	0.41%
France	14.14%	29.84%
Germany	21.13%	20.47%
Greece	1.08%	0.28%
Hungary	0.70%	0.16%
Ireland	1.25%	0.40%
Italy	10.63%	14.62%
Luxembourg	0.21%	0.72%
Netherlands	4.71%	2.06%
Norway	2.33%	1.61%
Poland	2.90%	0.36%
Portugal	1.13%	0.30%
Romania	1.16%	0.66%
Spain	7.51%	4.80%
Sweden	2.63%	1.53%
Switzerland	3.99%	3.40%
United Kingdom	15.04%	9.10%
Total MS	100.00%	99.21%
Latvia	0.11%	0.04%
Slovenia	0.17%	0.07%
Total Associate MS	0.28%	0.12%
Canada	4.05%	0.67%
Total Cooperating	4.05%	0.67%
ESA	104.33%	100.00%

Budgeted contributions from Member and Participating States in 2020 to mandatory activities and optional programmes

Workforce Management

2020 was the first year of the new flexibility period approved by Member States after Space19+. By means of a flexible utilisation of the staff ceiling, the Agency has been able to bring forward the first recruitments to address the departure wave coming up in the next few years and to manage workload peaks resulting from new activities decided at Space19+. Due to the COVID-19 pandemic, recruitment activities could not be implemented as originally planned after the successful Council meeting at ministerial level in Seville. This led the Agency to propose to shift the flexibility peak by one year to make full use of the 7% granted by Member States. The contractor workforce, both loan employment and service, has steadily grown and the Agency is further implementing actions to reinforce compliance with national labour legislation.

A breakdown of ESA Staff by nationality and grade is provided hereafter.

Member States	Hors Classe	A	L	B	C	Total Staff
Austria	1	34	1	2		38
Belgium	1	100		4		105
Czech Republic		8				8
Denmark		10				10
Estonia		2				2
Finland		18		5		23
France	2	458	9	59	4	532
Germany	3	418	4	39		464
Greece		25		2		27
Hungary		5				5
Ireland		16		3		19
Italy	1	388		19		408
Luxembourg		5				5
Norway		24		2		26
Poland		26		1		27
Portugal		29		2		31
Romania		24		1		25
Slovenia		1		1		2
Spain		200		3		203
Sweden		39		2		41
Switzerland	1	22		1		24
The Netherlands	1	65		9		75
United Kingdom	1	237	2	18		258
Total	11	2154	16	173	4	2358
Non-Member States						
Canada		20		1		21
Total		20		1		21
Grand Total	11	2174	16	174	4	2379

Legal Services

In 2020, the ESA Council took a large number of decisions, including the adoption of:

- Resolution on the setting up of a Council Working Group for the preparation of the Council meeting at ministerial level scheduled for November 2020 (Space Council) (ESA/C/CCXC/Res.1 (Final));
- Resolution on Temporary Adaptation of the Decision Making due to the COVID-19 outbreak (ESA/C/CCXC/ Res.2 (Final));
- Resolution on the Financial Statements of ESA for the year 2019 (ESA/C/CCXC/Res.3 (Final));
- Resolution on the Cooperation for the Civil Lunar Gateway (ESA/C/CCXC/Res. 4 (Final));
- Resolution on the Chairship of Council and subordinate bodies (ESA/C/CCXC/Res.5 (Final));
- Resolution on Industrial Policy Measures proposed for the Period 2020-2024, as stemming from the Formal Review of Geographical Return 2015 - 2019 Period (ESA/C/CCXCI/Res.1 (Final));
- Resolution on the Space Transportation Programme Board (ESA/C/CCXCI/Res. 2 (Final));
- Resolution on Orientations on the European contribution in establishing key principles for the global space economy (ESA/C-M/CCXCII/Res.1 (final));
- Resolution on the application of the common legal framework governing the exploitation of ESA developed launchers and Soyuz from the CSG (ESA/C/CCXCV/Res.1 (Final));
- Resolution on the Agency's budgets for 2021 (ESA/C/CCXCV/Res. 2 (Final)).

11 international agreements were also concluded with governments, international organisations and other institutions (see list in Annex), including two landmark cooperation agreements between ESA and NASA on the Lunar Gateway and the Mars Sample Return mission, respectively, continuing their decades-long successful cooperation on space exploration into the future.

ESA also welcomed Latvia as a new Associate Member with the ratification and entry into force of its Association Agreement with the Agency.

Throughout 2020, the ESA Legal Services Department not only accompanied the drafting and negotiation of, as well as the decision-making for those instruments, but also prepared a wealth of other legal documents for decision by Council and its subordinate bodies.

Relations with Member States

Relations with Member States

The Director General took the opportunity offered by various major events to regularly visit Member States, including during appointments with Ministers in charge of space matters, with delegations, and addressing parliaments, industry, the press, or universities whenever possible. As of March 2020, most meetings were held remotely, owing to the COVID-19 crisis, although discussions remained ongoing and fruitful thanks to the speed with which ESA has moved towards using secure and reliable digital tools.

Space for Earth (www.esa.int/spaceforearth)

Space supporting the UN Sustainable Development Goals (SDG)

(<https://www.esa.int/spaceforearth/sdg>)

Approximately 700 ESA activities supporting the Sustainable Development Goals (SDGs) are now listed in the unique ESA catalogue (<https://sdg.esa.int>). This ESA catalogue was presented and demonstrated on various occasions and was very well received, it being recognised that a tool of this kind was indeed necessary and did not previously exist. Over 40 projects from the UK Space Agency (UKSA) were included as a first contribution to extend the catalogue to also include projects supported at national level by the Member States.

Blue Worlds

(https://www.esa.int/Enabling_Support/Preparing_for_the_Future/Space_for_Earth/Blue_worlds)

The work of the ESA Blue Worlds Task Force continued for all the 19 Member States involved, with them being asked to review existing programmes and user requirements, analyse gaps and propose concrete actions before the next Council meeting at ministerial level. It covers the Atlantic Ocean, the Mediterranean, the Baltic, the Black Sea, the North Sea and the Arctic Ocean. A survey has been designed so as to gauge understanding and interest as well as current and potential involvement of the European maritime user community at large in the exploitation of space-based systems, data and products. The results will be made available in the course of 2021.

ESA_Lab@ initiative

The ESA_Lab@ network has continued to grow steadily (non-exchange of funds collaboration between ESA and research entity/university, ESA providing expertise in kind on research topics/technology areas with added value for both partners) . A dozen legally non-binding Memorandums of Collaboration were in force at the beginning of 2020, with that number now having risen to 15, and many more in the pipeline. The topics concerned cover a broad range from the space economy to engineering and technology, exploration and Earth observation with an emphasis on AI applications for space. In 2020, the Member States most interested were the United Kingdom, France and Italy, followed by Germany,

(https://www.esa.int/Enabling_Support/Preparing_for_the_Future/Space_for_Earth/The_ESA_Lab_initiative).

ESA and civil society

Space 4.0 calls for the integration of space into Europe's society and economy, of which the "third sector" is an important component. Comprising thousands of non-governmental organisations (NGOs), the sector mobilises governments and millions of citizens for community improvement, often providing much-needed societal balance between the private sector and government. An assessment has been made of the current landscape, identifying gaps in current implementation and proposing key recommendations and solutions to ESA's directorates with a view to improving how NGOs could benefit from ESA's programmes and activities and develop the conditions required to make Europe the leader in this space sector which is currently taking shape.

External Relations

Relations with EU Institutions

In 2020, the main focus of relations between ESA and the European Union was the preparation and negotiation of the ESA-EU Financial Framework Partnership Agreement (FFPA) with the European Union, necessary for the implementation by ESA of parts of the EU Space Programme under the 2021–2027 EU Multiannual Financial Framework (MFF).

Informal negotiations were initiated at the level of ESA's Director General and the Director-General of DG GROW in March, and continued throughout the year. In parallel, an FFPA Drafting Team was put in place to negotiate the incorporation of the high-level agreements reached at European Commission and ESA Director General level into an FFPA text. Progress was regularly reported to ESA Member States in the context of the AFC Advisory Group on the FFPA, AFC and Council.

On the EU side, the Council of the European Union and the European Parliament reached a provisional agreement on the EU Regulation for the Space Programme in December, and the MFF was adopted at around the same time. The EU Space Programme will have a financial envelope of €14.8 billion in current prices (€13.2 billion in 2018 prices), broken down by programme component as follows:

- Galileo and EGNOS: €9 billion
- Copernicus: €5.4 billion
- SSA and /GOVSATCOM: 442M€

The final adoption of the EU Space Regulation by Council is expected to take place on 19 April 2021, while the adoption in plenary session of the European Parliament is scheduled for May 2021.

In parallel, the usual, regular bilateral exchanges took place with the European Commission to ensure a smooth implementation by ESA of the EU flagship programmes and overcome challenges where necessary.

International Relations

The Agency now has formal relations with all the eight European Union members that are not members of ESA. The Agreement for a reinforced Association with Slovenia entered into force on 8 December. Latvia became an ESA Associate Member on 27 July. In December, Council approved the extension of the European Cooperating States (ECS) Agreement with Slovakia for a period of 18 months. The extension of the ECS Agreement with Bulgaria will be tabled for Council approval in March 2021 and followed by a second term of PECS under the reinforced ECS scheme. The End of PECS Period joint review with Cyprus was initiated on 17 November. The Implementing Arrangement with Croatia for Technical Assistance and Expertise was signed in May. ESA and Malta discussed hosting a second Maltese national trainee at ESA.

ESA and NASA concluded negotiations on the Memoranda of Understanding on the Flight Elements of the Mars Sample Return Campaign and on the Cooperation on the Civil Lunar Gateway, both signed in October. Initial meetings were held with NASA to address future cooperation on human lunar surface activities.

Sentinel 6-Michael Freilich, a key ocean observation satellite developed by ESA, NASA, EUMETSAT, NOAA and the European Commission, was launched on 21 November from Vandenberg Air Force Base.

In July, NASA and ESA tentatively agreed that the James Webb Space Telescope on Ariane 5 from the Guiana Space Centre (CSG) would be launched on 31 October 2021. This delay is mostly due to the impact of the COVID-19 pandemic.

ESA and Roscosmos decided in March to postpone the launch of the ExoMars lander and rover to the next launch window in 2022 to allow more time for testing. Cooperation progressed on ESA contributions to the Roscosmos Luna-25 and Luna-27 missions. Preparations of the launch of the Russian Multipurpose Laboratory Module with the European Robotic Arm (ERA) proceeded with ERA being transported to Baikonur.

The Arrangement concerning participation by the Canadian Government in ARTES was signed in April. In addition to their contribution to ARTES 4.0, Canada participated in Future EO Element 1, Copernicus Segment-4, Altius – Phase E, Arctic Weather Satellite, to GSTP and E3P, with a focus on Lunar and Mars Robotic Exploration. In December, Canada announced additional contributions to Mars Sample Return.

ESA and the National Space Science Centre of the Chinese Academy of Science made progress in the implementation of the joint SMILE mission. The two sides also worked on the European contributions to the Chinese Einstein Probe mission.

Following joint peer review selection, ESA and the National Remote Sensing Centre of China launched the fifth edition of the Dragon programme, which continues to yield high quality scientific data focusing on science and applications development in China using data from ESA and Chinese Earth observation missions.

ESA-JAXA space science collaboration progressed through joint activities related to XRISM, MMX, and BepiColombo. ESA negotiated an agreement with JAXA on HERA for which JAXA will contribute by providing a thermal infrared camera. The negotiation of an agreement on the use of SAR data, progress of the joint EarthCARE mission, and the setting up of a dashboard as part of the ESA-NASA-JAXA initiative on COVID-19 were Earth observation highlights. Cooperation on the development of the international habitation module (I-Hab) for the Lunar Gateway was prepared.

ESA and ISRO began identifying possible new areas of cooperation and negotiated a cross-support agreement for communication and mission operations support.

ESA and the Australian Space Agency met and identified possible joint projects and addressed the evolution of the ESA Deep Space Antenna in New Norcia with Australian contributions.

ESA and the South African National Space Agency signed a Cooperation Agreement on 17 November.

ESA continued cooperation with Argentinean entities to secure nominal operations of the ESA deep space antenna in Argentina.

ESA provided the Brazilian Space Agency and the National Institute for Space Research Sentinel images to cope with the Amazonia fires. ESA and the Mexican Space Agency (AEM) exchanged information on access to ESA data and expertise to address the COVID-19 pandemic in Mexico. ESA supported AEM with the monitoring of sargassum algae affecting the Mexican coast.

Education

In 2020 the ESA Education Programme focused on responding to the dramatic impact generated by the COVID-19 pandemic on Europe's education systems and daily practice, at all education levels.

Despite the magnitude of the challenge, the ESA Education Programme managed to respond in a timely way to this unexpected change of scenario and to deliver a rich portfolio of training programmes, projects and resources (<https://www.esa.int/Education>). While in-person events had to be cancelled from March onwards, with dedicated expert support provided by nearly all ESA directorates, a rapid conversion of activities into online training and remote support has been set up. Under strict health and safety conditions, a few in-person project campaigns for university students were able to take place successfully.

Among many other initiatives, in May 2020, the Education Office launched a new online platform called Expedition: Home (https://www.esa.int/Education/Expedition_Home). The website, aimed at children from 3 to 18 years old offered hands-on activities to be performed in a home setting; a new "Meet the Experts" video series featuring a variety of space topics and careers presented by ESA scientists and engineers; as well as specially designed home editions of school projects and two new Weekly Space Challenges.

Communication

ESA Member States increased the Agency's budget considerably at the Council meeting at ministerial level, Space19+, and urged the organisation to adapt to the fast-evolving ecosystem called New Space. Enforced by the global pandemic, the trend of converting physical face-to-face communication into digital-only has changed both societal communication and the communication profession itself.

As the world is forced to embrace accelerated digitalisation, ESA has maintained the ambition to promote its activities to the largest possible audience and raise awareness about European space activities carried out on behalf of its Member States. ESA is responding to this complex scenario in three ways.

First, the Newsroom and Editorial Board maintains an up-to-date publishing agenda, consistently prioritising internal and external communication opportunities. Under #SpaceCare, ESA's long-term lead campaign, key programmatic milestones across ESA programmes are communicated with the general public, allowing for engagement and ownership, despite being constrained to digital-only. Moving media relations fully into the digital universe, in terms of tools and media events, has resulted in greater media coverage of ESA. A newsjacking initiative, identifying daily news opportunities to align ESA's content publishing with trending societal news trends, has also contributed to the higher impact.

Second, to achieve the strategic objective of "OneESA talking with OneVoice" the Communication Department has reinforced the connection between internal and external communication, aligning priorities and synchronising messaging for both audiences. "This is ESA" is the embodiment of this: while available for the general public in all of ESA's Member State languages, it is also a set of

presentations for the ESA workforce to use in their professional and informal exchanges, together with supporting videos and social media guidelines.

In 2020 efforts to offer communication products in multiple Member State languages were further increased to allow “One Voice” to also go the last – national – mile. This paid off in the form of increased participation in various media events.

Creating an Advisory Committee on Communication (ACC) under the direct responsibility of the ESA Director General has brought ESA Communication closer to its Member States and has reinforced collaboration. The ACC supports ESA in defining and setting communication priorities. Member States’ content now features more often in ESA’s communication channels.

The rollout of a refreshed brand identity in 2020 and a simplified brand architecture have given “OneESA” a single and unique face. An online brand centre provides easy access to brand resources to ensure that ESA’s visual language can readily be used in all contexts. The Agency continues to promote its activities through partnerships and merchandising initiatives to multiply European citizens’ awareness of ESA in the context of popular music, computer games and children’s toys.

Third, the Communication Department has fully prioritised social, mobile, and video in its channel mix. Compared with 2019, ESA’s digital channels attracted 29% more users, with Instagram and YouTube contributing significantly to the growth, together with the fully mobile, responsive ESA home page.

ESA Web TV has delivered on its huge potential during the pandemic. Programmes such as “SpaceConnectsUs”, empowered astronauts, scientists, and other thought-leaders to deliver messages of encouragement to European audiences in six languages. An internal communication product of ESA Web TV, a video series entitled “Life at ESA during COVID-19” greatly helped in maintaining a sense of belonging amongst ESA’s population.

The public has also benefited from ESA Web TV providing an overview of European space activities for the non-English speaker.

Overall, all communication activities paid particular attention to a segment of individuals aged 12–35, focusing on women and diversity in general.

A more detailed report of the activities of the Communication Department in 2020 was presented to Council on 17 December (ESA/C(2020)154).

Corporate Development

Knowledge Management

ESA has continued to deploy knowledge management actions across the organisation, fostering the sharing culture and easing access to core knowledge and expertise. Processes, methods and tools have been deployed consistently across the different areas of the Agency.

In particular, a harmonised and coherent process to collect, analyse and make accessible Lessons Learned from ESA projects and activities was formalised with the publication of its related Policy and Process/Handbook. The ESA Lessons Learned Portal was successfully deployed ESA-wide and is now in operation, already containing more than 700 lessons proposed, validated and evaluated, coming from more than 21 ESA projects and activities and nine directorates.

To capture the tacit knowledge residing in ESA experts' heads before they retire, an Agency-wide process was formalised. 40 retiring key experts in seven directorates have been involved so far.

Corporate Social Responsibility

ESA achieved further progress in becoming a more sustainable and socially responsible organisation as well as a role model for the space sector. The Agency undertook a series of initiatives, following corporate social responsibility (CSR) principles and guidelines, developed in coordination with its stakeholders, and pursuing the sustainability of its activities through their application.

An Environmental Protection Working Group was set up to define action plans in response to environmental high-level objectives set for the Agency (30% of natural space in new buildings and refurbishment projects; zero disposable plastics by the end of 2020; by 2030 and compared to 2019, 40% fewer greenhouse gas emissions due to travel).

Meetings with CSR managers from the most important players in the European industrial space sector took place to define objectives towards a more sustainable and socially responsible European space sector. A white paper sharing the experiences and recommendations of the participants and calling policy- and decision-makers to action, will be presented at the next High-level Forum with industry.

ESA is also promoting a Charter for a Responsible Space Sector, ensuring that the purpose of the space sector's activities contributes to the sustainable development of society, without harming life on Earth or the space environment.

Advisory Bodies

Advisory Groups to the ESA Delegate Bodies

1. Technology Harmonisation Advisory Group (THAG)

The purpose of the Technology Harmonisation Advisory Group (THAG) is to advise the IPC on the harmonisation of European programmes in the technological research field and the preference to be given to technology developed within the Member States.

2. Technology Advisory Working Group (TA-WG)

The purpose of the Technology Advisory Working Group (TA-WG) is to advise the IPC on the Agency's technology policy and on the content of the Agency's Technology Programmes.

3. PB-HME advisory group: Exploration and Utilisation Board (EUB)

HME-EUB was established by the Human Spaceflight, Microgravity and Exploration Programme Board (PB-HME) to support PB-HME, in particular on matters related to science and research, including their technology aspects, and the utilisation aspects of the Agency programmes monitored by PB-HME. The main purpose of HME-EUB is to support the decision-making process at PB-HME in these areas.

4. PB-EO advisory group: Data Operations Scientific and Technical Advisory Group (DOSTAG)

The Earth Observation Programme Board has delegated to DOSTAG various functions limited to data, operations, scientific and technical aspects, for example reviewing the status and risks of all ESA Earth Observation programmes, including space, ground and exploitation segment aspects.

SEC advisory group: ESA INFOSEC Panel

The ESA Security Committee (SEC) has delegated to the ESA INFOSEC Panel all matters related to information and communication security.

In this capacity, the ESA INFOSEC Panel supports the ESA Security Office in its function as Security Accreditation Authority (SAA) and provides advice on an effective security programme for ESA in order to ensure the confidentiality, integrity and availability of classified information stored, processed and transmitted.

5. JCB Advisory Committee on 5G (5JAC)

The JCB 5G Advisory Committee (5JAC) is the advisory body to the JCB on matters related to the 5th Generation (5G) of communication networks.

The 5JAC mandate is currently under review, with the objective of aligning it with the ARTES 4.0 Strategic Programme Line “Space for 5G”, which Member States endorsed at Space19+ in November 2019. Its terms of reference are currently being revised.

6. Advisory Committee on Space Systems for Safety and Security (4S) (formerly on Governmental Secure Satcom)

The JCB Governmental Secure Satcom Advisory Committee is the advisory body to the JCB on matters related to ESA’s Govsatcom-related programmes and activities.

The Advisory Committee’s mandate is currently under review, with the objective of aligning it with the ARTES 4.0 Strategic Programme Line “Space Systems for Safety and Security” (4S), which Member States endorsed at Space19+ in November 2019.

7. Optical Communication - ScyLight Advisory Committee (SCOTT)

The JCB ScyLight Optical Communication Advisory Committee (SCOTT) is the advisory body to the JCB on matters related to the activities carried out under the ARTES 4.0 Strategic Programme Line “Optical Communication – ScyLight”.

It was proposed to establish the advisory committee following Member States’ endorsement of the Strategic Programme Line at Space19+ in November 2019. The advisory committee’s terms of reference are currently in the process of being defined.

Advisory Groups to the ESA Director General

1. Science advisory committees

The role of the science advisory committees is to advise on science and scientific activities taking into account the recommendations of the science working groups.

The science advisory committees are the Space Science Advisory Committee (SSAC), the Advisory Committee for Earth Observation (ACEO), the GNSS Science Advisory Committee (GSAC) and the Human Spaceflight and Exploration Science Advisory Committee (HESAC).

2. Space Science Advisory Committee (SSAC)

The Space Science Advisory Committee (SSAC) is the senior advisory body to the Director of Science (D/SCI), acting as the Director General’s representative on all matters concerning space science

included in the mandatory Science Programme of ESA. It is the main interpreter of the views and needs of the European scientific community as regards access to space experimentation and data exploitation in the mandatory Science Programme.

3. Advisory Committee for Earth Observation (ACEO)

The Advisory Committee for Earth Observation (ACEO) is the senior advisory body to the Director of Earth Observation (D/EOP), acting as the Director General's representative on all matters concerning Earth sciences included in the Earth Observation Programmes of ESA. It is the main interpreter of the views and needs of the European scientific community as regards access to space experimentation and data exploitation in the Earth science programme.

The ACEO also has the highest-level advisory capacity, through the Director of Earth Observation (D/EOP), on matters to be treated at the level of the Programme Board for Earth Observation (PB-EO), which may also request advice on particular issues of a scientific nature.

The transition from the existing ESAC (Earth Science Advisory Committee) to the newly-created ACEO was discussed during the 173rd meeting of PB-EO in February 2018, (ESAPB-EO(2018)10), including a description of the terms of reference of the ACEO.

4. GNSS Science Advisory Committee (GSAC)

The Global Navigation Satellite System (GNSS) Science Advisory Committee (GSAC) is the senior advisory body to the Director of Navigation (D/NAV) and the Director of Science (D/SCI), acting as the Director General's representatives on all matters concerning scientific exploitation of navigation programmes. It is the main interpreter of the views and needs of the European scientific community as regards uses of the GNSS Programme for research in Earth, navigational and physical sciences.

5. Human Spaceflight and Exploration Science Advisory Committee (HESAC)

The Human Spaceflight and Exploration Science Advisory Committee (HESAC) is the senior advisory body to the Directors of Human and Robotic Exploration (D/HRE) and Science (D/SCI), acting as the Director General's representatives on all matters concerning the scientific exploitation of human spaceflight, microgravity and exploration programmes. It is the main interpreter of the views and needs of the European scientific community as regards access to space experimentation and data exploitation in the human spaceflight, microgravity and exploration programmes.

6. Science working groups

Working groups report to the advisory committees on matters concerning their specific science areas. In order to provide a single set of science disciplinary working groups, avoiding duplication or conflicts of advice, the following are established to provide advice on their disciplines across the different science advisory committees.

The working groups support the work of all the science advisory committees and Directorates as appropriate. Any Directorate can request appropriate advice from any of the Working Groups.

7. Astronomy Working Group (AWG)

The Astronomy Working Group (AWG) advises on all science matters related to astronomy. It reports mainly to SSAC but also to HESAC and the ACEO (formerly ESAC).

8. Physical Sciences Working Group (PSWG)

The Physical Sciences Working Group (PSWG) advises on all aspects of material sciences, fundamental physics and experiments in microgravity conditions. It reports mainly to HESAC and SSAC, but also to the ACEO (formerly ESAC) and GSAC.

9. Solar System Exploration Working Group (SSEWG)

The Solar System Exploration Working Group (SSEWG) advises on all science matters related to solar system science and exploration, excluding Earth sciences. It reports mainly to SSAC but also to HESAC and the ACEO (formerly ESAC).

Accounts

Notes to the Financial Statements

As required by the ESA Financial Regulations, the Agency's annual Financial Statements have been prepared in accordance with International Public Sector Accounting Standards (IPSAS) since 2010.

The Financial Information presented herein was extracted from the audited ESA Financial Statements for the financial year 2020. This Financial Information consists of the following

extracted components:

- Statement of Financial Position;
- Statement of Financial Performance;
- Statement of Changes in Net Assets/Equity;
- Statement of Cash Flows.

In 2020 special attention was paid to the consolidation of earlier achievements. Following the Agency's Audit Commission recommendation to strengthen the internal control system of the Agency in order to meet the objectives of audit readiness, the following actions were undertaken and will continue to be pursued in the coming years:

- Continued optimisation of the interim and year-end closure process and further automation of the production of the Financial Statements;
- Creation of a formal HR risk matrix in relation to staff;
- Design of new internal controls and reports in fixed assets accounting (e.g. full automation of unplanned depreciation), thereby improving the audit trail in that domain;
- Continuation of the accounting risk assessment review of significant purchase contracts;
- Additional dedicated training delivered to relevant ESA staff on accruals and fixed assets accounting;
- Systematic monthly closure performed within 10 working days of month-end (for cost accounting);
- Revision of Financial Instructions;
- Automation of the process for collecting accruals.

The net surplus of 2020 amounts to 959.0 M€, after capitalisation of Assets Under Construction and work-in-progress, as compared to a surplus of 365.8 M€ in 2019. The breakdown is as follows:

- 1.193.1 M€ net surplus for ESA activities and programmes (326.2 M€ net surplus in 2019)
- 243.2 M€ net deficit for Pensions (32.0 M€ net surplus in 2019)
- 9.1 M€ net surplus for Social Security (7.7 M€ net surplus in 2019)

The increase of 593.2 M€ over the 2019 amount is mainly due to the increase for ESA programmes of 866.9 M€, partially offset by a decrease in respect of pensions of 275.2 M€.

The increased amount for ESA programmes is mainly due to increased contributions from Member States compared to 2019 reflecting the important commitments made at Space19+ on new and continued programmes and activities.

The Financial Statements for the Provident Fund are excluded from the accounts of the Agency and apply a separate procedure for their final approval.

The Financial Information presented hereinafter, was extracted from the ESA Financial Statements 2020. The ESA Financial Statements 2020 also include a summary of significant accounting policy and Notes to the Financial Statements and were audited by the Audit Commission of the European Space Agency. The Audit Commission conducted its audit in accordance with the International Standards on Auditing (ISA). As required by the Financial Regulations, this audit also included the aspect of regularity. In the Audit Commission's opinion, the 2020 audited Financial Statements give a true and fair view of the financial position of the European Space Agency as at 31 December 2020, and of its Financial Performance and Statement Cash Flows for the 12-month period ending 31 December 2020, in accordance with International Public Sector Accounting Standards (IPSAS).

Concerning the aspect of regularity, the Audit Commission reported that nothing came to its attention that would indicate that the transactions of ESA have not been made, in all significant respects, in accordance with the Regulations of the European Space Agency.

The audited Financial Statements and the Independent Auditor's report were submitted to the ESA Council in accordance with the ESA Financial Regulations, which were approved by the Council Meeting on 26 June 2021.

Statement of Financial Position

	31 December 2020 million €	31 December 2019 million €
Total Assets	14,135.5	13,112.9
Current assets	3,223.8	3,098.7
Cash and cash equivalents	1,573.4	1,302.7
Other financial assets	0.0	0.1
Recoverables from non-exchange transactions	258.2	412.2
Receivables from exchange transactions	1,044.1	801.2
Inventories	348.1	582.5
Non-current assets	10,911.7	10,014.2
Recoverables from non-exchange transactions	3.7	3.6
Receivables from exchange transactions	1,086.9	407.5
Financial assets	959.0	859.2
Property, plant and equipment	8,557.6	8,451.2
Intangible assets	304.5	292.7
Total Liabilities	12,399.2	11,783.0
Current liabilities	2,865.1	2,876.7
Payables under exchange transactions	2,378.9	2,302.4
Current payables	1,321.2	1,183.9
Pre-financing from exchange transactions	976.5	1,061.2
Deferred income	81.2	57.3
Contributions and other payables	212.5	358.5
Prepaid contributions and other amounts payable to Member/Participating States	164.4	277.2
Pre-financing from non-exchange transactions	3.3	5.2
Other payables	44.8	76.1
Provision for employee benefits	221.7	207.0
Provisions for risks and charges	52.0	8.8
Non-current liabilities	9,534.1	8,906.4
Provision for employee benefits	9,420.8	8,603.7
Other payables	0.0	302.1
Deferred income	95.1	0.0
Pre-financing from non-exchange transactions	0.5	0.6
Pre-financing from exchange transactions	17.7	0.0
Net Assets	1,736.3	1,329.8
Reserves	4,152.3	4,371.0
Amounts to be called from Member States	-3,375.0	-3,407.0
Net Surplus/ (Deficit) for the period	959.0	365.8
Net Assets	1,736.3	1,329.8

Statement of Financial Performance

	2020 million €	2019 million €
Operating Revenues	6,736.4	5,728.8
Contributions from Member/Participating States	4,800.2	4,043.6
Other contributions and grants	36.5	91.3
Revenues from exchange transactions	1,595.0	1,078.8
Staff contributions	49.1	272.4
Other operating income	255.6	242.6
Operating Expenses	-5,853.3	-5,437.9
Purchases	-2,814.1	-2,453.2
External services	-1,086.5	-1,063.9
Staff expenses	-697.1	-681.4
Other operating expenses	-254.1	-242.9
Expenses from non-exchange transactions	-165.2	-99.0
Depreciation and amortization	-803.0	-849.9
Impairment	56.0	7.9
Write-downs inventories and fixed assets, provisions for future losses	-89.3	-55.5
Net Surplus / (Deficit) from Operating activities	883.1	290.8
Financial activities	75.9	75.0
Financial revenue	116.0	79.4
Financial expenses	-40.1	-4.4
Net Surplus / (Deficit) for the period	959.0	365.8

Statement of Changes in Net Assets/Equity

Statement of Changes in Net Assets /Equity

		01 January 2019 as previously reported	Correction of transfers between reserves - 2018 movements	01 January 2019 restated	Correction of transfers between reserves - movements 2019*	Allocation of audited 2018 surplus/ (deficit)	Movements 2019	31 December 2019 restated
		million €				million €	million €	million €
RESERVES	A	5,240.4	0.0	5,240.4	0.0	752.0	-1,621.4	4,371.0
Member States Capital and Reserves		-980.8	30.0	-950.8	-11.6	-574.0	-1,622.1	-3,158.5
Revaluation Reserve		1,826.0	30.2	1,856.2	-11.6	-256.6	-1.1	1,586.8
Currency Exchange Gains Reserve		3.0	-0.1	2.9				2.9
Sundry Reserves		0.1		0.1				0.1
Other Reserves		-2,809.9		-2,809.9		-317.4	-1,621.0	-4,748.3
<i>Social security reserve</i>		<i>26.8</i>		<i>26.8</i>		<i>1.1</i>	<i>0.7</i>	<i>28.6</i>
<i>Buffer Fund fair value reserve</i>		<i>1.0</i>		<i>1.0</i>			<i>12.1</i>	<i>13.1</i>
Reserve Barter		35.4		35.4		-318.5		-283.1
Accumulated actuarial gains and losses		-2,873.1		-2,873.1			-1,633.8	-4,506.9
Accumulated Surplus		6,221.1	-30.0	6,191.1	11.6	1,326.0	0.7	7,529.4
Accumulated Surplus ESA programmes		6,067.6	-18.9	6,048.7	11.6	1,296.1	0.7	7,357.0
Accumulated Surplus Third Party programmes		139.1	-11.1	128.0		27.0		154.9
Accumulated Surplus PECS		14.4	0.0	14.4		3.0		17.4
AMOUNTS TO BE CALLED FROM MEMBER STATES	B	-3,158.3	0.0	-3,158.3	0.0	-248.7	0.0	-3,407.0
Member States pension scheme receivable		-3,158.3		-3,158.3		-248.7		-3,407.0
Accumulated Deficit								
Net Surplus / (Deficit) for the Period	C	503.3		503.3		-503.3	365.8	365.8
				0.0				
TOTAL NET ASSETS	A+B+C	2,585.4	0.0	2,585.4	0.0	0.0	-1,255.6	1,329.8

Statement of Changes in Net Assets /Equity

		31 December 2019 restated	31 December 2019 restated	Allocation of audited 2019 surplus/ (deficit)	Movements 2020	31 December 2020
		million €	million €	million €	million €	million €
RESERVES	A	4,371.0	4,371.0	333.8	-552.5	4,152.3
Member States Capital and Reserves		-3,158.5	-3,158.5	-616.2	-499.1	-4,273.8
Revaluation Reserve		1,586.8	1,586.8	-501.4	-0.2	1,085.2
Currency Exchange Gains Reserve		2.9	2.9	0.2		3.1
Sundry Reserves		0.1	0.1			0.1
Other Reserves		-4,748.3	-4,748.3	-115.0	-498.9	-5,362.1
Social security reserve		28.6	28.6	7.7	0.5	36.8
Buffer Fund fair value reserve		13.1	13.1		2.1	15.2
Reserve Barter		-283.1	-283.1	-122.6		-405.7
Accumulated actuarial gains and losses		-4,506.9	-4,506.9		-501.5	-5,008.5
Accumulated Surplus		7,529.4	7,529.4	950.0	-53.4	8,426.0
Accumulated Surplus ESA programmes		7,357.0	7,357.0	945.1	-0.8	8,301.3
Accumulated Surplus Third Party programmes		154.9	154.9	1.9	-52.6	104.2
Accumulated Surplus PECS		17.4	17.4	3.0		20.4
AMOUNTS TO BE CALLED FROM MEMBER STATES	B	-3,407.0	-3,407.0	32.0	0.0	-3,375.0
Member States pension scheme receivable		-3,407.0	-3,407.0	32.0		-3,375.0
Accumulated Deficit						
Net Surplus / (Deficit) for the Period	C	365.8	365.8	-365.8	959.0	959.0
TOTAL NET ASSETS	A+B+C	1,329.8	1,329.8	0.0	406.4	1,736.3

*The fixed asset first recognition reserve was created at 01.01.2010, as part of the “revaluation reserves”, to value the cost of fixed assets and assets under construction accrued before the first application of IPSAS. Since 01.01.2010, all transactions affecting the historical fixed assets and assets under construction are carefully tracked and the related reserve is decreased accordingly in accordance with ESA internal rules. In 2018 and 2019, the fixed asset revaluation reserve was improperly decreased and adjustments had to be consequently posted in 2020 to reinstate the correct balance of the reserve.

Statement of Cash Flow

	2020 million €	2019 million €
Net Cash Flows	270.6	379.1
Net Cash Flows from Operating Activities	1,283.2	1,520.7
Surplus/(deficit) from ordinary activities	959.0	365.8
<i>Reclassification to investing activities</i>	-73.3	-16.3
<i>Non-cash movements</i>	862.9	606.1
Depreciation and impairment	747.0	842.0
Provisions and Accruals	-2.3	-6.0
Write-downs / Adjustments inventories and fixed assets, provisions for future losses on inventories	88.6	55.0
Revaluation of financial assets at fair value	29.6	-60.6
Past service costs	0.0	-224.3
<i>Decrease (Increase) in Deferrals/accruals of past or future operating cash receipts or payments</i>	-465.4	565.0
Decrease/(increase) in inventories	284.8	-141.4
Decrease/(increase) in recoverables from Member States	154.9	65.5
Decrease/(increase) in other receivables	-709.0	-88.5
Decrease/(increase) in other current assets	-214.0	399.2
Increase/(decrease) in payable to Member/Participating States	-113.4	85.6
Increase/(decrease) in pre-financing	-68.9	93.8
Increase/(decrease) in other payables	-130.2	-169.3
Increase/(decrease) in provisions for employee benefits	330.3	320.2
Net Cash Flows from Investing Activities	-1,012.5	-941.6
Acquisition of fixed assets	-959.0	-893.4
Proceeds from sales of fixed assets	0.0	0.0
Purchases of financial investments (portfolio)	-517.0	-149.9
Proceeds on disposal of financial investments (portfolio)	455.7	93.6
Interests received on financial assets	7.8	8.0
Net Cash Flows from Financing Activities	0.0	-200.0
Proceeds from borrowings	0.0	200.0
Repayment of borrowings	0.0	-400.0
Net increase / (decrease) in cash and cash equivalents	270.6	379.1
Cash and cash equivalents at the beginning of the year	1,302.7	923.6
Cash and cash equivalents at year-end	1,573.4	1,302.7

ANNEX

Annex 1 – Agreements signed in 2020

- The Agreement between the United Kingdom Space Agency and the European Space Agency concerning the provision of Technical Assistance and Expertise in the framework of the Commercial Space Transportation Services and Support to Member States Programme was signed in Paris, France, on 27 February 2020 by the ESA Director General, Johann-Dietrich Wörner, and in Swindon, United Kingdom, on 10 March 2020 by the UKSA Chief Executive, Graham Turnock. Accordingly, this Agreement entered into force on 10 March 2020 for an initial period of five years. (LEG/506)
- The Arrangement between ESA and the Government of Canada concerning the participation by the Government of Canada in the Advanced Research in Telecommunication Systems (ARTES) 4.0 Programme was signed on 17 April 2020 by the ESA Director General, Johann-Dietrich Wörner, and on 20 April 2020 by Jean-Pierre Arsenault, Permanent Delegate of Canada at ESA. Accordingly, this Arrangement entered into force on 20 April 2020 for as long as Canada participates in the ARTES 4.0 Programme, or until completion of the ARTES 4.0 Programme, whichever comes first. (LEG/507)
- The Memorandum of Understanding between the National Aeronautics and Space Administration and the European Space Agency concerning the Flight Elements of the Mars Sample Return Campaign was signed on 23 September 2020 in Washington D.C. by the NASA Administrator, James F. Bridenstine, and on 5 October 2020 in Paris by the ESA Director General, Johann-Dietrich Wörner. Accordingly, this Memorandum of Understanding entered into force on 5 October 2020 for a period until 30 September 2033. (LEG/508)
- The Agreement between the European Space Agency and the International Space University was signed on 15 June 2020 in Illkirch-Graffenstaden by the President of the International Space University, Juan de Dalmau, and on 21 July 2020 in Paris by the Director General of ESA, Johann-Dietrich Wörner. Accordingly, this Agreement entered into force on 21 July 2020 for a period until 20 July 2023. (LEG/509)
- The Memorandum of Understanding between the National Aeronautics and Space Administration and the European Space Agency concerning Cooperation on the Civil Lunar Gateway was signed on 22 October 2020 in Darmstadt by the ESA Director General, Johann-Dietrich Wörner, and on 27 October 2020 in Tulsa, Oklahoma by the NASA Administrator, James F. Bridenstine. Accordingly, it entered into force on 27 October 2020. (LEG/510)
- The Association Agreement between the European Space Agency and the Government of the Republic of Latvia was signed in Paris on 24 June 2020 by the ESA Director General, Johann-Dietrich Wörner, and in Riga on 30 June 2020 by Ilga Šuplinska, Minister for Education and Science of the Republic of Latvia. Accordingly, it entered into force on 27 July 2020 for a period until 27 July 2027. (LEG/511)
- The Extension of the European Cooperating State (ECS) Agreement between the European Space Agency and the Government of the Republic of Lithuania was agreed via an exchange of letters signed in Paris on 23 July 2020 by the ESA Director General, Johann-Dietrich Wörner, and in Vilnius on 14 September 2020 by Rimantas Sinkevičius, Minister of Economy and Innovation of the

Republic of Lithuania. Accordingly, the Agreement has been extended until 27 September 2021. (LEG/449, add. 1)

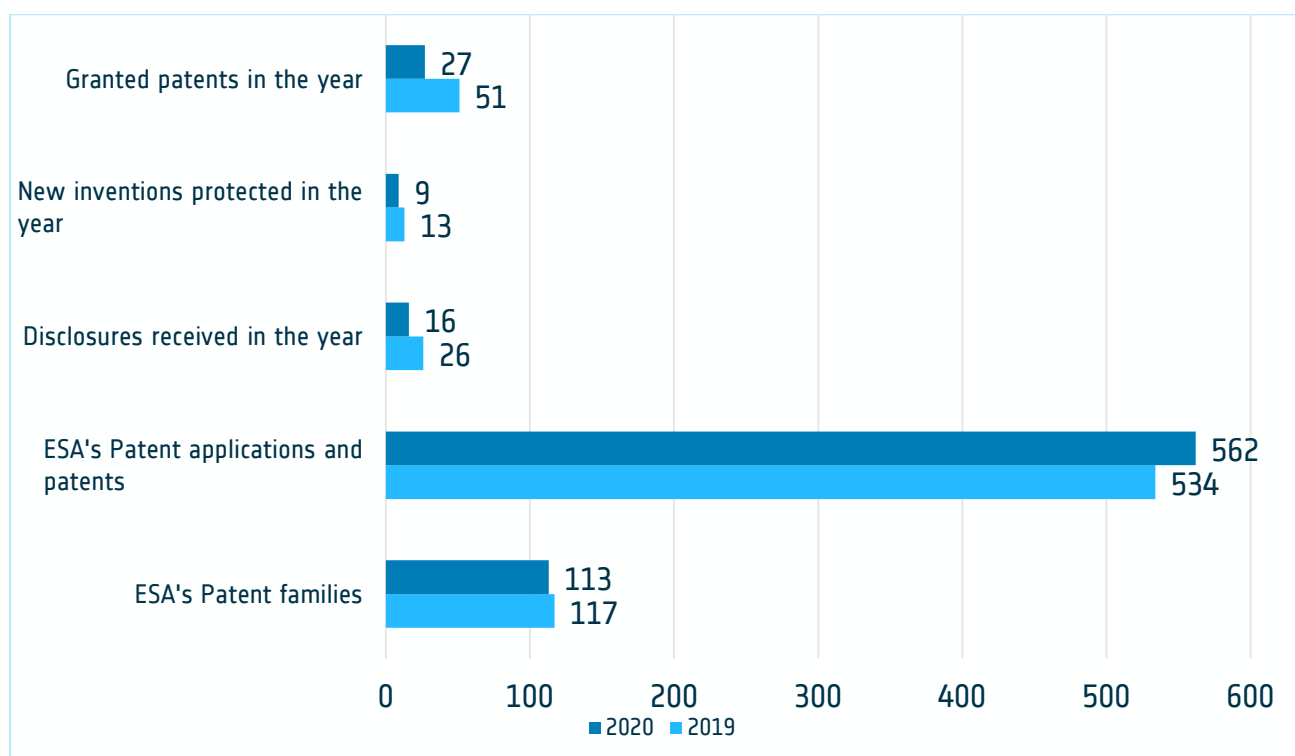
- The Agreement between the European Space Agency and the South African National Space Agency (SANSA) concerning space cooperation for peaceful purposes was signed on 10 November 2020 in Darmstadt by the ESA Director General, Johann-Dietrich Wörner, and on 17 November 2020 in Pretoria by the SANSA Chief Executive Officer, Valanathan Munsami. Accordingly, it entered into force on 17 November 2020 for a period of five years. (LEG/512)
- The Association Agreement between the European Space Agency and The Government of the Republic of Slovenia was signed in Paris on 8 October 2020 by Johann-Dietrich Wörner, ESA Director General, and in Ljubljana on 19 October 2020 by Zdravko Počivalšek, Minister for Economic Development and Technology of the Republic of Slovenia. Accordingly, it entered into force on 8 December 2020 for a period until 31 December 2024. (LEG/513)
- The Agreement between the European Space Agency (ESA) and the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) concerning cooperation in the fields of flight operations and related ground systems was signed in Darmstadt on 2 December 2020, by the ESA Director General, Johann-Dietrich Wörner, and by Alain Ratier, Director General of EUMETSAT. Accordingly, it entered into force on 2 December 2020 for a period until 31 December 2025. (LEG/514)
- The Memorandum of Understanding between the National Aeronautics and Space Administration and the European Space Agency concerning the Peregrine Ion Trap Mass Spectrometer (PITMS) Instrument was signed on 18 December 2020 in Paris by Frédéric Nordlund, Head of ESA's External Relations Department, and on 22 December 2020 in Washington D.C. by Gilbert R. Kirkham, Science Division Director of NASA's Office of International and Interagency Relations. Accordingly, it entered into force on 22 December 2020 for a period until 5 years from the date of launch of the Lander spacecraft, or until 30 June 2026, whichever is earlier. (LEG/515)

Annex 2 – Patent Groups

Introduction

At the end of 2020, ESA's portfolio contained 113 inventions protected by 562 patents.

- 36 patents were granted in 2020 by patent offices in different countries around the world.
- 16 new invention disclosures were received and processed by the Secretariat. The ESA Patents Group decided to protect six inventions in its own name and transferred two to industry partners.
- 60% of the portfolio is in use under ESA programmes or licensed.
- 17 new licenses were signed with industry.



2019-2020 Portfolio overview – Credit: ESA

List of Patents

Patents published:

- Variable Metasurface Antenna Structures ([WO2020244743A1](#))
- A Radio Frequency pass-band filter ([WO2020057722A1](#))
- Magnetic Damping for Space Vehicles after End-Of-Life ([EP3757021A1](#))
- Network for forming Multiple Beams from a Planar Array ([WO2019233565A1](#))
- Method and apparatus for performing joint channel and time estimation in a GNSS receiver ([WO2019223870A1](#))

Annex 3 – Chairs of Council and Delegate Bodies

Chair of Council	Ms Anna Rathsmann (SE)
Vice-Chairs:	Mr Renato Krpoun (CH)
	Mr Juan Carlos Cortés Pulido (ES)
Science Programme Committee (SPC)	Mr Werner Verschueren (BE)
Administrative and Finance Committee (AFC)	Mr Wouter Pont (NL)
Industrial Policy Committee (IPC)	Mr George Pritchard (UK)
International Relations Committee (IRC)	Ms Nicoletta Bini (IT)
Security Committee (SEC)	Mr Georgios Tsapanos (DE)
Programme Board on Satellite Navigation (PB-NAV)	Ms Sannah Plenaa Thorngreen (DK)
Joint Board on Communication Satellite Programmes (JCB)	Mr Alberto Tuozi (IT)
Earth Observation Programme Board (PB-EO)	Ms Juliette Lambin (FR)
Data Operations Scientific and Technical Advisory Group (PB-EO-DOSTAG)	Mr Ondřej Šváb (CZ)
Programme Board for Human Spaceflight, Microgravity and Exploration (PB-HME)	Ms Marianne Tantillo (NO)
Exploration and Utilisation Board (PB-HME-EUB)	Ms Libby Jackson (UK)
Space Transportation System Programme Board (PB-STs)	Mr Claus Lippert (DE)
Programme Board for Space Situational Awareness (PB-SSA)	Mr Stephan Mayer (AT)

Annex 4 – National Delegations to Council (as of 31/12/2020)

AUSTRIA

Mr Klaus Pseiner

Mr Andreas Geisler

Ms Margit Mischkulnig

BELGIUM

Mr Frank Monteny

Mr Pierre Coquay

Mr Hendrik Verbeelen

CZECH REPUBLIC

Mr Václav Kobera

Mr Ondřej Šváb

DENMARK

Mr Peter Sloth

Mr Gorm Kofoed Petersen

ESTONIA

Mr Madis Võõras

Mr Paul Liias

FINLAND

Mr Kimmo Kanto

Ms Maija Lönnqvist

Mr Pauli Stigell

FRANCE

Mr Jean-Yves Le Gall

Mr Jean-Pascal Le Franc

Mr Flavien Mie

Mr Lionel Suchet

GERMANY

Mr Walther Pelzer

Mr René Kleeßen

Mr Max Kroymann

Mr Klaus Steinberg

GREECE

Mr Dimitrios Kagklis

Ms Argyro Kagia

HUNGARY

Ms Orsolya Ildikó Ferencz

Mr Ferenc Horvai

IRELAND

Ms Ciara Bartley

Mr Tony McDonald

ITALY

Mr Giorgio Saccoccia

Mr S.E.M. Ambassador Antonio Bernardini

Mr Marco Maria Cerbo

Ms Maria Cristina Falvella

LUXEMBOURG

Mr Marc Serres

Mr Mathias Link

NETHERLANDS

Mr Eelco van der Eijk

Mr Harm van de Wetering

NORWAY

Mr Christian Hauglie-Hanssen

Mr Geir Hovmork

POLAND

Mr Piotr Zabadala

Ms Marta Kolibabska

PORTUGAL

Mr Ricardo Conde

Mr Hugo Costa

ROMANIA

Mr Marius-Ioan Piso

Ms Oana Maria Neagu

SPAIN

Mr Francisco Javier Ponce Martinez

Mr Juan Carlos Cortés Pulido

Mr Jorge Lomba Ferreras

SWEDEN

Mr Christer Nilsson (acting Head of
Delegation)

Mr Magnus Härviden

SWITZERLAND

Mr Renato Krpoun

Mr Raphaël von Roten

Mr Urs Frei

UNITED KINGDOM

Ms Alice Bunn

Ms Maria Cody

Mr Graham Turnock

ASSOCIATE MEMBERS

LATVIA

Mr Kaspars Karolis

Mr Mārtiņš Jansons

SLOVENIA

Ms Tanja Permozer

Ms Sabina Kolesa

COOPERATING STATE

CANADA

Ms Lisa Campbell

Ms Marie-Claude Guérard

Mr Jean-Pierre Arseneault